

Monochromatic finite sums and products in the positive integers

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Abstract

We prove Hindman’s finite sums and products conjecture: for every finite coloring of the positive integers and every positive integer k , there is a k -element set whose nonempty subset sums and nonempty subset products all have the same color.

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1 Introduction

Write $[n] = \{1, \dots, n\}$ for a positive integer n . For a finite set $A \subset \mathbb{N} = \{1, 2, \dots\}$, write

$$\text{FS}(A) = \left\{ \sum_{a \in B} a : \emptyset \neq B \subseteq A \right\}, \quad \text{FP}(A) = \left\{ \prod_{a \in B} a : \emptyset \neq B \subseteq A \right\}.$$

Thus each element of A may occur at most once in an individual sum or product, and singleton subsets are included. We prove the following.

Theorem 1.1. *Let $r, m \geq 1$ be integers, and fix real numbers $R \geq 2$ and $D \geq 1$. For every coloring $\chi : \mathbb{N} \rightarrow [r]$, there are distinct positive integers $a_1 < \dots < a_m$ and a color $c \in [r]$ such that*

$$\chi \left(\sum_{j \in J} a_j \right) = \chi \left(\prod_{j \in J} a_j \right) = c \quad \text{for every } \emptyset \neq J \subseteq [m].$$

Equivalently, $\text{FS}(A) \cup \text{FP}(A)$ is monochromatic for a set $A \subset \mathbb{N}$ of cardinality m . The elements can additionally be chosen to satisfy

$$a_1 > R, \quad a_d > R \left(\sum_{k < d} a_k + \prod_{k < d} a_k \right)^D \quad (2 \leq d \leq m). \quad (1.1)$$

Theorem 1.1 resolves Hindman's finite sums and products conjecture positively. The same assertion for colorings of $\mathbb{N}_0 = \{0, 1, 2, \dots\}$ follows by restriction to $\mathbb{N}$. The separation in (1.1) also eliminates every collision between expressions except the shared singleton values.

Corollary 1.2. *For every finite coloring of $\mathbb{N}$ and every $m \geq 1$, there is an m -element set $A \subset \mathbb{N}$ for which $\text{FS}(A) \cup \text{FP}(A)$ is monochromatic and*

$$|\text{FS}(A)| = |\text{FP}(A)| = 2^m - 1, \quad \text{FS}(A) \cap \text{FP}(A) = A.$$

In particular, the union has $2(2^m - 1) - m$ distinct elements.

The proof of the corollary is elementary and appears at the end of Section 2. Both conclusions concern prescribed finite sets; the separation does not assert the existence of an infinite simultaneous sequence.

1.1 History and significance

The separate additive and multiplicative problems have classical answers. Schur's theorem guarantees a monochromatic triple $\{x, y, x + y\}$ in every finite coloring of $\mathbb{N}$ [31]. The finite sums theorem of Folkman–Rado–Sanders extends this to the nonempty subset sums of arbitrarily large finite sets; Sanders's original argument appears in [30, Theorem 2 and Corollary 1.1]. Hindman proved the infinite finite sums theorem [19]. Applying an additive theorem to the coloring $n \mapsto \chi(2^n)$ gives its multiplicative counterpart. The simultaneous question is different: Hindman constructed a finite coloring of $\mathbb{N}$ with no infinite set whose terms, pairwise sums and pairwise products all have one color [21]. His finite conjecture asks for arbitrarily large finite sets instead [20]; see also [23, Question 17.18].

For two variables, computer-assisted arguments of Graham and Hindman already established the existence of a monochromatic quartet $\{x, y, x + y, xy\}$ in every two-coloring of $\mathbb{N}$; see the historical account in [22, p. 82]. Bowen gave a proof in which x and y are distinct and exceed any prescribed bound, and more generally controlled the individual variables, their prefix products and their total sum [8, Theorem 1.1]. For arbitrary finite colorings, Moreira proved the existence of monochromatic triples $\{x, x + y, xy\}$ as part of a wider theorem on prefix products and polynomial shifts [27, Theorem 1.4 and Corollary 1.5]. Alweiss gave a polynomial proof of the mixed triple and related polynomial-shift patterns [2, Theorem 1.2]. More recently, Alweiss, Bowen and Sabok proved that every two-coloring contains $\{x, y, xy, x + iy : 1 \leq i \leq k\}$ in one color for every prescribed k [3, Theorem 1.2].

Results over fields developed in parallel. Green and Sanders proved the four-term result in sufficiently large prime fields [11, Theorem 1.2], while Bergelson and Moreira developed affine ergodic methods for mixed additive–multiplicative patterns [7, 6]. Bowen and Sabok established $\{x, y, x + y, xy\}$ over $\mathbb{Q}$ [9, Theorem 1.1], and Alweiss proved the full finite sums and products theorem over $\mathbb{Q}$ [1, Theorem 1.3]. Richter obtained integer density theorems for the mixed pair $\{x + Q(y), xy\}$ using Fourier analysis, ergodic theory and logarithmic averages over products of primes [29, Theorems 1.5 and 1.7].

The passage from rational to integer configurations requires more than clearing denominators. A common dilation scales a sum by one power of the dilation and a product of j entries by its j th power; an arbitrary coloring need not respect these different changes. The ordered-block selection in our proof follows the Ramsey and finite-sums argument in [1, Section 5]. Our Alignment argument also adapts the compensating scale updates in [1, Section 4], which fix the current target product while retaining the requirements secured at earlier targets. The additional task is to control the additive shifts arithmetically while preserving all the required multiplicative colors.

1.2 Proof strategy

The finite combinatorial part of the argument uses ordered blocks of widely separated integer variables. A Ramsey argument and the finite sums theorem select block products whose

nonempty products already have one color. The remaining task is to ensure that their sums have that same color. We do this by constructing bounded predictions for the color indicators and then arranging that those predictions remain positive at the required sums.

The proof separates these tasks into a Prediction Principle and an Alignment Principle. Prediction replaces the color functions inside the relevant counts by piecewise nilsequences (Lipschitz functions evaluated along nilpotent orbits, separately on intervals and residue classes), while retaining the multiplicative color masks. It also ensures that a color occurring at a center is unlikely to have a small predicted value there. Alignment selects from a fixed finite list of rational scale vectors so that a positive prediction at every center remains positive after all the required additive shifts. Its success probability is bounded below independently of the complexity of the nilsequence models. This uniformity permits the final choice of small calibration errors.

Three constructions provide the analytic and arithmetic inputs.

1. *Weights for the distribution of a block product.* Each chosen element is a product of several raw variables, so its distribution differs from that of a single variable at the same largest scale. We encode divisibility by the earlier variables in nonnegative weights and attach the same weight to every sum ending at that block. Prime substitutions and weighted Cauchy–Schwarz remove the product-color conditions and reduce the counting error to one-variable additive tests.
2. *Predictions at two additive scales.* The counting estimate tests short additive shifts, whereas Alignment needs one model valid across a larger interval. We construct nilsequence models on nested intervals and use Ramsey selection to make their projections close. This supplies a common model for both requirements, with nilpotence step depending only on the requested configuration size.
3. *Realizing shifts through nilsequence states.* Adding an earlier block product may require a noninteger change in the largest-scale variable of that block. We first make an integer residue correction and then realize the remaining displacement as a transformation of the nilsequence model. Two separate arguments justify this transformation: comparison of averages on arithmetic progressions, and a lifting theorem that makes transformations for different models act on one common state. These transformations generate a nilpotent group whose step is independent of model complexity. Finite polynomial recurrence in this group gives the required uniform alignment probability.

The rough-step comparison and the passage from marginal cube symmetries to joint lifts are stated separately in Propositions 6.3 and 7.5. Their hypotheses make explicit the joint polynomial dependence and the cube information needed to control otherwise unavailable shifts. The proof uses the inverse theorem of Green, Tao and Ziegler [17, 18], the concatenation theorem of Tao and Ziegler [34], quantitative polynomial equidistribution [12, 13], and nilpotent polynomial recurrence [37]. We state the required forms when they are used. The intervening transference, progression-comparison and lifting arguments are proved below.

Section 2 gives the two principles and derives Theorem 1.1 from them, including integrality and separation. Sections 3–5 develop Prediction; Sections 6–8 develop Alignment. The missing-corner Lemma 7.2, proved independently in Section 7, is also used in Prediction. All parameters are qualitative: their finiteness and order of choice are essential, but no numerical bound for the smallest configuration is claimed.

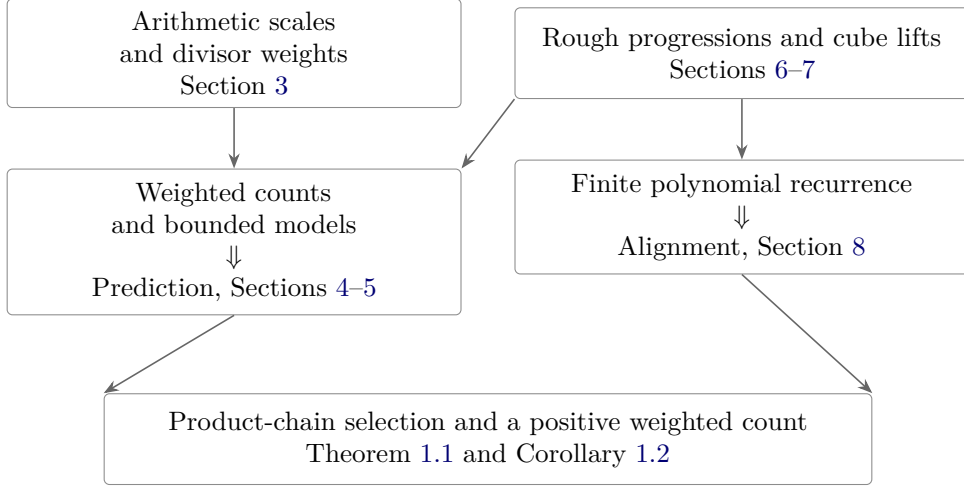


Figure 1: The main analytic routes meet in the combinatorial deduction. The diagonal arrow records the shared missing-corner reconstruction from Lemma 7.2. Alignment fixes a positive mass before Prediction fixes model complexity; this order permits the counting error to be made small enough.

2 Two principles and the combinatorial deduction

We first state the two analytic principles in the precise forms used to prove Theorem 1.1. This also fixes the order of all parameters. The proof of the Prediction Principle occupies Sections 3–5, with the independent missing-corner Lemma 7.2 proved in Section 7. Sections 6 and 7 develop the nilsequence tools for the Alignment Principle, proved in Section 8.

2.1 Ordered blocks and admissible scales

For nonempty subsets $A, B \subseteq [n]$, $A < B$ means $\max A < \min B$; we write $A < i$ for $A < \{i\}$. For a list of multiplicative quantities $u_1, \dots, u_n$, put $u_A = \prod_{j \in A} u_j$. In particular this convention applies to t_A, h_A, b_A , and x_A . Empty products, when they occur, equal one.

A *block* is a set $B = T \cup \{i\}$ with $\emptyset \neq T < i$. We call i its *pivot* and T its *tail*. For such a block let

$$\mathcal{E}_B = \{P \cup \{j\} : \emptyset \neq P < T < j < i\}. \quad (2.1)$$

These are the blocks whose values will be added at the anchor B . A *chain of length m* is a list $B_d = T_d \cup \{i_d\}$, $d \in [m]$, satisfying

$$\emptyset \neq T_1 < T_2 < \dots < T_m < i_1 < \dots < i_m. \quad (2.2)$$

All blocks in a chain are disjoint. Moreover, if $k < d$, then $B_k \in \mathcal{E}_{B_d}$. This last property is why we use this particular ordering of tails and pivots.

All asymptotic parameters below are indexed by positive integers $w \rightarrow \infty$, and

$$W = W(w) = \prod_{p \leq w} p,$$

where the product is over primes. An integer is *w-smooth* if all its prime factors are at most w . We say that a positive quantity A *dominates powers* of a quantity $B \geq 2$ if $A/B^C \rightarrow \infty$ for every fixed $C > 0$. Every finite index set, complexity bound and tolerance is held fixed in this convention unless additional uniformity is stated. Sampling an interval always means sampling its integer points.

Fix once and for all a nonprincipal ultrafilter $\mathcal{U}$ on the positive integers w , and write $\lim_{\mathcal{U}}$ for its limit. Every bounded real sequence has such a limit, and it agrees with the ordinary limit when that exists. Every set belonging to $\mathcal{U}$ meets every sufficiently late range of w . These are the limit and extraction properties used below.

Definition 2.1 (Admissible parameters). For a fixed n , an admissible family consists of positive integers $M, h_1, \dots, h_n, H_1, \dots, H_n$, positive powers of two $X_1, \dots, X_n$, and independent random variables $t_1, \dots, t_n$, all depending on w , with the following properties.

1. M and each h_i are w -smooth multiples of W^w . For every block B , and also for every singleton $\{i\}$, $h_B \leq M$. For every $A \in \mathcal{E}_B$, the ratio h_A/h_B is an integer multiple of W^w .
2. Each H_i is a multiple of M and dominates powers of $2 + M + \prod_{\ell < i} X_\ell$. The quantity $\log X_i$ dominates powers of H_i .
3. The law of t_i is the harmonic W -unit probability measure

$$\mu_i(y) = \frac{\mathbf{1}_{X_i \leq y < X_i^2} \mathbf{1}_{(y, W)=1}}{Z_i y}, \quad Z_i = \sum_{\substack{X_i \leq y < X_i^2 \\ (y, W)=1}} \frac{1}{y}. \quad (2.3)$$

Fixed positive rational factors have only finitely many denominator primes and fixed denominator valuations. Consequently they can be absorbed by W^w for all sufficiently large w . In particular, every color argument used below is eventually a positive integer. For a chain and fixed positive rational $a_1, \dots, a_m$, the ratios $h_{B_k} a_k / (h_{B_d} a_d)$, $k < d$, are eventually positive integers: the pair belongs to (2.1). These facts allow us to use rational scale lists without ever evaluating the coloring on a noninteger.

2.2 Piecewise nilsequence models

A *nilsequence of step at most s* is a sequence

$$k \longmapsto F(g^k x), \quad k \in \mathbb{Z}, \quad (2.4)$$

where G is a connected simply connected nilpotent Lie group of step at most s , $\Gamma \subset G$ is a lattice, $x \in G/\Gamma$, $g \in G$, and F is a Lipschitz function on G/Γ . Throughout this paper a family has *bounded complexity* if its nilmanifolds belong to a fixed finite list and its observables have uniformly bounded supremum and Lipschitz norms for fixed smooth metrics. No bound is imposed on the translating elements g or the points x . Constants are included. Products and Lipschitz combinations of any fixed finite number of such families remain bounded-complexity nilsequences of the same maximal step, by taking product nilmanifolds.

Definition 2.2 (Piecewise models). Fix finite sets $\mathcal{A} \subset \mathbb{Q}_{>0}$ and $[r]$, and an admissible family. A system of step-at-most- s models consists of functions $S_{B,a,c} : \mathbb{Z} \rightarrow [0, 1]$, one for each block B , $a \in \mathcal{A}$, and $c \in [r]$. If $i = \max B$, then on each interval $[kH_i, (k+1)H_i)$ and each residue class modulo M , the function is a nilsequence in the progression index. The complexity is bounded uniformly over w , all pieces, and all model indices. The representing observables may be, and are, chosen $[0, 1]$ -valued on their entire nilmanifolds.

The sequence and its nilmanifold may change from piece to piece within the fixed finite list. This freedom is necessary in the inverse-theorem argument. Real parts followed by clipping to $[0, 1]$ justify the final sentence of Definition 2.2 without increasing the nilpotence step.

For a block $B = T \cup \{i\}$ define its divisor weight on all of $\mathbb{Z}$ by

$$\nu_B(y) = \mathbb{E}_{\sigma=t_T} \sigma \mathbf{1}_{\sigma|y}. \quad (2.5)$$

The expectation uses the raw laws of the tail variables only. For a fixed tail product σ , we have $(\sigma, W) = 1$, so the conditional mass of σt_i at y is

$$\frac{\sigma \mathbf{1}_{\sigma|y} \mathbf{1}_{(y,W)=1}}{Z_i y} \quad \text{on } [\sigma X_i, \sigma X_i^2).$$

Thus ν_B averages the divisibility factor inserted by multiplication by the tail. Corollary 3.2 controls the change of endpoints back to $[X_i, X_i^2]$ and shows that the sum of the absolute differences between the masses of $\text{Law}(t_B)$ and $\nu_B \mu_i$ tends to zero. Whenever several copies of a weight are expanded, they receive independent divisor samples. In particular, two occurrences of ν_B do not share a sample unless this is expressly stated.

2.3 Prediction and alignment

We use finite lists $\mathcal{B} \subset \mathbb{Q}_{>0}^n$ and $\mathcal{A} \subset \mathbb{Q}_{>0}$ satisfying

$$b_B \in \mathcal{A} \quad \text{for all } b \in \mathcal{B} \text{ and all blocks } B. \quad (2.6)$$

The coloring is a map $\chi : \mathbb{N} \rightarrow [r]$.

Principle 2.3 (Prediction). *For every $m \geq 2$ there is an integer $s = s(m)$ with the following property. Given n, r, χ , finite lists satisfying (2.6), and tolerances $0 < \tau < 1/4$, $\eta > 0$, there are admissible parameters and step-at-most- s models such that the following conclusions hold along the fixed ultrafilter $\mathcal{U}$ in w .*

1. For each B, a, c ,

$$\lim_{\mathcal{U}} \mathbb{P}(\chi(h_{B,a} t_B) = c, S_{B,a,c}(t_B) \leq 2\tau) \leq 3\tau + \eta. \quad (2.7)$$

2. Fix a chain $B_1, \dots, B_m$, $b \in \mathcal{B}$ and a color $c \in [r]$. Put

$$c_d = h_{B_d} b_{B_d}, \quad f_d(y) = \mathbf{1}_{\chi(c_d y) = c}, \quad \nu_d = \nu_{B_d}, \quad S_d = S_{B_d, b_{B_d}, c}.$$

Here f_d is defined on positive integers and may be extended by zero elsewhere. For $\emptyset \neq J \subseteq [m]$ write $d(J) = \max J$ and

$$L_J(z) = \sum_{k \in J} \frac{c_k}{c_{d(J)}} z_k, \quad (2.8)$$

$$U(z) = \prod_{\emptyset \neq J \subseteq [m]} \mathbf{1}_{\chi(\prod_{k \in J} c_k z_k) = c}. \quad (2.9)$$

Then the nonnegative weighted count

$$\mathcal{I}_{b,c,\mathbf{B}} = \mathbb{E}_{z \sim \bigotimes_{d=1}^m \mu_{i_d}} U(z) \prod_{d=1}^m \nu_d(z_d) \prod_{\substack{J \subseteq [m] \\ |J| \geq 2}} (f_{d(J)} \nu_{d(J)})(L_J(z)) \quad (2.10)$$

satisfies, for $z(t)_d = t_{B_d}$,

$$\lim_{\mathcal{U}} \left| \mathcal{I}_{b,c,\mathbf{B}} - \mathbb{E}_t U(z(t)) \prod_{\substack{J \subseteq [m] \\ |J| \geq 2}} S_{d(J)}(L_J(z(t))) \right| \leq \eta. \quad (2.11)$$

The step bound depends only on the requested chain length m . The complexity bound may depend on all the fixed data and tolerances. The extra weights on the sum forms in (2.10) are deliberate: they put the function at each sum under the same majorant as the function at its last summand. Positivity of this weighted count still gives an actual monochromatic configuration.

Principle 2.4 (Alignment). *For every n, r, s there are finite lists $\mathcal{B}, \mathcal{A}$ satisfying (2.6) and a constant $\delta > 0$, depending only on n, r, s , such that the following holds. For any admissible family, any system of step-at-most- s models, and any fixed $\tau > 0$, define $\mathbf{A}_w$ to be the event that some $b \in \mathcal{B}$ satisfies, simultaneously for every block B , color $c \in [r]$, and subset $D \subseteq \mathcal{E}_B$,*

$$S_{B,b_B,c}(t_B) > 2\tau \implies S_{B,b_B,c} \left(t_B + \sum_{A \in D} \frac{h_A b_A}{h_B b_B} t_A \right) > \tau. \quad (2.12)$$

Then

$$\liminf_{w \rightarrow \infty} \mathbb{P}(\mathbf{A}_w) \geq \delta. \quad (2.13)$$

In particular, δ is independent of τ and of the model complexity bound.

There is no assertion of a convergence rate uniform over all complexities in (2.13). Its positive lower bound is uniform. This distinction allows us to choose the small calibration tolerance before constructing the models.

2.4 A finite combinatorial selection

We use the finite sums theorem in its following finite form: for every m, r there is $F = F(m, r)$ such that every coloring of $[F]$ with r colors contains positive integers $u_1, \dots, u_m$ whose nonempty subset sums all lie in $[F]$ and have one color. This is the finite sums theorem of Folkman–Rado–Sanders [30, Theorem 2 and Corollary 1.1]; it also follows from Hindman’s finite sums theorem and compactness [19, 23]. Indeed, if the finite form failed, the finitely branching tree of bad finite colorings, ordered by restriction, would have an infinite branch. Its limiting coloring of $\mathbb{N}$ would contradict the finite sums theorem. We require no distinctness of the auxiliary u_d here.

The following selection uses the Ramsey-by-cardinality and finite-sums argument of [1, Section 5, Steps II–IV]; the ordered tails and pivots are chosen to match the additive shifts in (2.1).

Lemma 2.5 (Selection of a product chain). *For every m, r there is $n = n(m, r)$ such that, for every coloring $\chi : \mathbb{N} \rightarrow [r]$ and every list $x_1, \dots, x_n \in \mathbb{N}$, a chain $B_1, \dots, B_m$ has all nonempty products of $x_{B_1}, \dots, x_{B_m}$ in a single color class.*

Proof. Choose F as above and put $n_0 = 2F$. Repeated finite Ramsey gives an n such that every coloring of the nonempty index subsets of $[n]$ has an n_0 -element subset on which the color depends only on cardinality, when the number of colors is r . More precisely, apply the finite Ramsey theorem successively to the subset sizes $1, \dots, n_0$, choosing the successive ambient bounds backward. Apply this to the coloring $A \mapsto \chi(x_A)$, and call the resulting cardinality color $\kappa(q)$, $1 \leq q \leq n_0$.

Color $a \in [F]$ by $\kappa(2a)$. Choose $u_1, \dots, u_m$ by the finite sums theorem and put $q_d = 2u_d$. Then $q_d \geq 2$, $\sum_d q_d \leq n_0$, and all nonempty subset sums of the q_d have the same κ -color. In the homogeneous ordered index set, choose successive groups of $q_1 - 1, \dots, q_m - 1$ indices for $T_1, \dots, T_m$, followed by m indices for the pivots $i_1 < \dots < i_m$. This uses $\sum_d q_d$ indices and satisfies (2.2). A product of the corresponding block products is $x_{\bigcup_{d \in J} B_d}$, whose index-set size is $\sum_{d \in J} q_d$. All these sizes have the chosen color. $\square$

2.5 Deduction of the main theorem

Proof of Theorem 1.1, assuming the two principles. The case $m = 1$ follows by choosing any integer $a_1 > R$. For $m \geq 2$, fix $s = s(m)$ from Principle 2.3 and $n = n(m, r)$ from Lemma 2.5. Apply Principle 2.4 to fix $\mathcal{B}, \mathcal{A}$ and $\delta > 0$. Let C be the number of triples (B, a, c) indexing calibration events, and let K be the number of triples consisting of a scale vector, a chain of length m , and a color. These are fixed finite numbers depending on data already chosen.

Choose $0 < \tau < 1/4$ and then $\eta > 0$ so that

$$3C\tau < \frac{\delta}{4}, \quad C\eta < \frac{\delta}{4}, \quad K\eta < \frac{\delta\tau^{2^m}}{4}. \quad (2.14)$$

Now apply Principle 2.3. Define $\mathbf{G}_w$ to be the event that none of its calibration failures occurs. The union bound and (2.7) give

$$\lim_{\mathcal{U}} \mathbb{P}(\mathbf{G}_w^c) \leq C(3\tau + \eta) < \frac{\delta}{2}.$$

Since an ordinary lower limit bounds every ultrafilter limit, (2.13) yields

$$\lim_{\mathcal{U}} \mathbb{P}(\mathbf{A}_w \cap \mathbf{G}_w) \geq \frac{\delta}{2}. \quad (2.15)$$

On this intersection choose a witnessing $b \in \mathcal{B}$ and apply Lemma 2.5 to $x_i = h_i b_i t_i$, which are positive integers for all sufficiently large w . It supplies a chain and a color c for which $U(z(t)) = 1$. In particular, the singleton product masks say $\chi(h_{B_d} b_{B_d} t_{B_d}) = c$ for every d . Because $\mathbf{G}_w$ holds, all centers $S_d(t_{B_d})$ exceed 2τ .

For a nonsingleton $J \subseteq [m]$, let $d = \max J$. Each B_k , $k \in J \setminus \{d\}$, belongs to $\mathcal{E}_{B_d}$. The corresponding instance of (2.12) therefore gives

$$S_d(L_J(z(t))) > \tau.$$

There are $q = 2^m - m - 1$ such factors. Thus at least one of the K nonnegative model integrands in (2.11) is greater than $\tau^q \geq \tau^{2^m}$ on $\mathbf{A}_w \cap \mathbf{G}_w$. Summing expectations and using (2.15) shows that their sum has ultrafilter limit at least $\delta\tau^{2^m}/2$. It follows from (2.11) and (2.14) that

$$\lim_{\mathcal{U}} \sum_{b,c,\mathbf{B}} \mathcal{I}_{b,c,\mathbf{B}} > 0. \quad (2.16)$$

In particular, the finite sum is bounded below by a positive number on a set belonging to $\mathcal{U}$. We did not need a fixed choice of witnessing chain across different samples or values of w .

For some sufficiently large w , one weighted count is positive. Since it is a finite nonnegative average, there is an actual tuple z in its support for which every factor is positive. Put $a_d = c_d z_d$. The product mask gives every nonempty product of the a_d in color c , including their singleton values. For every nonsingleton J , positivity of its color factor gives

$$\chi\left(c_{d(J)} L_J(z)\right) = \chi\left(\sum_{k \in J} a_k\right) = c.$$

The extra divisor factors cannot create a false positive; they only restrict which tuples can contribute to the count.

It remains to obtain the prescribed separation. The finite rational scale lists give a constant C_0 such that $c_d \leq C_0 M$ for all these counts, and $c_d \geq 1$ once it is a positive integer. If $d < e$, then

$$a_d < C_0 M X_{i_d}^2, \quad a_e \geq X_{i_e}.$$

For $e \geq 2$, put $V_e = 2 + M + \prod_{j < i_e} X_j$. There is a constant C_1 , depending only on the fixed finite data, such that every support tuple satisfies

$$\sum_{d < e} a_d + \prod_{d < e} a_d \leq C_1 V_e^{3m}.$$

Indeed each preceding a_d is at most $C_0 V_e^3$, and there are at most $m - 1$ such terms. Admissibility makes X_{i_e} dominate every fixed power of V_e . Therefore, for the prescribed R, D , all support tuples at sufficiently large w satisfy (1.1); the first element exceeds R since $a_1 \geq X_{i_1} \rightarrow \infty$. The positive-count set in (2.16) belongs to the nonprincipal ultrafilter and hence meets every sufficiently late range of w . Choose w in that intersection. This proves both monochromaticity and separation, and completes Theorem 1.1. $\square$

Proof of Corollary 1.2. Apply Theorem 1.1 with $R = 2$ and $D = 1$. Every a_j exceeds the sum and the product of all its predecessors, and every element is at least two. If two subset sums were equal, cancel their common terms. The greatest remaining index on either side would exceed the sum on the other side, a contradiction. For products, cancel common factors and compare the greatest remaining factor with the product of all preceding factors. An empty side after cancellation equals one, which is also smaller than every remaining factor. Thus both subset maps are injective.

To compare a subset sum with a subset product, first compare their greatest indices. If these differ, the expression with the larger index exceeds the other expression. If both have greatest index j , the sum lies in $[a_j, 2a_j)$, whereas a nonsingleton product is at least $2a_j$. A singleton product equals a_j , which equals a sum with greatest index j only for the singleton sum. Hence the intersection is exactly A , giving the stated cardinalities. $\square$

Corollary 2.6 (Finite interval form with prescribed divisibility). *Fix integers $r, m, q \geq 1$ and real numbers $R \geq 2, D \geq 1$. There exists N such that every r -coloring of $[N]$ admits $A = \{a_1 < \dots < a_m\} \subset q\mathbb{N}$ with $\text{FS}(A) \cup \text{FP}(A) \subseteq [N]$ monochromatic and*

$$a_1 > R, \quad a_d > R \left(\sum_{k < d} a_k + \prod_{k < d} a_k \right)^D \quad (2 \leq d \leq m).$$

The subset sums and products have the distinctness properties of Corollary 1.2.

Proof. First color all of $\mathbb{N}$, refining the given color by the residue modulo q . Apply Theorem 1.1 to this finite coloring with $\max(m, 2)$ elements. Their common residue v also equals the residue of the sum of the first two elements, so $v = 2v$ modulo q and $v = 0$. Retain the first m elements. The separation and monochromaticity persist, and the preceding proof gives the distinctness assertions.

If no N worked, the finite colorings avoiding such a configuration would form a finitely branching tree, closed under restriction and with a node at every depth. An infinite branch would color $\mathbb{N}$ without the configuration just proved to exist. Indeed each finite configuration and all its sums and products lie in some initial interval. This contradiction proves the finite interval assertion. $\square$

This compactness argument gives a finite bound, but the proof does not supply a numerical estimate for N .

3 Arithmetic scales and divisor weights

Prediction compares a count sampled at the pivots with a count sampled at the block products. For a block $B = T \cup \{i\}$, conditioning on $\sigma = t_T$ changes the harmonic density of t_i to $\sigma \mathbf{1}_{\sigma|y}/(Z_i y)$ on $[\sigma X_i, \sigma X_i^2]$, with the same W -unit restriction. Averaging the divisibility factor gives ν_B . We first show that returning the endpoints to $[X_i, X_i^2]$ has negligible total mass.

We then construct scales and prime parameters for the correlation argument. The prime laws have enough harmonic mass to permit prime insertion, and their residue uniformity makes products of divisor weights have mean one on the required linear systems. A separate coprimality estimate for independent polynomial values will allow Section 5 to combine cube tests with different prime-dependent step sizes.

We use *rough* to mean coprime to $W = \prod_{p \leq w} p$, and put

$$|a|_{>w} = \prod_{p > w} p^{v_p(a)} \quad (a \in \mathbb{Z} \setminus \{0\}).$$

All finite templates in this Section are fixed before the limit $w \rightarrow \infty$. A template specifies numbers of variables and rows, block sizes, and integer polynomials, but no color functions.

3.1 Harmonic sampling identities

We give explicit total-mass bounds to distinguish multiplication of an argument from conditioning on divisibility. For a signed measure on $\mathbb{Z}$, write $\|\xi\|_1 = \sum_{n \in \mathbb{Z}} |\xi(n)|$. Thus probability total variation is one half of this norm.

Lemma 3.1 (Sampling and changes of variables). *Let $X \geq 2$ be an integer, set*

$$\vartheta_W = \frac{\phi(W)}{W}, \quad L_X = \log X, \quad Z_X = \sum_{\substack{X \leq n < X^2 \\ (n, W)=1}} \frac{1}{n}, \quad \mu_X(n) = \frac{\mathbf{1}_{[X, X^2]}(n) \mathbf{1}_{(n, W)=1}}{n Z_X},$$

and suppose $L_X > W/X$. Let Y have law μ_X .

1. For $(k, W) = 1$, every residue $a \pmod{k}$, and $0 < A < B$,

$$\left| \sum_{\substack{A \leq n < B \\ (n, W)=1 \\ n \equiv a \pmod{k}}} \frac{1}{n} - \frac{\vartheta_W}{k} \log \frac{B}{A} \right| \leq \frac{\phi(W)}{A}. \quad (3.1)$$

In particular $|Z_X - \vartheta_W L_X| \leq \phi(W)/X$, and

$$|k\mathbb{P}(Y \equiv a \pmod{k}) - 1| \leq E_X(k) := \frac{W(k+1)}{X(L_X - W/X)}. \quad (3.2)$$

The total-mass distance of the residue law from uniform measure is also at most $E_X(k)$.

2. If $h \in W\mathbb{Z}$, then

$$\|\text{Law}(Y + h) - \mu_X\|_1 \leq \min \left(2, \frac{2|h|}{X(L_X - W/X)} \right). \quad (3.3)$$

3. If $1 \leq k \leq X$ and $(k, W) = 1$, let $\eta_k(n) = k \mathbf{1}_{k|n} \mu_X(n)$, an unnormalized positive measure. Then

$$\|\text{Law}(kY) - \eta_k\|_1 \leq \frac{2 \log k + (Wk/X)(1 + 1/X)}{L_X - W/X}, \quad (3.4)$$

and $|\eta_k(\mathbb{Z}) - 1| \leq E_X(k)$.

For the asymptotic conclusions, let $K, H, V \geq 1$ be auxiliary quantities depending on w . Each residue and translation error is $o(V^{-C})$, for every fixed C , uniformly for $k \leq K$ and $|h| \leq H$, if X dominates every fixed power of $2 + W + K + H + V$. The dilation error has the same conclusion if $\log X$ dominates every fixed power of $2 + W + K + V$. These conclusions still hold in expectations of functions bounded by any fixed power of V .

Proof. The indicator in (3.1) is periodic modulo Wk and occupies exactly $\phi(W)$ residue classes. On any interval, its counting function differs from ϑ_W/k times length by at most $\phi(W)$: apply the discrepancy bound of one to each occupied class. Partial summation against $1/t$ gives (3.1). Apply it with $k = 1$ to obtain the bound on Z_X . For a general class the error in its unnormalized mass is at most $\phi(W)/X$; subtracting Z_X/k and using $Z_X \geq \vartheta_W(L_X - W/X)$ proves (3.2). Sum the absolute class errors to obtain the total-mass assertion.

For translation put $H = |h|$. Translation invariance of the norm reduces to $h = H \geq 0$. On the overlap of the supports, $1/(n - H) \geq 1/n$, and $W \mid H$ preserves the unit condition. The negative part of $\text{Law}(Y + H) - \mu_X$ is therefore precisely the original mass on $[X, \min(X + H, X^2))$. Both measures have mass one, so the norm is twice that mass. If $H \leq X^2 - X$, the interval $[X, X + H)$ has exactly $\vartheta_W H$ units modulo W , since its integer length is a multiple of W ; its harmonic mass is at most $\vartheta_W H/X$. If $H > X^2 - X$, the norm is at most two, and the second bound in (3.3) is already at least two. This proves the translation estimate.

For dilation, at a multiple n of k the pushed-forward mass is $k/(nZ_X)$, supported on $[kX, kX^2)$ and on $(n, W) = 1$. Thus it agrees exactly with η_k on the overlap. Changing variables $n = ku$ in the two boundary pieces, and using $k \leq X$, gives the exact formula

$$\|\text{Law}(kY) - \eta_k\|_1 = \frac{1}{Z_X} \left(\sum_{\substack{X/k \leq u < X \\ (u, W)=1}} \frac{1}{u} + \sum_{\substack{X^2/k \leq u < X^2 \\ (u, W)=1}} \frac{1}{u} \right).$$

Each main term in (3.1) is $\vartheta_W \log k$, and the two errors are at most $\phi(W)k/X$ and $\phi(W)k/X^2$, respectively. Dividing by the lower bound for Z_X proves (3.4). Its mass assertion follows from (3.2) with $a = 0$. Finally, integration against a function of supremum norm B costs at most B times the total-mass error. The stated growth conditions absorb every fixed choice of $B = V^{O(1)}$. $\square$

For later use, uniform sampling on an integer interval of length T has residue law modulo k at total-mass distance at most $2k/T$ from uniform measure. Each residue occurs either $\lfloor T/k \rfloor$ or $\lceil T/k \rceil$ times, up to the harmless endpoint convention. The same bound is uniform in the interval's location. Translating the interval by an integer u changes its probability law in total-mass norm by at most $2 \min(1, |u|/T)$. Product laws incur the sum of the coordinate errors, by telescoping their tensor products. These facts and Lemma 3.1 apply conditionally when the interval endpoints, dilations, or translations have been fixed by outside variables and the displayed bounds hold uniformly in those variables.

3.2 Recovering the product sampling law

The next Corollary identifies the center weights in (2.10) with the actual block-product sampling. Once the shifted factors have bounded models, this identity will remove the remaining weights from the count.

Corollary 3.2 (Product law). *Let $B = T \cup \{i\}$ be a block, and let t_j be the independent harmonic samples at its raw cutoffs. Let $V = V(w) \geq 1$ be an auxiliary scale, and put $K_T = \prod_{j \in T} X_j^2$. If $\log X_i$ dominates every fixed power of $2 + W + K_T + V$, then*

$$\|\text{Law}(t_B) - \nu_B \mu_i\|_1 = o(V^{-C}) \quad \text{for every fixed } C > 0. \quad (3.5)$$

For any fixed family of disjoint blocks the corresponding joint law differs in the same sense from the product of their weighted pivot measures, when these hypotheses hold at each pivot.

Proof. Condition on $\sigma = t_T$. It is coprime to W , independent of t_i , and bounded by K_T . Apply (3.4) with $k = \sigma$, uniformly in this range, and average. The comparison measure is exactly

$$\mu_i(y) \mathbb{E}_\sigma \sigma \mathbf{1}_{\sigma|y} = \mu_i(y) \nu_B(y),$$

proving (3.5). In particular its mass is $1 + o(V^{-C})$. Disjoint blocks use disjoint raw variables, so their original laws are independent. Telescoping a fixed tensor product bounds its total-mass error by the sum of the individual errors times the masses of the other factors, all of which are $1 + o(1)$. This proves the joint assertion. $\square$

3.3 A sequential construction of the master scales

We now work on a master index set $[N]$. Here R_l are auxiliary gap lengths and X_j are cutoffs for the raw variables. Section 5 will select the final n indices and choose each H_i from these gap lengths. The smooth factors h_j make all required shifts integral. The prime pools supply the harmonic mass and residue uniformity described above; the gap lengths absorb the finitely many polynomial divisibilities needed by the subsequent changes of variables. The raw cutoffs are chosen last in each gap so that the preceding sampling estimates apply.

Lemma 3.3 (Master scales). *Fix N , a bound on block sizes, a finite set $\mathcal{A} \subset \mathbb{Q}_{>0}$, and a finite list $\mathcal{D}$ of nonzero integer polynomials in finitely many prime-parameter slots. There are choices of $h_j, M, X_j, R_l, P_l^-, P_l^+$, for $1 \leq j, l \leq N$, with the following properties.*

1. $h_j = W^{w2^{N-j}}$, and M is a power of W , divisible by W^w , with $h_j \leq M$ for every index and $h_B \leq M$ for every block under consideration. Every permitted adding pair satisfies $h_A/h_B \in W^w \mathbb{N}$. For a master chain and $c_d = h_{B_d} a_d$, $a_d \in \mathcal{A}$, all c_d are integers and $c_u/c_d \in W\mathbb{N}$ when $u < d$, for sufficiently large w .
2. There is an integer $e_0 = e_0(w)$ such that, for independent uniform units in the slots modulo W^{e_0} ,

$$\mathbb{P}(p^{e_0} \mid D(\mathbf{u}) \text{ for some } p \leq w, D \in \mathcal{D}) = o(1). \quad (3.6)$$

Moreover $W^{e_0+1} c_d \mid M$ for every possible c_d .

3. Writing

$$V_l = 2 + M + \prod_{j < l} X_j^2, \quad Q_l = W^{e_0} \prod_{w < p \leq V_l} p,$$

the pool $\mathcal{P}_l$ consists of all primes in a finite union of consecutive complete dyadic intervals $[P_l^-, P_l^+)$. Its law and harmonic mass are

$$H_l^{\text{pr}} = \sum_{p \in \mathcal{P}_l} \frac{1}{p}, \quad \lambda_l(p) = \frac{1}{p H_l^{\text{pr}}}. \quad (3.7)$$

Both P_l^- and H_l^{pr} dominate every fixed power of V_l . The residue law of a sample modulo Q_l differs from uniform measure on $(\mathbb{Z}/Q_l\mathbb{Z})^\times$ by $o(V_l^{-C})$ in total variation, for every fixed C . All slots are sampled independently, including slots in different gaps.

4. R_l dominates every fixed power of $P_l^+ + V_l$. It is divisible by M , by every earlier R_j , and by $M|D(\mathbf{p})|$ for all the nonzero values required for divisibility in gap l , with these polynomial slots drawn from that gap's pool. The integer X_l is a power of two, and $\log X_l$ dominates every fixed power of R_l .

One can include any prescribed finite family of polynomial nonvanishing conditions in $\mathcal{D}$. Repeated entries and zero values then have probability $o(V_l^{-C})$ for every fixed C , whereas the small-prime exceptions in (3.6) have probability $o(1)$.

Proof. Set $e_j = 2^{N-j}$. The strict inequality $e_j > \sum_{k>j} e_k$ shows that the sign of $\sum_{j \in A} e_j - \sum_{j \in B} e_j$, for distinct subsets, is determined by the smallest index in their symmetric difference. For an adding pair that index belongs to A ; for two chained blocks it belongs to the earlier block. The positive difference is an integer, so the corresponding ratio of h -products is divisible by W^w . Fixed rational multipliers have bounded valuations and only finitely many denominator primes. Consequently they are absorbed by these powers for all sufficiently large w , leaving at least one factor of every prime dividing W in each required ratio of the c_d 's.

For completeness, a nonzero polynomial over $\mathbb{Q}_p$ has a null zero set in a product of p -adic unit groups. In one variable it has only finitely many roots. In several variables, expand in a variable on which it depends and choose a nonzero coefficient polynomial. By induction, the set on which that coefficient vanishes has measure zero; off that set, each fiber has finitely many roots. Fubini's Theorem proves the assertion. The decreasing events $p^e \mid D$ therefore have probabilities tending to zero as $e \rightarrow \infty$. At any fixed w , the list of polynomials and primes is finite. Choose a common e_0 making their union probability at most $1/w$. Reduction of uniform units modulo W^{e_0} has exactly the required distribution modulo p^{e_0} . We can now choose the power M large enough for all the finitely many stated smooth divisibilities and bounds.

Proceed in increasing order of l . At this stage Q_l is a fixed integer. The prime number theorem in each reduced arithmetic progression modulo Q_l [32, Equation (1.1)], followed by partial summation, says that the harmonic prime law on $[Y, 2Y)$ tends to the uniform law on its unit classes as $Y \rightarrow \infty$. Since there are finitely many classes, choose a dyadic lower endpoint so large that the total variation error is at most V_l^{-w} on every subsequent dyadic interval, and also $P_l^- \geq wV_l^w$. Increasing the upper endpoint gives $H_l^{\text{pr}} \geq wV_l^w$, since the harmonic prime series diverges. A mixture of measures having the same error bound has that error bound. Thus this finite pool has the asserted residue distribution, however long the union of intervals is.

Only finitely many polynomial values occur in a finite pool. Take their nonzero absolute values, the earlier gap lengths, and M , form the required least common multiple, and choose a multiple R_l at least $w(P_l^+ + V_l)^w$. Finally choose a power of two X_l with $\log X_l \geq wR_l^w$. There is no upper bound in any of these choices, and no later requirement changes an earlier choice.

Here is the elementary zero-value estimate used for the final assertion. If independent variables each have maximum atom at most a , a nonzero polynomial of total degree d vanishes with probability at most da . Expand in its last relevant variable: outside the zero set of its leading coefficient there are at most as many roots as its degree in that variable, and induction bounds the leading-coefficient exception by its degree times a . For our pool, $a \leq 1/(P_l^- H_l^{\text{pr}})$; the same estimate for $x_i - x_j$ handles repeats. These bounds are smaller

than every fixed inverse power of V_l . Finally, the residue-law approximation modulo Q_l , together with the Chinese remainder theorem (CRT), transfers (3.6) to the actual prime slots. $\square$

For a block $B = T \cup \{i\}$ at these scales, its tail bound K_T and W are at most V_i . Since $\log X_i$ dominates every fixed power of V_i , Corollary 3.2 applies with $V = V_i$. In particular, the joint law comparison holds for every chain and hence for bounded expectations of all its block variables.

Remark 3.4 (Countably many fixed tests). Later, a fixed test may be raised to any fixed moment or combined with any fixed number of other tests. This does not require a moment order growing inside an estimate. Enumerate the resulting integer-polynomial templates and positive integer auxiliary parameters. At stage j impose the requirements for the first j templates. Each stage has finitely many requirements, so the preceding proof applies. Choose increasing thresholds w_j so that the finitely many error bounds at that stage, including constants in the estimates below, are at most $1/j$ once $w \geq w_j$. At a given w use stage $\max\{j : w_j \leq w\}$, with the growth exponents also tending to infinity. Every fixed finite test family then satisfies all the conclusions eventually. This diagonal choice can be made before the functions to be tested are chosen, because its templates depend only on their algebraic forms. All moment limits below mean first $w \rightarrow \infty$ at a fixed moment order.

3.4 Coprimality beyond the CRT cutoff

The next estimate treats prime factors larger than V_l , as well as those controlled by the residue distribution of a pool. The linear-forms estimate below uses residue uniformity separately. Coprimality has a different purpose: in a cyclic group, the subgroups of multiples of two coprime integers together generate the whole group. Section 5 applies this fact to combine cube tests with different step sizes.

Lemma 3.5 (Rough coprimality). *Let F and G be fixed nonzero integer polynomials in disjoint tuples of independent prime variables. Each variable is sampled with probability proportional to $1/p$ in a dyadic interval $[Y, 2Y)$. Let L be the smallest of these lower endpoints. For all sufficiently large w, L ,*

$$\mathbb{P}(FG = 0) \ll_{F,G} \frac{\log L}{L}, \quad (3.8)$$

$$\mathbb{P}(FG \neq 0, \gcd(|F|_{>w}, |G|_{>w}) > 1) \ll_{F,G} \frac{1}{w} + \frac{\log L}{\sqrt{L}}. \quad (3.9)$$

The constants are independent of the ratios between the dyadic endpoints. Consequently two independent tuples from any of the constructed pools have coprime rough polynomial values with probability $1 - o(1)$. The same assertion holds after restricting each tuple to a prime-only event of probability $1 - o(1)$.

Proof. The prime number theorem gives a maximum atom $O(\log Y/Y)$ in the harmonic prime law on $[Y, 2Y)$. The polynomial zero estimate in the proof of Lemma 3.3 proves (3.8), uniformly in the endpoints.

We shall also use the following consequence of the interval Brun–Titchmarsh inequality. If $p \leq \sqrt{Y}$ is prime, a harmonically sampled prime in $[Y, 2Y)$ belongs to any prescribed nonzero class modulo p with probability $O(1/p)$. Indeed Theorem 2 of [36], together with $\log(Y/p) \geq \frac{1}{2} \log Y$, bounds the number of primes in a reduced class in that interval by $O(Y/(p \log Y))$. Divide by the prime count $\gg Y/\log Y$. Harmonic and uniform sampling

on that interval have densities within a factor of two. The zero class contains no prime in $[Y, 2Y)$ when $p \leq \sqrt{Y}$.

Remove variables on which the polynomials do not depend, and argue by induction on the total number of remaining variables. A fixed nonzero constant has no prime factors larger than w once w is large, so it gives the base case. In each nonconstant polynomial choose as its main variable one whose interval has largest lower endpoint in that tuple. Denote those endpoints by Y_F, Y_G , and the corresponding nonzero leading-coefficient polynomials by A, B . The events $A = 0$ or $B = 0$ cost $O_{F,G}(\log L/L)$. If a common prime divisor of F, G also divides A , apply the induction hypothesis to A, G ; if it divides B , apply it to F, B . The two tuples in each application remain disjoint, and their total number of variables strictly decreases. It remains to consider common primes dividing neither leading coefficient.

Put $Y_0 = \min(Y_F, Y_G)$. For $w < p \leq \sqrt{Y_0}$, condition on all variables except the two main variables. When the respective leading coefficient is nonzero modulo p , each polynomial has at most its fixed degree many roots modulo p . The preceding Brun–Titchmarsh consequence bounds the two independent tests by $O_{F,G}(p^{-2})$. Their sum is $O_{F,G}(1/w)$.

For the remaining primes, expose the entire tuple whose main endpoint is Y_0 , say the F -tuple. Every variable in that tuple is at most $2Y_0$, so, when $F \neq 0$, $1 \leq |F| \leq C_F Y_0^{\deg F}$. It follows that F has only $O_F(1)$ distinct prime divisors exceeding $\sqrt{Y_0}$. Test each such prime p against the independent G -tuple, whose main endpoint $Y = Y_G$ satisfies $Y \geq Y_0$. If $p \leq \sqrt{Y}$, the cost is $O_G(1/p) = O_G(Y_0^{-1/2})$, again excluding the already treated event $p \mid B$. If $p > \sqrt{Y}$, each of the boundedly many root classes contains at most $Y/p + 1$ integers of the interval. Dividing by $\gg Y/\log Y$ gives

$$O_G\left(\frac{\log Y}{\sqrt{Y}}\right) = O_G\left(\frac{\log Y_0}{\sqrt{Y_0}}\right).$$

The last inequality holds for all sufficiently large Y_0 , since $\log Y/\sqrt{Y}$ is then decreasing. Summing over the bounded number of large prime factors proves the induction and (3.9).

Conditioning pool samples on their individual dyadic intervals leaves independent harmonic prime laws. The established bounds are uniform over these choices, so averaging proves the pool assertion. Restricting to events of probabilities $1 - o(1)$ changes any probability by $o(1)$. $\square$

The prime pools now have both residue uniformity and rough coprimality. We turn to the mean-one estimate that permits repeated Cauchy–Schwarz with divisor weights.

3.5 The weighted linear-forms estimate

Recall that for a block $B = T \cup \{i\}$ and independent tail product $\sigma = t_T$, the divisor weight is

$$\nu_B(y) = \mathbb{E}_\sigma \sigma \mathbf{1}_{\sigma|y}. \quad (3.10)$$

In expanding a product of weights, every occurrence of a weight receives a fresh independent draw, even when its block label repeats. The next Proposition records precisely the row hypotheses needed below. In particular, restrictions on prime parameters are permitted.

The mean-one linear-forms estimate below plays the role of the pseudorandomness conditions in Green and Tao’s transference method [15, Definition 3.1 and Section 5]. We prove the estimate for the present divisor weights and prime-dependent forms.

Proposition 3.6 (Weighted linear forms). *Fix the number q of rows, the number d of base variables, and a bound b on the number of raw factors in each divisor. For each $u \in [q]$, let σ_u be a product of at most b independent harmonic W -unit variables at the master cutoffs*

of Lemma 3.3, and write $\nu_u(y) = \mathbb{E}\sigma_u \mathbf{1}_{\sigma_u|y}$. All these raw draws are independent across occurrences. Suppose $\sigma_u \leq V$ surely, where $V \geq M \rightarrow \infty$. Let $\mathbf{p}$ consist of independent prime slots from finitely many gaps, each having $V_l \geq V$. These slots are independent of all divisor draws.

Let $\mathcal{G}$ be a domain depending only on $\mathbf{p}$, and let $\ell_u(\mathbf{x}; \mathbf{p})$ be homogeneous linear rows. On $\mathcal{G}$ their arguments are integers on the base sampling support. Assume the following conditions.

1. Given $\mathbf{p}$ and any possible divisor draws, the joint residue law of $\mathbf{x}$ modulo $K = \prod_{u=1}^q \sigma_u$ differs in total variation by at most ϵ_{base} from the uniform law on $(\mathbb{Z}/K\mathbb{Z})^d$. The bound is uniform on $\mathcal{G}$.
2. For every $w < p \leq V$, each row is nonzero modulo p on $\mathcal{G}$. There is a fixed finite list of nonzero integer polynomials in the prime slots such that, if all their values are nonzero modulo p , every pair of rows is linearly independent over $\mathbb{F}_p$.

The CRT law of all prime slots at the primes $w < p \leq V$ is within total variation ϵ_{CRT} of independent uniform units, with independence also across these primes. Then, uniformly in every further prime-only event $\mathcal{E} \subseteq \mathcal{G}$,

$$\mathbb{E}_{\mathbf{p}, \mathbf{x}} \mathbf{1}_{\mathcal{E}}(\mathbf{p}) \prod_{u=1}^q \nu_u(\ell_u(\mathbf{x}; \mathbf{p})) = \mathbb{P}(\mathcal{E}) + O\left(\frac{1}{w} + V^q(\epsilon_{\text{base}} + \epsilon_{\text{CRT}})\right). \quad (3.11)$$

The error is absolute, so the statement also applies when $\mathbb{P}(\mathcal{E})$ tends to zero. The implied constant depends only on the fixed row and divisor templates and polynomial tests. In particular the error is $o(1)$ with the scales of Lemma 3.3 and base residue errors smaller than every fixed inverse power of V .

Rows may have rational coefficients if their denominators are units modulo every possible divisor: interpret their residues by inverting those denominators. The integer-value and row hypotheses must still hold. Clearing such denominators leaves the congruence calculation unchanged. In particular this permits smooth denominators and prime pool denominators larger than V .

Proof. We first express the normalized divisibility count as a product of local kernel counts. Their excess over one is nonnegative; this lets us bound it without retaining the prime-only event. We then dominate the divisor draws by laws with independent prime valuations and sum the local excesses.

Local kernel counts. Expand each weight with its independent divisor draw. Replacing the base residue law by exact uniform measure modulo K costs at most $2\epsilon_{\text{base}} \prod_u \sigma_u \leq 2V^q \epsilon_{\text{base}}$. The CRT theorem now factors the normalized divisibility count as

$$\prod_{w < p \leq V} \alpha_p, \quad \alpha_p = p^{\sum_u a_u} \mathbb{P}_{\mathbf{x} \bmod p^A}(p^{a_u} \mid \ell_u(\mathbf{x}) \text{ for every } u),$$

where $a_u = v_p(\sigma_u)$ and $A = \max_u a_u$. Primes with all $a_u = 0$ contribute one. Homogeneity makes the divisibility conditions the kernel of a homomorphism into $\prod_u \mathbb{Z}/p^{a_u}\mathbb{Z}$. Its image has size at most $p^{\sum a_u}$, so $\alpha_p \geq 1$. With a single positive valuation, the corresponding primitive row is surjective modulo every power of p , giving $\alpha_p = 1$. In general one primitive row with valuation A bounds the kernel probability by p^{-A} . If the two rows having the largest valuations $A \geq B > 0$ are independent modulo p , one of their two-column minors is a unit. Fixing the other coordinates, the two selected coordinates are uniformly and bijectively

mapped to the pair of row values modulo p^A . Their required divisibilities consequently have probability p^{-A-B} . Additional rows can only reduce the probability.

We bound the nonnegative excess uniformly, so that an arbitrary prime-only domain can subsequently be retained. Let $\mathcal{T}_p$ be the test that at least one of the prescribed polynomial values vanishes modulo p . Set $\beta_p = 0$ when fewer than two valuations are positive; otherwise set

$$\beta_p = \begin{cases} p^{\sum a_u - A - B} - 1, & \mathcal{T}_p \text{ fails,} \\ p^{\sum a_u - A} - 1, & \mathcal{T}_p \text{ holds.} \end{cases}$$

On $\mathcal{G}$, the preceding kernel bounds give $0 \leq \alpha_p - 1 \leq \beta_p$. For every choice of the parameters and the true, bounded divisor draws,

$$0 \leq \mathbf{1}_{\mathcal{E}} \left(\prod_p \alpha_p - 1 \right) \leq \prod_{w < p \leq V} (1 + \beta_p) - 1 \leq \prod_u \sigma_u \leq V^q. \quad (3.12)$$

In particular, we can first replace the prime residues by independent CRT-uniform units at cost at most $2V^q \epsilon_{\text{CRT}}$. This replacement is made while all divisor draws are still bounded.

Independent comparison laws. We next compare those draws with laws whose prime valuations are independent. For a raw factor at cutoff X_j , put $\epsilon_j = 1/\log X_j$ and define

$$\tilde{\mu}_j(n) = \frac{\mathbf{1}_{(n, W)=1} n^{-1-\epsilon_j}}{\zeta(1+\epsilon_j) \prod_{p \leq w} (1 - p^{-1-\epsilon_j})} \quad (n \geq 1).$$

Its denominator is $(1 + o(1)) \vartheta_W \log X_j$. Indeed $\epsilon \zeta(1+\epsilon) \rightarrow 1$, and

$$0 \leq \log \prod_{p \leq w} \frac{1 - p^{-1-\epsilon}}{1 - p^{-1}} \leq \epsilon \sum_{p \leq w} \frac{\log p}{p-1} \leq \epsilon \log W = o(1).$$

Lemma 3.1 gives the same asymptotic normalization for the original harmonic law on $[X_j, X_j^2)$, and $n^{\epsilon_j} \leq e^2$ on that interval. Hence $\mu_j(n) \leq C_0 \tilde{\mu}_j(n)$ pointwise, with an absolute constant C_0 for sufficiently large w . For at most bq independent raw draws this incurs one overall constant C_0^{bq} , rather than a separate constant for each prime.

The Euler product defining $\tilde{\mu}_j$ shows that its valuations at different primes are independent, with

$$\tilde{\mathbb{P}}(v_p(n) = a) = (1 - p^{-1-\epsilon_j}) p^{-a(1+\epsilon_j)} \leq p^{-a}.$$

Thus a product of at most b raw factors has valuation mass at a at most $(a+1)^b p^{-a}$. This bound is uniform in the individual cutoffs. Divisors remain independent of each other and of all prime parameters.

Summing the local excesses. For sufficiently large w , none of the fixed nonzero test polynomials is identically zero modulo any prime $p > w$. The elementary polynomial root bound on the product grid $(\mathbb{F}_p^\times)^k$ gives $\mathbb{P}(\mathcal{T}_p) = O(1/p)$. Under the CRT-uniform law the tests at different primes are independent. Combining the valuation bounds with the definition of β_p , the contribution when $\mathcal{T}_p$ fails is at most a constant times

$$\sum_{A \geq B \geq 1} (A+1)^{C_1} (B+1)^{C_1} p^{-A-B} \ll p^{-2}.$$

To see this sum explicitly, choose the rows with the two largest valuations; all other valuations lie between zero and B , and summing their polynomial factors is bounded by a fixed power

of $B + 1$. The factors $p^{-\sum a_u}$ cancel the normalizing power in β_p , leaving p^{-A-B} . On $\mathcal{T}_p$ the same argument, with all lesser valuations bounded by A , gives

$$\sum_{A \geq 1} (A + 1)^{C_2} p^{-A} \ll p^{-1}.$$

Its additional $O(1/p)$ test probability again gives $\mathbb{E}\beta_p \ll p^{-2}$. The two displayed series bounds follow by factoring out respectively p^{-2} and p^{-1} and bounding the remaining geometrically convergent series using $p \geq 2$.

We have now obtained independence across primes for every quantity in the nonnegative right side of (3.12). Consequently its expectation, after the single global divisor-domination constant, is at most

$$C_0^{bq} \left\{ \prod_{w < p \leq V} (1 + C_3 p^{-2}) - 1 \right\} \ll \sum_{p > w} p^{-2} \ll \frac{1}{w}.$$

The original local product has lower bound one on $\mathcal{E}$. Integrating that lower bound and the estimated excess, and restoring the two residue-approximation errors, proves (3.11). Rational denominators which are units modulo the divisors induce the same homomorphisms after inversion, so the proof covers the stated rational-row extension. $\square$

Several consequences of Proposition 3.6 will be used without altering its hypotheses. A fixed product of factors $1 + \nu_u$ has main term $2^q \mathbb{P}(\mathcal{E})$, by expansion over subsets of rows. If at least one factor is $\nu_u - 1$, and the remaining factors are each ν_v , $1 + \nu_v$, or $\nu_v - 1$, its average is $o(1)$, provided every subsystem in that expansion meets the same row conditions. This is cancellation of the constant main terms. In particular a prime-only exceptional event of probability $o(1)$, contained in a domain of primitive rows, has weighted mass $o(1)$. One may also condition on a prime-only event whose probability is bounded below, dividing (3.11) by that probability.

These statements include systems assembled from different gaps. Use the common divisor bound V ; every relevant pool has $V_l \geq V$, so projecting its CRT law supplies the required accuracy. Independent replicas give independent slots. For coefficients involving $M(\mathbf{p}) = M|D(\mathbf{p})|_{>w}$, a prime $w < p \leq V$ dividing such a coefficient must divide $D(\mathbf{p})$, so the indicated polynomial tests cover this possible loss of a minor. Fixed integer contents are w -smooth eventually. Primitivity and separation of each actual row system still have to be checked; those checks are given in Sections 4 and 5.

To verify the base-law hypothesis in those applications, every divisor product K is at most V^q . Uniform auxiliary intervals of lengths dominating all powers of V , and harmonic pivot variables whose cutoffs dominate all powers of $W + V$, are therefore jointly uniform modulo K , with error smaller than every fixed inverse power of V . Their conditional interval locations do not matter. A short translation changes such a box by the sum of the displacement-to-side-length ratios; the translation and dilation bounds above cover harmonic variables. For example, $R_l/(J_0 M(\mathbf{p}))$ dominates all powers of the common tail bound for each fixed J_0 , since $M(\mathbf{p})$ has a fixed-power bound in $P_l^+ + V_l$. Taking square roots of these lengths preserves that domination and gives negligible relative translations. These observations supply uniform bounds for the shifted boxes used later, as well as for unshifted boxes.

4 Removing multiplicative masks and detecting a shifted error

The purpose of this Section is to turn a correlation containing arbitrary multiplicative masks into a test of one function on an additive cube. The number of Cauchy–Schwarz steps will depend only on the number of blocks in the chain. This independence from the master index count is needed when the analytic tolerances are chosen before the Ramsey bound.

Fix a master chain

$$B_k = T_k \cup \{i_k\}, \quad T_1 < \cdots < T_m < l < i_1 < \cdots < i_m,$$

where each tail is nonempty, and fix positive scales $c_k = h_{B_k} a_k$ from the finite list allowed in Section 3. Write $V = V_l$, $R = R_l$, and let $\mathcal{P}_l$ be the prime pool in this gap, with law

$$\lambda(p) = \frac{1}{pS_l}, \quad S_l = \sum_{p \in \mathcal{P}_l} \frac{1}{p}.$$

In this Section z_k has law μ_{i_k} , independently for different k . We abbreviate $\nu_k = \nu_{B_k}$, so that $0 \leq \nu_k \leq V$ on all integers. For $\emptyset \neq J \subseteq [m]$ put

$$a(J) = \max J, \quad L_J(z) = \sum_{k \in J} \frac{c_k}{c_{a(J)}} z_k.$$

For sufficiently large w , all the displayed coefficients are integers, and every coefficient other than the anchor coefficient is divisible by W . The scales c_k are positive integers with no prime factor greater than w . We consider

$$\mathcal{C} = \mathbb{E}_z \left[\prod_{\emptyset \neq U \subseteq [m]} b_U(z_U) \prod_{\emptyset \neq J \subseteq [m]} g_J(L_J(z)) \right], \quad z_U = \prod_{k \in U} z_k, \quad (4.1)$$

where the functions are real, $|b_U| \leq 1$, and $|g_J(y)| \leq 1 + \nu_{a(J)}(y)$ for every integer y . Fix J_* with $|J_*| \geq 2$, and write $g_* = g_{J_*}$ and $a_* = a(J_*)$. Bounded extensions outside the positive integers are permitted whenever needed.

We will keep one copy of g_* throughout: first remove the product masks, then eliminate the other linear factors, and finally identify the sampling law of the surviving cube root.

All estimates below are uniform over these functions. An $o(1)$ may depend on the fixed master data and on a fixed positive integer J_0 , but not on the functions. Uniformity also holds over any fixed finite set of J_0 's. In particular, it does not assert a convergence rate uniform in a growing master index count.

4.1 Prime insertion and weighted mask removal

Lemma 4.1 (Prime insertion). *Let $i > l$ and let Y have law μ_i . For every fixed $A > 0$, uniformly over functions F with $|F| \leq V^A$,*

$$\mathbb{E}_Y F(Y) = \mathbb{E}_{p,Y} F(pY) + o(1). \quad (4.2)$$

For a fixed positive integer k , coprime to W and bounded by a fixed power of $P_l^+ + V$, one also has

$$\mathbb{E}_Y F(kY) = \mathbb{E}_Y k \mathbf{1}_{k|Y} F(Y) + o(1), \quad (4.3)$$

uniformly in k . The total-mass error before multiplication by F is smaller than every fixed negative power of $P_l^+ + V$. These assertions remain valid with other independent variables as parameters.

Proof. The dilation assertion is Lemma 3.1, applied at a cutoff whose logarithm dominates all fixed powers of $P_l^+ + V$. The weighted law on the right has the same harmonic density as the law on the left; only the endpoints $[X_i, X_i^2)$ and $[kX_i, kX_i^2)$ differ. The logarithmic boundary mass is $O(\log k / \log X_i)$, with the residue counting error from that Lemma. The scale separation makes their sum smaller than any prescribed negative power of $P_l^+ + V$.

Averaging the multiplier in (4.3) gives

$$A_l(Y) = \mathbb{E}_p p \mathbf{1}_{p|Y} = \frac{1}{S_l} \sum_{p \in \mathcal{P}_l} \mathbf{1}_{p|Y}.$$

Periodic counting for the moduli p and pq , including the W -unit restriction, gives their reciprocal probabilities with uniformly negligible relative errors. Therefore

$$\begin{aligned} \mathbb{E} A_l &= 1 + o(V^{-C}), \\ \mathbb{E} A_l^2 &= \frac{1}{S_l^2} \left(\sum_p \frac{1}{p} + \sum_{p \neq q} \frac{1}{pq} \right) + o(V^{-C}) \\ &= 1 + \frac{1}{S_l} - \frac{1}{S_l^2} \sum_p \frac{1}{p^2} + o(V^{-C}) \end{aligned}$$

for every fixed C . Thus $\|A_l - 1\|_{L^2(\mu_i)} \leq S_l^{-1/2} + o(V^{-C})$. Cauchy–Schwarz bounds the error in replacing A_l by 1 by $V^A(S_l^{-1/2} + o(V^{-C})) = o(1)$, since S_l dominates every fixed power of V . The estimates are uniform in the parameters of F , so one may integrate over those parameters afterwards. $\square$

A row template will mean a vector $A_R = (A_{R,1}, \dots, A_{R,m})$ whose nonzero entries are monomials in finitely many formal prime variables. No prime variable occurs in two different columns of a single row. For $a(R) = \max\{k : A_{R,k} \neq 0\}$, its associated linear form and weight are

$$\ell_R(z) = \sum_k \frac{c_k}{c_{a(R)}} A_{R,k} z_k, \quad W_R(y) = 1 + \nu_{a(R)}(y). \quad (4.4)$$

Parallelity of templates always means parallelity over the rational function field in the formal prime variables. This qualification is essential: accidental equalities after numerical substitution are exceptional events, not identities of templates.

Reciprocal dilations preserving a product, followed by Cauchy–Schwarz to remove its bounded factor, also occur in the proof of [11, Proposition 2.4] over finite fields. Here the divisibility weights allow the corresponding substitutions on integer variables, and we carry out the elimination for the full family of product masks.

Lemma 4.2 (Weighted removal of multiplicative masks). *Put $q_{\text{mask}} = 2^m - 1$ and $K_m = q_{\text{mask}}^{2^{q_{\text{mask}}}}$. Starting from (4.1), one obtains a family $\mathcal{R}$ of at most K_m pairwise nonparallel row templates, with one distinguished row $*$, and functions f_R that may depend on the introduced primes, such that*

$$|\mathcal{C}|^{2^{q_{\text{mask}}}} \leq C_m \left| \mathbb{E}_{p,z} \prod_{R \in \mathcal{R}} f_R(\ell_R(z)) \right| + o(1). \quad (4.5)$$

Here $|f_R| \leq W_R$, the distinguished function is exactly $f_* = g_*$, its support is J_* , and every prime parameter has the law λ , independently before deletion of exceptional tuples. The row templates, the number of parameters, and C_m depend only on m and the choice of J_* . They are independent of the scales and functions.

Proof. Initially the rows are the indicator vectors of the nonempty subsets J of $[m]$. They are pairwise nonparallel. We remove the masks in a fixed order, maintaining (4.4), the weight bounds, and a distinguished copy of g_* .

A substitution fixing the mask. Consider the mask with support U . If the distinguished support has an index $u \notin U$, use the change $D(p)z$ given by $z_u \mapsto pz_u$. Otherwise choose distinct u, v in the distinguished support, necessarily both in U , and use

$$z_u \mapsto z_u/p, \quad z_v \mapsto pz_v, \quad \eta_p(z) = p\mathbf{1}_{p|z_u}. \quad (4.6)$$

The first case has $\eta_p = 1$. Lemma 4.1 justifies the first case directly. For the second, first insert p in coordinate v and then use (4.3) in coordinate u with the function read at z_u/p . Thus division by p is used only on the support of the displayed indicator, and its multiplier is retained. At this stage all unweighted functions and all products of row weights are bounded by a fixed power of V ; even the intermediate fixed-prime multipliers are absorbed by the stronger endpoint error in (4.3). Both substitutions therefore have $o(1)$ error. In either case z_U is unchanged.

Call a row invariant at this step when the coefficient vector of $\ell_R(D(p)z)$ is a scalar multiple of that of $\ell_R(z)$. The scalar is $1, p$, or p^{-1} . All divisors in $\nu_{a(R)}$ are smaller than the pool primes, so multiplication or admissible division by p preserves their divisibility conditions. Consequently

$$W_R(\ell_R(D(p)z)) = W_R(\ell_R(z))$$

on the support in use. Put $\Omega(z) = \prod_{R \text{ invariant}} W_R(\ell_R(z))$. Divide each invariant function inside the fresh prime average by its corresponding weight, and call the remaining integrand H_p . Let $\mathbf{r}$ denote all prime slots introduced at earlier steps. The current mask and row functions may depend on $\mathbf{r}$. All these previous slots are integrated together with z in the outer expectation. Thus the substituted correlation is

$$I = \mathbb{E}_{\mathbf{r}, z} b_U(\mathbf{r}; z_U) \Omega(\mathbf{r}, z) \mathbb{E}_p \eta_p(z) H_p(\mathbf{r}, z).$$

Here the new slot p is independent of $(\mathbf{r}, z)$. Weighted Cauchy–Schwarz on the joint outer probability space gives

$$|I|^2 \leq (\mathbb{E}_{\mathbf{r}, z} \Omega(\mathbf{r}, z)) \mathbb{E}_{\mathbf{r}, z, p, q} \Omega(\mathbf{r}, z) \eta_p(z) \eta_q(z) H_p(\mathbf{r}, z) H_q(\mathbf{r}, z). \quad (4.7)$$

The old tuple $\mathbf{r}$ is shared by the two branches, while p and q are fresh independent slots. This inequality applies to the full prime average.

If there are s_{inv} invariant rows, the first factor equals $2^{s_{\text{inv}}} + o(1)$, and hence is bounded in terms of the number of rows alone. Indeed, expand Ω as a sum of products of ν 's and apply Proposition 3.6 with $\mathbf{r}$ among its prime parameters. Every anchor coefficient is a monomial of pool primes, hence a unit at every prime $w < \pi \leq V$. A nonzero minor of two templates is multiplied in the actual rows by smooth row and column factors, also units at such π , so its possible vanishing is covered by a fixed polynomial test. This bound uses the average over $\mathbf{r}$.

The two prime branches. In the one-coordinate case the two branches have multipliers p and q on u . In the two-coordinate case, first discard $p = q$. Since all remaining factors are bounded by $V^{O_m(1)}$, its contribution is at most

$$V^{O_m(1)} \sum_p \lambda(p)^2 p^2 \mathbb{P}(p \mid z_u) \leq (1 + o(1)) V^{O_m(1)} \sum_p p \lambda(p)^2 = \frac{V^{O_m(1)}}{S_l} + o(1) = o(1).$$

For $p \neq q$, the multiplier is $pq\mathbf{1}_{pq|z_u}$. Absorb it by (4.3), replacing z_u by pqz_u . The two branch multipliers on (u, v) are then

$$(q, p) \quad \text{and} \quad (p, q). \quad (4.8)$$

Every surviving multiplicative mask is a bounded function of a scalar multiple of its original monomial; its two copies can be combined into one mask on that same support.

For an invariant row the two scalar-changed arguments are $\alpha_p \ell_R(z)$ and $\alpha_q \ell_R(z)$, after absorption if necessary. Its new function is

$$t \mapsto \frac{f_R(\alpha_p t) f_R(\alpha_q t)}{W_R(t)},$$

on the arguments actually used, and it is bounded by $W_R(t)$. This formula records why only one weight survives. It is extended elsewhere with the same bound. In particular, in the two-coordinate case an invariant singleton on u acquires arguments qt and pt ; the weight originally evaluated at pqt equals $W_R(t)$. Every noninvariant row instead gives two functions, each with its original weight bound, on the two substituted rows.

Preserving the row structure. The new templates are pairwise nonparallel. For descendants of different old rows, a purported symbolic parallelity specializes at $p = q = 1$ to parallelity of the old rows, a contradiction. For the two descendants of one row, use the exponent description of the substitution: a column is multiplied by p^{e_k} , with $e_k \in \{0, 1\}$ or $\{-1, 0, 1\}$. The p and q branches are parallel precisely when all e_k on the row support are equal. That is exactly the definition of an invariant row. The common column multiplication by pq in (4.8) is invertible over the rational function field and does not affect this argument. The distinguished row is noninvariant: it contains both an affected and a differently affected coordinate. Track a fixed one of its two copies. Its function is still g_* and its support is unchanged.

The new slots occur in at most one column per row, as is explicit in (4.8). Thus the template invariant persists. Repeated prime entries and exact polynomial coincidences have probability smaller than every fixed negative power of V . After absorption the integrands are bounded by $V^{O_m(1)}$, so these events can be deleted or reinstated at $o(1)$ cost. The reinstated substitution-form terms on $p = q$ have this same harmless bound; the larger, pre-absorption diagonal was estimated separately above. Consequently fresh slots can always be sampled independently. At the next step the entire existing tuple, including this step's p and q , is the shared outer tuple $\mathbf{r}$. The deletion and reinstatement argument restores its independent product law before that next step. Any temporarily retained prime-only domain is covered by the restricted form of Proposition 3.6; the base variables and the independently expanded divisor draws retain their original laws.

Each step removes one mask, uses one square, and at most doubles the number of rows. There are q_{mask} masks and initially q_{mask} rows, giving at most K_m rows. Iterating (4.7), with its bounded prefactors and uniform errors, proves (4.5). Fixing the order of masks, the choices of coordinates, and the tracked branch makes every template depend only on m and J_* . The original and intermediate correlations are bounded by constants, by the same linear-forms estimates, so raising the inequalities to these fixed powers preserves $o(1)$ errors. $\square$

4.2 Directions with one vanishing row response

The product masks have been removed. To eliminate a nontarget row by Cauchy–Schwarz, we need a translation that fixes that row and moves every other row, including the target. We choose these translations so that all target responses are equal.

Let $\mathcal{R}$ now denote the output family of Lemma 4.2, and let $d = |\mathcal{R}| - 1$. For a nonzero integer n , write $|n|_{>w}$ and $|n|_{\leq w}$ for its rough and smooth parts, including multiplicity.

Lemma 4.3 (Polynomial directions and integer translations). *For each $R \neq *$ there is an integer polynomial vector w_R such that*

$$A_R w_R = 0, \quad A_I w_R \neq 0 \quad (I \neq R).$$

There is also an integer polynomial vector w_0 with $A_ w_0 = 0$ and $A_I w_0 \neq 0$ for $I \neq *$. These choices can be made from a finite list determined solely by the row templates. Set*

$$d_R = A_* w_R, \quad D = \prod_{R \neq *} d_R, \quad M(\mathbf{p}) = M|D(\mathbf{p})|_{>w}. \quad (4.9)$$

On the good prime tuples planned in Lemma 3.3, the vectors

$$v_{R,k} = M(\mathbf{p}) \frac{c_{a_*}}{c_k} \frac{w_{R,k}}{d_R}, \quad v_{0,k} = M \frac{c_{a_*}}{c_k} w_{0,k} \quad (4.10)$$

belong to $W\mathbb{Z}^m$ and have sizes bounded by a fixed power of $P_l^+ + V$. Their responses satisfy

$$\ell_*(v_R) = M(\mathbf{p}), \quad \ell_R(v_R) = 0, \quad \ell_*(v_0) = 0. \quad (4.11)$$

Every other response is nonzero. At primes $w < \pi \leq V$ it is a unit unless a member of a fixed list of nonzero integer polynomials in the prime slots vanishes modulo π .

Proof. Here is an explicit finite choice of a kernel vector. For a row A_R , choose a column j with $A_{R,j} \neq 0$. The polynomial vectors

$$b_k = A_{R,j} e_k - A_{R,k} e_j \quad (k \neq j)$$

span its kernel over the rational function field. Enumerate them as $b_0, \dots, b_{m-2}$ and set $w_R(t) = \sum_{h=0}^{m-2} t^h b_h$. For $I \neq R$, the polynomial $A_I w_R(t)$ is not identically zero: otherwise A_I would vanish on the kernel of A_R and hence be parallel to it. The product of these polynomials has degree at most $(|\mathcal{R}| - 1)(m - 2)$ in t . At least one of the first $(|\mathcal{R}| - 1)(m - 2) + 1$ positive integers is therefore not a root over this field. Use the first such integer. This is a deterministic choice, gives integer polynomial coordinates, and has every asserted nonzero response. Applying the same construction to A_* gives w_0 .

Include D , all nonzero response polynomials, and the required row minors in the finite polynomial list used to choose M and the gap scales. There is no dependence on the functions in this list. Delete tuples with a zero polynomial value or a repeated prime entry, and tuples for which $\pi^{e_0} \mid D$ for some $\pi \leq w$. The deleted probability is $o(1)$; the first two parts even have superpolynomially small probability in V . Deletion at this point remains legitimate for weighted expectations: for any product of distinct row weights its integral over a prime-only domain E equals $2^t \mathbb{P}(E) + o(1)$, where t is the number of weights, by expansion and the restricted-domain assertion of Proposition 3.6. A domain of probability $o(1)$ therefore has $o(1)$ weighted mass. We henceforth normalize the prime law on the remaining good tuples, whose probability tends to one.

For integrality, write $D = \epsilon S Q$, where $\epsilon \in \{-1, 1\}$, $S = |D|_{\leq w}$, and $Q = |D|_{>w}$. On the good tuples $S \mid W^{e_0-1}$. Since M/c_k is an integer divisible by W^{e_0+1} , the expression in (4.10) can be written

$$v_{R,k} = \epsilon \frac{M/c_k}{S} c_{a_*} w_{R,k} \frac{D}{d_R}.$$

All its factors are integers, and $(M/c_k)/S$ is divisible by W . The assertion for v_0 follows directly from its formula. The degree and coefficient bounds for the polynomial vectors give $|v_{R,k}| + |v_{0,k}| \leq (P_l^+ + V)^{O_m(1)}$; all the scales c_k are at most $M \leq V$.

Direct substitution proves (4.11). More generally,

$$\ell_I(v_R) = M(\mathbf{p}) \frac{c_{a_*}}{c_{a(I)}} \frac{A_I w_R}{d_R}, \quad \ell_I(v_0) = M \frac{c_{a_*}}{c_{a(I)}} A_I w_0.$$

Outside primes dividing $D(A_I w_R)$, respectively $A_I w_0$, the first, respectively second, expression is a unit at $w < \pi \leq V$. The smooth factors are units there. This proves both the response claim and the finite-polynomial nature of all exceptional tests. $\square$

4.3 Additive elimination and removal of the remaining weights

Lemma 4.4 (Weighted additive elimination). *With the rows, good-tuple law, and modulus of Lemma 4.3, fix $J_0 \in \mathbb{N}$ and put*

$$L(\mathbf{p}) = \left\lfloor \frac{R}{J_0 M(\mathbf{p})} \right\rfloor. \quad (4.12)$$

*Independently conditional on the primes, let every u_R^0, u_R^1 be uniform on $[0, L(\mathbf{p})) \cap \mathbb{Z}$, for $R \neq *$. Then*

$$\left| \mathbb{E}_{\mathbf{p}, z} \prod_{I \in \mathcal{R}} f_I(\ell_I(z)) \right|^{2^d} \leq C_m \left| \mathbb{E}_{\mathbf{p}, z, u} \prod_{\omega \in \{0,1\}^d} g_* \left(\ell_*(z) + M(\mathbf{p}) \sum_{R \neq *} u_R^{\omega_R} \right) \right| + o(1). \quad (4.13)$$

The constants are independent of J_0 ; the error is for fixed J_0 .

Proof. Weighted Cauchy–Schwarz will leave a cube of g_* multiplied by retained row weights. We first form that cube, then use the direction v_0 to remove the retained weights in a weighted second moment.

Every interval in (4.12) has length dominating all fixed powers of V , uniformly over the prime tuples. Indeed $M(\mathbf{p})$ is bounded by a fixed power of $P_l^+ + V$, whereas R dominates all such powers. Insert independent uniform u_R on these intervals and translate

$$z \mapsto z + \sum_{R \neq *} v_R u_R.$$

The translations are in $W\mathbb{Z}^m$. Their sizes are at most $R(P_l^+ + V)^{O_m(1)}$, and hence are negligible at every pivot cutoff, even after multiplying the total-variation errors by $V^{O_m(1)}$. Lemma 3.1 therefore changes the average by $o(1)$.

Eliminate the nondistinguished rows in a fixed order. At the step assigned to row R , all its current copies are independent of u_R , because $\ell_R(v_R) = 0$. Let H_{out} be their product, and let Ω_R be the product of their bounds W_R . All other factors, including weights retained at earlier steps, belong to H_{in} . With all variables other than u_R outside, the exact inequality is

$$\begin{aligned} & |\mathbb{E}_{\text{out}} H_{\text{out}} \mathbb{E}_{u_R} H_{\text{in}}|^2 \\ & \leq (\mathbb{E}_{\text{out}} \Omega_R) \mathbb{E}_{\text{out}, u_R^0, u_R^1} \Omega_R H_{\text{in}}(u_R^0) H_{\text{in}}(u_R^1). \end{aligned} \quad (4.14)$$

It follows by applying Cauchy–Schwarz with density Ω_R and using $|H_{\text{out}}| \leq \Omega_R$. In particular, the assigned functions disappear and their weights occur once, while every factor inside the average is duplicated. All prime parameters remain outside throughout this procedure.

We verify every use of Proposition 3.6 in (4.14). A current copy of a base row I has the form

$$\ell_I(z) + \sum_Q \ell_I(v_Q) u_Q^{\eta_Q},$$

where η_Q is one of the already duplicated choices or the single current variable. Distinct base rows are distinguished by their z -coefficient vectors. For two copies of the same row, some duplicated coordinate $Q \neq I$ has different choices, and its response $\ell_I(v_Q)$ is nonzero. Their difference therefore has a nonzero coefficient on an independent variable u_Q^0 or u_Q^1 . Taking a minor with an anchor z -column proves pairwise independence at every relevant prime outside the response-polynomial tests of Lemma 4.3. A row's anchor coefficient is still a monomial of pool primes, so each row is primitive at all $w < \pi \leq V$. Finally, conditional on the primes the base variables are independent harmonic variables or independent interval variables whose lengths dominate powers of V . Their joint residues modulo the product of any fixed number of sampled divisors are uniform with error smaller than every fixed negative power of V . These facts check primitivity, pair independence with polynomial exceptions, and the sampling hypotheses. Expanding the weights now shows that every prefactor in (4.14) is bounded by a constant depending only on the number of row copies.

After all d steps, the nondistinguished functions have disappeared. The remaining target factor is

$$G(z, u) = \prod_{\omega \in \{0,1\}^d} g_* \left(\ell_*(z) + M(\mathbf{p}) \sum_R u_R^{\omega_R} \right).$$

For each eliminated row I , there is one retained weight for each choice of the duplicated variables in the directions $R \neq I$. Thus the retained product is

$$\Psi(z, u) = \prod_{I \neq *} \prod_{\eta \in \{0,1\}^{[d] \setminus \{I\}}} W_I \left(\ell_I(z) + \sum_{R \neq I} \ell_I(v_R) u_R^{\eta_R} \right), \quad (4.15)$$

after identifying the nondistinguished rows with $[d]$. There are $t = d2^{d-1}$ weight factors. Iteration has proved the left side of (4.13) is bounded by a constant times $|\mathbb{E}G\Psi| + o(1)$.

It remains to justify replacing Ψ by 2^t in this correlation; its mean alone would not justify that replacement. Translate z once more by $v_0 u_0$, with u_0 uniform on $[0, R) \cap \mathbb{Z}$. The sampling error is again $o(1)$, and the target G is unchanged because $\ell_*(v_0) = 0$. Put

$$B(z, u) = \prod_{\omega \in \{0,1\}^d} \left(1 + \nu_{a_*} \left(\ell_*(z) + M(\mathbf{p}) \sum_R u_R^{\omega_R} \right) \right), \quad H(z, u) = \mathbb{E}_{u_0} \Psi(z + v_0 u_0, u).$$

Then $|G| \leq B$. The same row check gives

$$\mathbb{E}B = 2^{2^d} + o(1), \quad \mathbb{E}BH = 2^{2^d+t} + o(1), \quad \mathbb{E}BH^2 = 2^{2^d+2t} + o(1). \quad (4.16)$$

For completeness, in the last expression duplicate u_0 independently as u_0^0, u_0^1 . Copies from distinct base rows are distinguished on z ; copies from one base row and different old shift choices are distinguished on an old u_R^η ; copies with identical old choices but different u_0 branches are distinguished by the nonzero response $\ell_I(v_0)$. Target rows have distinct cube shift choices and no u_0 response. All forms in every expanded moment are thus distinct and satisfy exactly the primitivity and polynomial-minor conditions checked above. Expanding each $1 + \nu$ and replacing every product of ν 's by its main term 1 gives all three powers of 2 in (4.16).

Consequently

$$\mathbb{E}B(H - 2^t)^2 = o(1).$$

Cauchy–Schwarz with density B now yields

$$|\mathbb{E}G(H - 2^t)| \leq (\mathbb{E}B)^{1/2}(\mathbb{E}B(H - 2^t)^2)^{1/2} = o(1).$$

This is the required weighted replacement. Combining it with the d inequalities (4.14) proves (4.13). $\square$

4.4 A one-variable cube estimate

Proposition 4.5 (Uniform correlation test). *For every $m \geq 2$ and every nonsingleton target support J_* , the construction above determines a finite family of integer polynomial templates D , with good prime-tuple laws, such that the following holds for every fixed master chain and valid gap l . There are an integer $1 \leq d \leq K_m - 1$, a constant C_m , and $\theta > 0$, all bounded in terms of m alone, for which*

$$|C| \leq o(1) + C_m \left| \mathbb{E}_{\mathbf{p}, y, u} \prod_{\omega \subseteq [d]} g_* \left(y + M(\mathbf{p}) \sum_{R \in \omega} (u_R^1 - u_R^0) \right) \right|^\theta. \quad (4.17)$$

Here y has law $\mu_{i_{a_*}}$, independently of the primes and shifts; the primes have the normalized good-tuple law of Lemma 4.3; $M(\mathbf{p})$ is given by (4.9); and the shifts have the conditional laws in (4.12). One may take $\theta = 2^{-(q_{\text{mask}} + d)}$. The gap length R_l is divisible by every $M(\mathbf{p})$ that occurs.

The estimate is uniform over all the masks and functions in (4.1), including g_* , and over each fixed finite set of positive integers J_0 . Neither d , C_m , nor θ^{-1} depends on N , on the scales, or on the functions. The polynomial templates and direction choices are fixed before M and the prime pools are chosen. In particular, the same tests can be used both for a bounded error and for an error bounded by $1 + \nu_{a_*}$.

Proof. Only the pushforward of the cube root remains after Lemmas 4.2 and 4.4. For fixed primes and shifts, write

$$Y_* = \ell_*(z) + M(\mathbf{p}) \sum_R u_R^0 = kz_{a_*} + h, \quad k = A_{*, a_*}. \quad (4.18)$$

The target support remains J_* , so it contains some $j < a_*$. The coefficient $b = (c_j/c_{a_*})A_{*, j}$ is coprime to k : the prime slots in different columns are disjoint, their numerical entries are distinct on the good tuples, and all scale ratios are w -smooth whereas all pool primes exceed $V \geq w$. Also k is coprime to W and $W \mid h$.

Condition first on every z -variable except z_{a_*} . Put $X = X_{i_{a_*}}$ and $\delta_W = \phi(W)/W$. Uniformly in the variables being conditioned on, one may take

$$0 \leq h \leq H := M(P_l^+)^{B_m} \sum_{j < a_*} X_{i_j}^2 + dR/J_0$$

for a fixed B_m . The logarithm of X dominates every fixed power of $H + k + W$, and in particular $H < X/2$ eventually. The dilation and translation calculation of Lemma 3.1 gives

$$\mathcal{L}(kz_{a_*} + h) = k \mathbf{1}_{y \equiv h \pmod{k}} \mu_{i_{a_*}}(dy) + \mathcal{E}_h, \quad \|\mathcal{E}_h\|_{\text{TV}} = o(V^{-C}) \quad (4.19)$$

for every fixed C . More explicitly, the total-mass error is at most

$$O \left(\frac{\log(2k)}{\log X} + \frac{H}{X} + \frac{Wk^2}{\delta_W X \log X} \right). \quad (4.20)$$

To see this directly, let Z_i be the normalizing constant of μ_i , so $Z_i \asymp \delta_W \log X$. The exact image density is $k/(Z_i(y-h))$ on $[kX+h, kX^2+h)$ in the class $h \pmod k$, with $(y, W) = 1$. The comparison density is $k/(Z_i y)$ on $[X, X^2)$ in the same class. The W -unit restriction agrees because $W \mid h$ and $(k, W) = 1$. On the intersection of the intervals the relative discrepancy is at most H/X . Harmonic progression counting on the two remaining boundary intervals gives a logarithmic main term $O(\delta_W \log(2k))$ and error $O(Wk^2/X)$ before normalization. This proves (4.20). All three terms are smaller than every fixed negative power of V by scale separation; the crude bound $\delta_W^{-1} \leq W$ already suffices here.

Now condition on all remaining variables except z_j . Since $i_j > l$, Lemma 3.1 makes z_j uniform modulo k ; its relative error is at most

$$O\left(\frac{Wk^2}{\delta_W X_{i_j} \log X_{i_j}}\right) = o(V^{-C})$$

for every fixed C . The same holds for $h = bz_j + h_0$ modulo k , because $(b, k) = 1$. Averaging the density in (4.19) therefore gives

$$\mathbb{E}_{z_j} k \mathbf{1}_{y \equiv h \pmod k} = 1 + o(V^{-C})$$

uniformly in y and in the conditioned primes and shifts. Thus the conditional law of Y_* , after integrating all z 's, differs from $\mu_{i_{a*}}$ by $o(V^{-C})$ in total mass, uniformly in the primes and shifts. This argument also covers $k = 1$, when there is no residue condition to average.

Every remaining cube argument can be expressed using this root as

$$Y_* + M(\mathbf{p}) \sum_{R \in \omega} (u_R^1 - u_R^0).$$

There is no other z -dependent factor left. The product of target bounds is at most $(1+V)^{2^d}$, so the total-mass error just proved is still $o(1)$ in the cube expectation. Substitution in (4.13) and then (4.5) gives (4.17) with the stated exponent. All constants arise from fixed counts of row weights and squares, hence depend only on m . At least one other row exists, so $d \geq 1$.

Finally, $M(\mathbf{p})$ divides $M|D(\mathbf{p})|$, whose divisibility into R_l was arranged in Lemma 3.3. The finite template assertion follows from the deterministic mask and kernel-vector choices. Every sampling estimate and every restricted linear-forms estimate used above was uniform over the functions; taking the maximum of the errors over finitely many J_0 's proves the last uniformity assertion. $\square$

Remark 4.6 (Tests used in the next section). The base vertex in (4.17) is $g_*(y)$. All its other vertices may therefore be regarded as inputs in a dual test

$$\mathcal{D}(y) = \mathbb{E}_{\mathbf{p}, u} e(\mathbf{p}) \prod_{\emptyset \neq \omega \subseteq [d]} h_{\omega, \mathbf{p}} \left(y + M(\mathbf{p}) \sum_{R \in \omega} (u_R^1 - u_R^0) \right), \quad |e| \leq 1, \quad |h_{\omega, \mathbf{p}}| \leq 1 + \nu_{a*}.$$

In particular, choosing every $h_{\omega, \mathbf{p}} = g_*$ and $e = 1$ recovers its cube expectation as $\mathbb{E}_{\mu_{i_{a*}}} g_* \mathcal{D}$. The enlarged test class, including independently chosen inputs and bounded prime-dependent signs, is fixed without reference to a dense model. Proposition 4.5 itself asserts the inequality for the repeated target function; the pseudorandomness of the enlarged test class is proved in Section 5.

5 Dense models and prediction

We prove Principle 2.3. The correlation estimate of Section 4 first allows the unbounded color weights to be replaced by bounded functions. We then choose nilsequence approximations at coarse scales. A Ramsey argument compares their projection energies with those at the finer scales required by the correlation estimate. All analytic moduli in this last comparison will be independent of the number of master indices.

Fix m . Choose an integer $s \geq 3$ such that

$$s + 1 \geq 2(2^d) - 1 \quad (5.1)$$

for every cube order d occurring in Proposition 4.5. The order bound in that Proposition makes this a choice depending only on m . The later Cauchy–Schwarz argument will bound a d -cube test by a subgroup norm of order $k = 2^d$; concatenation then uses order $2k - 1$, whose inverse theorem gives nilsequences of step $2k - 2$. Fix also the data $n, r, \chi, \mathcal{B}, \mathcal{A}$ in Principle 2.3. For the moment fix a master index count N and the parameters of Lemma 3.3; only blocks with at most n members are used. For $B = T \cup \{i\}$, $a \in \mathcal{A}$, and $c \in [r]$, write

$$\nu = \nu_B, \quad f(y) = \mathbf{1}_{\chi(h_B a y) = c}, \quad \rho(y) = \nu(y)f(y).$$

Here f is zero on nonpositive integers. All sufficiently large w make its positive color arguments integral. We make two replacements, $\rho \rightarrow F \rightarrow S$. The bounded function F must approximate ρ against the cube tests from Section 4; the piecewise nilsequence S must additionally admit alignment at the coarse scale. The gap-energy comparison will make these two demands compatible.

5.1 An algebra of dual tests

A cube type at B means one of the types supplied by Proposition 4.5, at a gap valid for its chain. We also include, at every gap $\max T < l < i$, the type of dimension $s + 1$ with modulus M and no prime variables. Fix an integer $J_0 \geq 1$. For any one of these types put $M_{\mathbf{p}} = M(\mathbf{p})$ and

$$L_{\mathbf{p}} = \lfloor R_l / (J_0 M_{\mathbf{p}}) \rfloor.$$

Primes have the good-tuple law of Proposition 4.5; the variables u_j^0, u_j^1 are independent and uniform in $[0, L_{\mathbf{p}}] \cap \mathbb{Z}$. Its dual tests are the real functions

$$\mathcal{D}(y) = \mathbb{E}_{\mathbf{p}, u e(\mathbf{p})} \prod_{\emptyset \neq \omega \subset [d]} g_{\omega, \mathbf{p}} \left(y + M_{\mathbf{p}} \sum_{j \in \omega} (u_j^1 - u_j^0) \right), \quad |e| \leq 1, \quad |g_{\omega, \mathbf{p}}| \leq 1 + \nu. \quad (5.2)$$

The functions, as well as the bounded prime factor, may vary arbitrarily with w . The finitely many cube templates and any fixed finite set of J_0 values are included in each uniform assertion below.

To construct F , we need more than a mean-one estimate for ν . The next Lemma shows that $\nu - 1$ is asymptotically orthogonal to the algebra generated by the dual tests, and controls the tests sufficiently to replace them by bounded ones.

Lemma 5.1 (Dual-test pseudorandomness). *For every fixed $b \geq 0$ and dual tests $\mathcal{D}_1, \dots, \mathcal{D}_b$ at the same block,*

$$\mathbb{E}_{\mu_i}(\nu - 1) \prod_{q=1}^b \mathcal{D}_q = o(1) \quad (5.3)$$

uniformly over their inputs. The tests may use different permissible gaps and different cube types. If d_* bounds the dimensions of these types and $A_* = 2^{2^{d_*}-1}$, then for every fixed positive integer b ,

$$\mathbb{E}_{\mu_i}(1+\nu)|\mathcal{D}|^b \leq 2A_*^b + o(1). \quad (5.4)$$

For any fixed $K > A_*$, replacing every test by its clipping to $[-K, K]$ changes it by $o(1)$ in every fixed $L^p((1+\nu)\mu_i)$ norm, and (5.3) remains true for the clipped tests.

Proof. Expand the product using independent prime and shift replicas. In replica q , of dimension d_q and modulus M_q , the nonroot rows are

$$Y_{q,\omega} = y + M_q \sum_{j \in \omega} (u_{q,j}^1 - u_{q,j}^0), \quad \emptyset \neq \omega \subset [d_q].$$

The root row is y . For a fixed nonempty ω , choose integers $a_0, a_1, \dots, a_{d_q}$ such that

$$a_0 \neq 0, \quad a_0 + \sum_{j \in \omega} a_j = 0, \quad a_0 + \sum_{j \in \omega'} a_j \neq 0 \quad (\omega' \neq \omega).$$

Indeed, the hyperplane defined by the equality is not contained in any of the finitely many hyperplanes excluded by the inequalities: their normal vectors $(1, \mathbf{1}_{\omega'})$ are pairwise nonparallel. A rational point avoiding their intersections exists, and clearing denominators gives the indicated integers. Insert a translation parameter for this row, acting by

$$y \mapsto y + a_0 M_q v, \quad u_{q,j}^1 \mapsto u_{q,j}^1 + a_j v,$$

and leaving all other shift variables unchanged. It fixes the chosen row. Every other row in its replica has a nonzero response, the root has response $a_0 M_q$, and a row in another replica has the same nonzero response $a_0 M_q$.

These directions are compatible even when the gaps differ. Put $V_B = 2 + M + \prod_{b \in T} X_b^2$. The shortest shift interval in each replica dominates every fixed power of V_B . The translation parameters for replica q may, for example, have lengths comparable to the square root of its shortest shift length. Their lengths still dominate powers of V_B , whereas the resulting displacements of its shifts are negligible relative to their intervals, even after multiplication by any fixed power of V_B . The sum of the root displacements is likewise negligible at X_i . The parameters depend on the primes only through the lengths, which is permitted by the uniform box version of Proposition 3.6. Thus inserting all these averaged translations changes the original expression by $o(1)$: use the crude bound $1 + \nu \leq 1 + \prod_{b \in T} X_b^2$ and the translation estimates of Lemma 3.1. Fixed direction constants do not affect these separations.

Eliminate the nonroot rows one at a time by the weighted Cauchy–Schwarz procedure of Lemma 4.4. At the step assigned to a row, all its current copies are independent of that row’s translation parameter. Bound those copies by their $1 + \nu$ weights and use their product as the outside density. Squaring duplicates that row’s assigned translation parameter; all other variables remain in the outside expectation, as in that procedure. Prime-only bounded factors can be discarded outside the inner average. Every prefactor is bounded, and one weight for each eliminated copy is retained in the final expression. The root, which is never eliminated, contributes copies of $(\nu - 1)$ indexed by all choices of the duplicated translation parameters.

We verify the linear-forms hypotheses at every such application. Every form has coefficient 1 at y . Different base rows have distinct coefficients on the original shifts. Copies of one base row are distinguished by a duplicated direction with nonzero response; for an eliminated row these are precisely directions other than its own. Root copies are distinguished because

every inserted direction has nonzero root response. A distinguishing minor is a fixed nonzero integer times a single replica modulus M_q ; no difference of two moduli is required. At a rough prime, its possible vanishing is therefore included among the polynomial exceptional tests defining that modulus. Small-prime factors and fixed nonzero integer factors are covered by the smooth modulus and, eventually, by W . All replicas use gaps after the common tail T , so the multi-gap form of Proposition 3.6 applies with V_B . The arbitrary prime-only restrictions in that Proposition include the separate good-tuple restrictions in each replica.

Expand the retained $1 + \nu$ weights and every root difference. Each resulting product has the main term obtained by replacing each ν by 1, with a uniform $o(1)$ error. In that substitution at least one root difference becomes zero. Thus the final Cauchy–Schwarz expression is $o(1)$, proving (5.3). When $b = 0$ this is simply $\mathbb{E}\nu = 1 + o(1)$.

For the moment bound, Jensen’s inequality and independent replicas give

$$|\mathcal{D}(y)|^b \leq \mathbb{E} \prod_{q=1}^b \prod_{\omega \in \omega \subset [d]} (1 + \nu)(Y_{q,\omega}).$$

Include the root weight $1 + \nu(y)$ and expand. These rows are already distinct without the additional translations. Proposition 3.6 gives $2^{1+b(2^d-1)} + o(1)$, proving (5.4).

Let $\mathcal{D}^{[K]} = \max(-K, \min(K, \mathcal{D}))$. For integers $b > p$,

$$\limsup_{w \rightarrow \infty} \mathbb{E}(1 + \nu) |\mathcal{D} - \mathcal{D}^{[K]}|^p \leq 2K^{p-b} A_*^b.$$

This estimate holds for every fixed b ; sending b to infinity after the limit proves that the left side is zero. Telescoping a fixed product and applying Hölder’s inequality with respect to $(1 + \nu)\mu_i$ now transfers (5.3) to clipped tests, since $|\nu - 1| \leq 1 + \nu$ and all required fixed moments are bounded. No moment order growing with w has been used. $\square$

We use the separation and polynomial-approximation proof of the dense model theorem developed independently by Gowers [10, Section 4] and Reingold, Trevisan, Tulsiani and Vadhan [28, Section 2]. The preceding Lemma supplies the pseudorandomness needed for our particular weighted test class.

Proposition 5.2 (Bounded dense models). *There are functions $F_{B,a,c} : \mathbb{Z} \rightarrow [0, 1]$ such that, for every fixed J_0 and every dual test at B ,*

$$\mathbb{E}_{\mu_i}(\rho - F_{B,a,c})\mathcal{D} = o(1) \tag{5.5}$$

uniformly over its inputs and over the finitely many master blocks, scale labels, and colors.

Proof. First fix a finite collection of J_0 values, a positive desired error, and a common clipping bound K from Lemma 5.1. The support of μ_i is finite. Let $\mathcal{C}$ be the closed convex hull of the signed clipped tests, viewed as vectors on that support. It is compact and lies in $[-K, K]$ coordinatewise. The candidate models form the compact convex cube $\mathcal{F} = [0, 1]^{\text{supp } \mu_i}$. Finite-dimensional minimax for the bilinear pairing gives

$$\inf_{F \in \mathcal{F}} \sup_{G \in \mathcal{C}} \langle \rho - F, G \rangle = \sup_{G \in \mathcal{C}} (\langle \rho, G \rangle - \langle 1, G_+ \rangle), \quad G_+ = \max(G, 0),$$

because $\sup_{F \in \mathcal{F}} \langle F, G \rangle = \langle 1, G_+ \rangle$. Since $0 \leq \rho \leq \nu$, the right-hand side is at most $\sup_G \langle \nu - 1, G_+ \rangle$.

To bound it uniformly, approximate $t \mapsto \max(t, 0)$ on $[-K, K]$ within δ by a polynomial $P(t)$. Every fixed power of a convex combination of signed tests is a convex combination of their signed products. Lemma 5.1, followed by continuity for the closed convex hull,

therefore gives $\langle \nu - 1, P(G) \rangle = o(1)$ uniformly in $G \in \mathcal{C}$. The polynomial approximation costs at most $\delta \mathbb{E}(1 + \nu) = 2\delta + o(1)$. First take $w \rightarrow \infty$ and then $\delta \rightarrow 0$. Minimax produces models with the desired error against the clipped tests; the clipping estimate gives the same conclusion for the original tests, since $|\rho - F| \leq 1 + \nu$.

Apply this argument successively to errors $1/q$ and $J_0 \leq q$. For each fixed q it works for all sufficiently large w , uniformly over the finite master data. Choosing the stage $q = q(w)$ to increase sufficiently slowly gives (5.5) for every fixed J_0 . This diagonal choice is made together with the fixed-template diagonal in Lemma 3.3; it asserts no estimate for an arbitrary growing moment order. Extend the models outside the support of μ_i with values in $[0, 1]$. $\square$

Here is the first consequence for a chain with a valid gap. In its weighted count, replace one factor $\rho_{B_d, a_d, c}(L_J)$, $|J| \geq 2$, by $F_{B_d, a_d, c}(L_J)$. The difference has target $h = \rho - F$, with $|h| \leq 1 + \nu_B$; all other linear factors still satisfy the bounds required by Proposition 4.5. The product masks and the center weights are retained. In the resulting cube, its nonroot copies of h are admissible inputs in (5.2); its root pairing is therefore $o(1)$ by (5.5). Equation 4.17 and a finite telescoping sum show that all nonsingleton color weights can be replaced by their bounded models with total error $o(1)$.

5.2 Nilsequence tests and nested projections

The bounded functions F now replace the weighted color factors in the counts. Calibration requires another comparison: ρ and F must agree against bounded Lipschitz functions of the eventual prediction S , so that small prediction values cannot carry much actual color mass. We first establish this nilsequence testing property, using missing-corner reconstruction to express the tests through the dual tests already controlled. We then choose coarse models whose differences from F have small projections at the finer chain scales. The next subsection turns those projection bounds into counting estimates.

Lemma 5.3 (Testing piecewise nilsequences). *Fix $B = T \cup \{i\}$ and a gap $\max T < l < i$. Let S' be any family of bounded real piecewise nilsequences on intervals of length R_l and residues modulo M , of step at most s and bounded complexity in the sense of Definition 2.2. Then*

$$\mathbb{E}_{\mu_i}(\rho - F_{B, a, c})S' = o(1). \quad (5.6)$$

The assertion is uniform for each fixed finite list of nilmanifolds and fixed observable bounds.

Proof. Lemma 7.2, proved independently in Section 7, supplies a compact set of nilmanifold cubes containing every linear orbit cube

$$(x_\omega)_{\omega \subset [s+1]} = \left(g^{k + \sum_{j \in \omega} v_j x} \right)_{\omega \subset [s+1]}, \quad k, v_1, \dots, v_{s+1} \in \mathbb{Z}.$$

On this set the nonroot vertices determine $x_\emptyset$ continuously. Its Stone–Weierstrass consequence gives the following concrete input. For any accuracy $\delta > 0$, the observable at the root is uniformly approximated by a finite sum

$$\sum_{t=1}^q \lambda_t \prod_{\emptyset \neq \omega \subset [s+1]} \phi_{t, \omega}(x_\omega)$$

with $|\phi_{t, \omega}| \leq 1$. The same Lemma gives finitely many such recipes, with fixed total coefficient bounds, for our finite nilmanifold list and uniformly bounded Lipschitz observables. The recipes are valid for every g and x , so they can be used on all pieces even though the

translating elements and base points are unrestricted. It remains to realize these recipes as our dual tests and control the cubes that cross piece boundaries.

Use the extra dual test of dimension $s + 1$, modulus M , and a fixed large J_0 . If y is farther than $(s + 1)R_l/J_0$ from the endpoints of its R_l interval, all its cube vertices lie in that interval and have the same residue modulo M . The recipe for that piece then reconstructs $S'(y)$. Recipes can be selected through the input functions of the dual test: for a term belonging to recipe r , put in every nonroot input the indicator that its piece selected r , multiplied by its single-vertex factor evaluated on that piece's orbit. Sum over the finite recipes and terms. On cubes staying in one piece this selects exactly its recipe. Each resulting input is bounded by 1, and hence is allowed in (5.2).

We record the weighted boundary estimate needed for the discarded roots. If E_l is the union of these boundary strips, then

$$\mathbb{E}_{\mu_i}(1 + \nu_B)\mathbf{1}_{E_l} \leq O_s(J_0^{-1}) + o(1). \quad (5.7)$$

For a fixed divisor draw $\sigma = t_T$, divisibility by σ and coprimality to W are periodic with period σW . This period is negligible compared with R_l , uniformly in $\sigma \leq \prod_{b \in T} X_b^2$. Counting in each full R_l interval shows that a union of end strips of relative length $O_s(J_0^{-1})$ has that proportion of its weighted mass, with relative error $O(\sigma W/R_l)$. The harmonic factor varies by $o(1)$ within a cell, since $R_l/X_i = o(1)$; the two partial cutoff cells have $o(1)$ mass by the same counting estimate. This proves the assertion for $\sigma \mathbf{1}_{\sigma|y\mu_i}$, uniformly in σ . Average over σ and also take $\sigma = 1$ to obtain (5.7).

The recipe approximation costs $O(\delta)$ against $|\rho - F|\mu_i \leq (1 + \nu_B)\mu_i$. For fixed recipes and J_0 , Proposition 5.2 makes every dual pairing $o(1)$. The terms outside the good roots are bounded by a fixed constant (depending on the recipes) times the quantity in (5.7). First fix sufficiently accurate recipes, then take J_0 sufficiently large for those recipes, and finally take $w \rightarrow \infty$. Since δ was arbitrary, this proves (5.6). $\square$

Use the fixed nonprincipal ultrafilter $\mathcal{U}$ on the positive integers w . For each pivot i , form the real Hilbert space of families of vectors in $L^2(\mu_i)$ with uniformly bounded norms, with pairing $\lim_{\mathcal{U}} \langle \cdot, \cdot \rangle$, after quotienting out zero-norm families and completing. For $l < i$, let $\mathcal{N}_{i,l}$ be the closed span in this space of bounded-complexity piecewise step-at-most- s nilsequence families on the R_l intervals and residues modulo M . Complexity is allowed to vary from one family to another but is bounded within a family. Denote the orthogonal projection by $P_{i,l}$.

When $l < l' < i$, the divisibility $R_l \mid R_{l'}$ aligns the smaller intervals with the larger ones. Restricting an orbit to a smaller interval only changes its base point. Consequently

$$\mathcal{N}_{i,l'} \subseteq \mathcal{N}_{i,l}, \quad \|P_{i,l'}v - P_{i,l}v\|_2^2 = \|P_{i,l}v\|_2^2 - \|P_{i,l'}v\|_2^2. \quad (5.8)$$

Products and Lipschitz real combinations of finitely many representing families remain representing families, using product nilmanifolds. This includes clipping to $[0, 1]$.

Lemma 5.4 (Ramsey selection of gap energies). *For every $\varepsilon > 0$, a master count N depending only on $n, r, |\mathcal{A}|, \varepsilon$ can be chosen so that there are indices*

$$k_1 < j_1 < k_2 < j_2 < \cdots < k_n < j_n$$

and $[0, 1]$ -valued piecewise models $S_{B,a,c}$ at pivot $i = j_u$ on intervals of length R_{k_u} such that

$$\|S_{B,a,c} - P_{j_u,k_u}F_{B,a,c}\|_2 \leq \varepsilon. \quad (5.9)$$

For a chain on the principal indices $j_1, \dots, j_n$, let l be the padding index immediately before its first pivot. At every block $B = T \cup \{j_u\}$ of the chain,

$$\|P_{j_u,l}(F_{B,a,c} - S_{B,a,c})\|_2 \leq 2\varepsilon. \quad (5.10)$$

The families S have bounded complexity after N and ε are fixed.

Proof. The energies $\|P_{i,l}F_{T \cup \{i\},a,c}\|_2^2$ lie in $[0, 1]$. Color each ordered tuple consisting of the increasing members of T , followed by l and i , by their bins of width ε^2 , jointly for all a, c . There are finitely many colors, with a bound independent of N . Apply finite Ramsey successively for all tuple lengths $3, \dots, n+1$ to obtain a homogeneous set of size $2n$. Label it alternately by padding and principal indices as displayed.

At each selected block approximate its coarse projection by a finite linear combination T' of representing families, to accuracy ε . Its clipping S to $[0, 1]$ is still a representing family. Pointwise clipping decreases the distance to $F \in [0, 1]$. Since both T' and S lie in the coarse subspace, orthogonality gives

$$\|F - S\|_2^2 = \|F - PF\|_2^2 + \|PF - S\|_2^2,$$

and the analogous identity for T' . Thus clipping also decreases the distance to PF , proving (5.9). There are only finitely many selected blocks and labels, so their representing families share a finite nilmanifold list and uniform observable bounds.

All tails of the chain precede its first pivot and hence precede l ; also $l \leq k_u < j_u$. If $l < k_u$, the tuples (T, l, j_u) and (T, k_u, j_u) lie in the homogeneous set and have the same color. Their energies differ by at most ε^2 . Equation (5.8) bounds the distance between the two projections by ε . As S belongs to the coarse and hence the fine space, (5.10) follows by the triangle inequality. The case $l = k_u$ follows directly from (5.9). $\square$

5.3 Subgroup cubes and the inverse theorem

The energy selection makes $F - S$ have small projection onto the fine nilsequence space. We must turn that conclusion into a small cube test in Proposition 4.5, with a threshold independent of the master count N . Finite subgroup norms supply this link.

For a finite subgroup Q acting by measure-preserving transformations T^q on a probability space $\mathcal{X}$, and a real bounded function h , use the convention

$$\|h\|_{U_Q^t(\mathcal{X})}^{2^t} = \mathbb{E}_{x \in \mathcal{X}} \mathbb{E}_{v_1, \dots, v_t \in Q} \prod_{\omega \subset [t]} h\left(T^{\sum_{j \in \omega} v_j} x\right).$$

This is the usual subgroup Gowers seminorm; for complex functions the usual alternating conjugations are inserted.

We use exactly two external inverse statements. First, the subgroup case of the finitary qualitative Bessel inequality [34, Theorem 1.23] says that for each k there is a function $b_k(\delta) \rightarrow 0$ as $\delta \downarrow 0$ such that

$$\mathbb{E}_{\alpha, \alpha'} \|h\|_{U_{Q_\alpha + Q_{\alpha'}}^{2k-1}(\mathcal{X})} \leq \delta \implies \mathbb{E}_\alpha \|h\|_{U_{Q_\alpha}^k(\mathcal{X})} \leq b_k(\delta), \quad |h| \leq 1. \quad (5.11)$$

The bound is independent of the finite family, the group, and the probability system. Finite subgroups are rank-zero coset progressions in that Theorem. Its dilations leave them unchanged, and its multiset sum of two subgroups induces uniform measure on their subgroup sum. The statement consequently has exactly the form (5.11). Rational probability weights on indices are implemented by repetition of indices; arbitrary finite weights follow by approximation. We choose b_k nondecreasing and enlarge its argument by a factor of two to absorb the approximation slack; the resulting modulus, still denoted b_k , tends to zero.

Second, the Green–Tao–Ziegler inverse theorem, in its finite-list linear-nilsequence formulation [17, Conjecture 1.2 and Theorem 1.3], with the correction [18], gives the following statement. For each integer $t \geq 2$ and $\delta > 0$, a 1-bounded function on an integer interval with U^t norm at least δ correlates by at least $c_t(\delta) > 0$ with a sequence $\Phi(g^k x)$

on a nilmanifold of step at most $t - 1$. There is a finite list of such nilmanifolds, with connected simply connected groups, from which the manifold is chosen; $\|\Phi\|_\infty \leq 1$ and its Lipschitz bound depend only on t, δ . The cases $t = 2, 3$ are the established lower-order cases recalled in the Introduction of [17]; Theorem 1.3 supplies the higher orders. No bound on g is asserted or needed. We use only this inverse conclusion, not the original auxiliary Proposition 8.3 addressed by the erratum. The interval norm here is the normalized cube mean with all vertices in the interval, equivalently the zero-extension norm divided by the norm of the interval indicator in a sufficiently large auxiliary cyclic group.

Lemma 5.5 (From subgroup cubes to a fine nilsequence projection). *Fix a cube type of dimension d , a gap $l < i$, and $J_0 \geq 1$. Assume $2(2^d) - 2 \leq s$. For every $\gamma > 0$ there is $\kappa = \kappa(d, J_0, \gamma) > 0$, independent of the master count, the gap, the cutoffs, and the cell probabilities, with the following property. If $h : \mathbb{Z} \rightarrow [-1, 1]$ is a family with $\|P_{i,l}h\|_2 < \kappa$, then*

$$\lim_U \left| \mathbb{E}_{p,y,u} \prod_{\omega \subset [d]} h \left(y + M_p \sum_{j \in \omega} (u_j^1 - u_j^0) \right) \right| \leq \gamma + O_d(J_0^{-1}). \quad (5.12)$$

Here $y \sim \mu_i$, and primes and shifts have the laws in (5.2). The constant in the last term depends only on d .

Proof. We pass from the short increments to subgroup norms, combine the subgroups using rough coprimality, and apply the inverse theorem in each cyclic cell. We keep the cell probabilities throughout, so every threshold is independent of the number of cells and the master count.

From short increments to subgroup norms. Discard the two partial R_l intervals at the ends of $[X_i, X_i^2]$; their μ_i mass is $o(1)$. Within every remaining interval and every residue modulo M , harmonic measure differs from conditional uniform measure by relative $o(1)$, uniformly in the cell, because $R_l/X_i = o(1)$. Only residues coprime to W occur, and translation by M preserves them.

Write $q_l = R_l/M$. Each cell has exactly q_l points. Give it its original probability, normalized after discarding the partial cells, and identify its progression indices with $G_l = \mathbb{Z}/q_l\mathbb{Z}$. Their disjoint union, with these probabilities and conditional Haar measure in each cell, is a finite probability system $\mathcal{X}_l$. Translation by $1 \in G_l$ means translation by M with wrapping inside that cell. Replacing the original cube by this periodized cube costs $O_d(J_0^{-1}) + o(1)$: every vertex moves by at most dR_l/J_0 from its root, so only roots within that distance of an interval endpoint can wrap. All functions in this argument are bounded by 1.

For a prime tuple put

$$q_p = M_p/M, \quad Q_p = q_p G_l.$$

The divisibility provisions in Lemma 3.3 give $q_p \mid q_l$. Its subgroup has order q_l/q_p , while $L_p = \lfloor q_l/(J_0 q_p) \rfloor$ tends to infinity uniformly over the pool. The pushforward of a uniform $u \in [0, L_p)$ to $q_p u \in Q_p$ has density at most $2J_0$ relative to subgroup Haar measure. The difference of two independent such variables has no larger density. Thus the d independent increments of the periodized cube have joint density at most $(2J_0)^d$ on Q_p^d .

For completeness, this density can be removed with a bounded number of Cauchy–Schwarz steps. Regard it as one factor, independent of the root, on Haar variables $(x, v_1, \dots, v_d)$, where $x \in \mathcal{X}_l$ and $v_j \in Q_p$. Insert one uniform subgroup direction translating the root alone; the density is invariant in that direction. For each nonempty $\omega \subset [d]$, choose $j \in \omega$ and insert a direction

$$x \mapsto T^a x, \quad v_j \mapsto v_j - a, \quad a \in Q_p,$$

which leaves that vertex fixed. These invariant changes of Haar variables commute. Successive Cauchy–Schwarz inequalities outside the assigned direction discard first the density and then each nonroot vertex function, including its copies from earlier steps. The root has response a in every direction. After $k = 1 + (2^d - 1) = 2^d$ steps its retained copies are a Haar subgroup k -cube. Taking the 2^k -th root proves

$$|\text{periodized cube at } \mathbf{p}| \leq C_{d,J_0} \|h\|_{U_{Q_{\mathbf{p}}}^k}(\mathcal{X}_l). \quad (5.13)$$

The argument acts inside each cell and integrates its probability throughout; its constant is independent of the number of cells.

Combining the subgroups. For independent tuples, Lemma 3.5, also after the good-tuple restrictions, gives $\gcd(q_{\mathbf{p}}, q_{\mathbf{p}'}) = 1$ with probability $1 - o(1)$. On this event Bézout’s identity implies $Q_{\mathbf{p}} + Q_{\mathbf{p}'} = G_l$. Consequently

$$\mathbb{E}_{\mathbf{p}, \mathbf{p}'} \|h\|_{U_{Q_{\mathbf{p}} + Q_{\mathbf{p}'}}}^{2k-1}(\mathcal{X}_l) \leq \|h\|_{U_{G_l}^{2k-1}}(\mathcal{X}_l) + o(1).$$

Together with (5.11) and (5.13), this says: for every $\gamma > 0$ a bound on the global U^t norm, $t = 2k - 1$, sufficiently small in terms of d, J_0, γ makes the periodized cube mean at most γ . This assertion is uniform in all finite system data.

From the global norm to a nilsequence projection. We next show that a fixed positive global norm forces a fixed positive fine projection. We first justify use of the interval inverse theorem on one cyclic cell, without requiring the resulting nilsequence to be periodic. Let v be 1-bounded on $\mathbb{Z}/q\mathbb{Z}$ and extend it periodically to $[0, Kq)$. For $L = Kq$ define the integer cube domain

$$A_L = \{(x, a_1, \dots, a_t) \in \mathbb{Z}^{t+1} : 0 \leq x + \sum_{j \in \omega} a_j < L \text{ for every } \omega \subset [t]\}.$$

The 2^t -th power of the interval norm is the mean of the periodic cube product over A_L . Averaging a parameter translate $r \in \{0, \dots, q - 1\}^{t+1}$ gives exactly the cyclic cube mean at every parameter point. The discrepancy is therefore bounded by $\mathbb{E}_r |(A_L + r) \triangle A_L| / |A_L|$. For a vertex ω , its displacement is $r_0 + \sum_{j \in \omega} r_j \leq (t+1)(q-1)$. The boundary discrepancy at that vertex has size at most this quantity times $2L^t$: once that vertex and its t neighboring vertices are specified the integer cube is determined, and each neighbor has at most L choices on the relevant side of the symmetric difference. Summing over vertices bounds the numerator by $O_t(qL^t)$. Nonnegative increments and root with total less than L already give $|A_L| \geq \binom{L+t}{t+1} \geq L^{t+1}/(t+1)!$. Thus, uniformly in q, K, v ,

$$\|v^{\text{per}}\|_{U^t[Kq]}^{2^t} = \|v\|_{U^t(\mathbb{Z}/q\mathbb{Z})}^{2^t} + O_t(K^{-1}). \quad (5.14)$$

This is an estimate for the powers of the norms.

If the cyclic norm is at least δ , choose a sufficiently large fixed $K = K(t, \delta)$ in (5.14). The interval norm is then at least $\delta/2$. The inverse theorem gives a uniformly bounded complexity correlator on those K periods. Splitting the correlation into its K period averages shows that one period has correlation of at least the same modulus. Translating that period back to $[0, q)$ only replaces the nilsequence base point by a translate $g^{jq}x$. It leaves its manifold, step and observable bounds unchanged. We have proved a cyclic-to-interval correlation statement on fixed representatives of $\mathbb{Z}/q\mathbb{Z}$; the correlator itself need not have period q .

Now write the cell weights of $\mathcal{X}_l$ as α_C . The exact identity

$$\|h\|_{U_{G_l}^t}^{2^t}(\mathcal{X}_l) = \sum_C \alpha_C \|h_C\|_{U_{G_l}^t}^{2^t}$$

shows that, if the left-hand norm is at least δ , cells with $\|h_C\|_{U^t(G_l)} \geq \delta/2$ have total probability at least $\delta^{2^t}/2$. On each such cell choose its inverse-theorem correlator, rotate its complex phase and take the real part so that its pairing with the real function h_C is positive. Put zero on the remaining cells. Every chosen sequence comes from the same finite nilmanifold list and has the same uniform observable bounds. They form one permitted real piecewise nilsequence family V at gap l , with $|V| \leq 1$. Translating from conditional uniform measure back to μ_i costs $o(1)$. Thus along any set of w on which the global norm is at least δ ,

$$\langle h, V \rangle \geq \frac{1}{2} \delta^{2^t} c_t(\delta/4) + o(1).$$

The harmless smaller inverse threshold allows slack in all these inequalities. The step is at most $t - 1 = 2(2^d) - 2 \leq s$, so V represents an element of $\mathcal{N}_{i,l}$. If the stated norm bound holds on a set belonging to $\mathcal{U}$, select the correlators there and set the family to zero on its complement. Orthogonality and $\|V\|_2 \leq 1$ give the positive lower bound

$$\|P_{i,l}h\|_2 \geq \frac{1}{2} \delta^{2^t} c_t(\delta/4).$$

Choose δ small enough that Bessel and (5.13) give the prescribed γ , and choose κ smaller than the displayed positive projection bound. If the projection is less than κ , the global norm cannot exceed that threshold along $\mathcal{U}$. Restoring the periodization error proves (5.12). $\square$

5.4 Completion of the Prediction Principle

Proof of Principle 2.3. Fix the data $n, r, \chi, \mathcal{B}, \mathcal{A}, \tau, \eta$ of the Principle, with $s = s(m)$ as in (5.1). There are $q_m = 2^m - m - 1$ nonsingleton factors in each count. We spell out the order of choices before selecting master indices.

Choose a positive target cube error ζ so small that

$$C_m(2\zeta)^\theta < \eta/(2q_m)$$

for every exponent and constant in Proposition 4.5. There are only finitely many orders and exponent choices, all controlled by m ; increasing the uniform constant if necessary makes this one finite set of requirements. Choose a fixed J_0 so large that every periodization term $O_d(J_0^{-1})$ in Lemma 5.5 is smaller than ζ . For the finitely many correlation-test types, apply that Lemma with $\gamma = \zeta$, and choose

$$0 < \varepsilon < \eta, \quad 2\varepsilon < \min_d \kappa(d, J_0, \zeta).$$

These choices depend on the requested tolerances and on m , and are independent of N . Now choose N by Lemma 5.4, construct its arithmetic scales and dense models, and carry out that Lemma. Restrict h_i, X_i, t_i to the principal indices $j_1, \dots, j_n$, relabel them by $[n]$, and set $H_u = R_{k_u}$. The intervening padding gaps and Lemma 3.3 give all conditions in Definition 2.1; the models S have exactly the pieces required by Definition 2.2.

For calibration at a selected block choose a Lipschitz function $\psi : [0, 1] \rightarrow [0, 1]$ equal to 1 on $[0, 2\tau]$ and to 0 on $[3\tau, 1]$. The bounded distribution replacement in Corollary 3.2 gives

$$\Pr(\chi(h_{Bat_B}) = c, S_{B,a,c}(t_B) \leq 2\tau) \leq \mathbb{E}_{\mu_i} \rho \psi(S_{B,a,c}) + o(1).$$

The function $\psi(S)$ is a bounded-complexity coarse piecewise nilsequence. Lemma 5.3 replaces ρ by F . In the Hilbert limit it also lies in the coarse subspace, so, writing $P = P_{i,k_u}$ and using (5.9),

$$\langle F, \psi(S) \rangle = \langle PF, \psi(S) \rangle \leq \langle S, \psi(S) \rangle + \varepsilon \leq 3\tau + \varepsilon.$$

This is the first assertion of the Principle, since $\varepsilon < \eta$.

For the count comparison, fix a chain on the principal indices, $b \in \mathcal{B}$, and a color. Its block scale b_{B_d} belongs to $\mathcal{A}$, so all its models have already been constructed. As shown after Proposition 5.2, telescope the nonsingleton color weights $\rho_d(L_J)$ to $F_d(L_J)$ at error $o(1)$, retaining the center weights $\nu_d(z_d)$ and every product mask, including the singleton masks.

Next telescope $F_d(L_J)$ to $S_d(L_J)$. Choose as fine gap the padding index immediately before the chain's first pivot. It is valid for every block of the chain. Each individual difference has bounded target $h = F_d - S_d \in [-1, 1]$, and (5.10) gives $\|P_{i_d, l} h\|_2 \leq 2\varepsilon$. All other factors, including the unbounded center weights, still satisfy the bounds of Proposition 4.5. The parameter choices and Lemma 5.5 bound its target cube mean by 2ζ in the ultrafilter limit. Equation 4.17 therefore bounds each telescoping error by less than $\eta/(2q_m)$. Summing the finitely many differences gives a limit error at most $\eta/2$ between the original count and

$$\mathbb{E}_{z \sim \bigotimes_d \mu_{i_d}} U(z) \prod_d \nu_d(z_d) \prod_{|J| \geq 2} S_{d(J)}(L_J(z)).$$

All factors except the center weights are now bounded by 1. The blocks of the chain are disjoint. Corollary 3.2 thus replaces this last expectation, with error $o(1)$, by

$$\mathbb{E}_t U(z(t)) \prod_{|J| \geq 2} S_{d(J)}(L_J(z(t))), \quad z(t)_d = t_{B_d}.$$

This proves the required count comparison with the stated tolerance η . Every limit in this construction may be taken along the same fixed ultrafilter $\mathcal{U}$, as allowed in Principle 2.3. In particular the auxiliary ε was chosen before the master Ramsey count N , whereas the resulting model complexity was allowed to depend on all those fixed choices. The proof of the Principle is complete. $\square$

6 Removing rough progression steps

At an alignment pivot i , the input of a model has the form $t_T t_i$, where the earlier variables in t_T are temporarily fixed. Consequently, averaging t_i samples only one progression of step t_T in the model input. We will need to remove the factors of this step while retaining polynomial dependence on the factor being removed. The comparison proved here permits the sampling box, residue and observable to depend arbitrarily on that factor. This last uniformity is necessary because the actual pivot cell is determined only after sampling.

All nilmanifolds in this Section are quotients G/Γ , with G connected, simply connected and nilpotent and Γ a lattice. We fix rational coordinates on $\mathfrak{g} = \log G$ and a smooth metric on the compact quotient. “Polynomial in logarithmic coordinates” means that the coordinates of $\log P$ are ordinary real polynomials. Their degrees, but not their coefficients, will be bounded. For a scalar a , write $\|a\|_{\mathbb{R}/\mathbb{Z}} = \inf_{k \in \mathbb{Z}} |a - k|$.

The orbit theory underlying this comparison begins with Leibman's qualitative equidistribution theorem [26]. We use the quantitative form of Green and Tao, with its multiparameter correction, in the precise version stated next.

6.1 The corrected quantitative equidistribution input

A rational filtration is a finite decreasing sequence of connected rational subgroups $G_\bullet = (G_j)_{j \geq 0}$ with $G_0 = G_1 = G$ and $[G_i, G_j] \subseteq G_{i+j}$. A map on $\mathbb{Z}^d$ is polynomial for this filtration if every k -fold group difference takes values in G_k . Fix a rational Mal'cev basis

adapted to the filtration. A horizontal character is a continuous homomorphism $\xi : G \rightarrow \mathbb{R}$ such that $\xi(\Gamma) \subseteq \mathbb{Z}$; its size is the norm of its integer coefficient vector on the horizontal torus. In particular, characters of bounded size form a finite set.

Theorem 6.1 (Quantitative Leibman theorem, corrected box form). *Fix a rational filtered nilmanifold as above, a number d of variables, and $0 < \delta < 1$. There is a constant A , depending only on these data, with the following property. If $P : \mathbb{Z}^d \rightarrow G$ is polynomial for this filtration and $(P(v)\Gamma)_{v \in \{0, \dots, N-1\}^d}$ is not δ -equidistributed, then there is a nonzero horizontal character ξ of size at most A such that, on writing*

$$\xi(P(v)) = \sum_I \alpha_I \binom{v}{I}, \quad \binom{v}{I} = \prod_{j=1}^d \binom{v_j}{I_j},$$

one has

$$N^{|I|} \|\alpha_I\|_{\mathbb{R}/\mathbb{Z}} \leq A \quad (|I| > 0). \quad (6.1)$$

Here δ -equidistribution means comparison with Haar probability to error at most δ against every observable of Lipschitz norm at most one. The constant A is independent of the coefficients of P and of N .

The following consequence will be used. Fix also $c, C, B, \eta > 0$. Let $\mathcal{Q} \subseteq [-CZ, CZ]^d$ be an axis-parallel box with each side at least cZ , sampled at its integer points. If

$$|\mathbb{E}_{x \in \mathcal{Q} \cap \mathbb{Z}^d} F(P(x)\Gamma) - \int_{G/\Gamma} F| \geq \eta, \quad \|F\|_{\text{Lip}} \leq B, \quad (6.2)$$

then, for sufficiently large Z , there is a nonzero horizontal character in a fixed finite list such that, writing $\xi(P(x)) = \sum_I \theta_I x^I$ in ordinary monomials,

$$\|\theta_I\|_{\mathbb{R}/\mathbb{Z}} \leq A' Z^{-|I|} \quad (|I| > 0). \quad (6.3)$$

The list and A' depend only on the fixed filtered nilmanifold, d, c, C, B, η . The same assertions hold with the nilmanifold and its filtration chosen from a fixed finite list.

The equal-side statement is the corrected version of [12, Theorem 8.6]; see the expanded erratum [13, p. 3 and Section 3, arXiv:1311.6170v3]. The one-variable case is [12, Theorem 2.9], which is unaffected by the correction. The published quantitative statement bounds the rationality of the adapted basis by the reciprocal accuracy. Here the basis is fixed; decreasing the accuracy below the reciprocal of its fixed rationality bound absorbs that hypothesis into A . We use the equal-side case of Theorem 6.1 and now verify the stated box consequence; no assertion about unrestricted unequal side lengths is needed.

Derivation of the box consequence. Normalize the observable by its Lipschitz bound. Choose a small positive λ , depending only on d, c, C, B, η , and tile the integer box by cubes of side $\lfloor \lambda Z \rfloor$, leaving strips along its boundary. The strips contain at most an $O_{d,c}(\lambda + Z^{-1})$ fraction of its integer points. Choose λ so that their contribution to (6.2) is less than $\eta/4$. Expressing the remaining average as a convex combination shows that one cube has Haar discrepancy at least $\eta/2$. Apply the equal-side case of Theorem 6.1 to $P(a + v)$ on this cube. Integer translations preserve filtered polynomiality because a group difference of a translated map is the corresponding translate of its group difference.

There is a fixed bound D_0 on the scalar degree. Multiply the resulting character by $(D_0!)^d$. Indeed, on expanding the products $\binom{v}{I}$, this multiplication clears every denominator, so each ordinary monomial coefficient is an integer plus an error $O(Z^{-|I|})$: a coefficient of degree I receives contributions only from binomial degrees $J \geq I$, and $Z^{-|J|} \leq Z^{-|I|}$.

Finally expand $(x - a)^J$. The translation has integer entries and $|a| = O_{C,d}(Z)$; hence its integer coefficient contributions stay integral, while its errors at degree I are bounded by

$$\sum_{J \geq I} O(Z^{|J|-|I|}) O(Z^{-|J|}) = O(Z^{-|I|}).$$

This proves (6.3). The bounded multiplication of the character still leaves a finite list. The bound on the location of $\mathcal{Q}$ is used precisely in this last translation estimate. $\square$

We explain why Theorem 6.1 applies to the logarithmic polynomials used below. Let $G^{(j)}$ be the lower central series, with $G^{(1)} = G$, and suppose G has step at most s . For an ordinary degree bound $D \geq 1$, put

$$G_0 = G, \quad G_i = G^{(\lceil i/D \rceil)} \quad (i \geq 1). \quad (6.4)$$

These are connected rational subgroups, the filtration ends after sD , and $\lceil i/D \rceil + \lceil j/D \rceil \geq \lceil (i+j)/D \rceil$ gives the bracket condition; the condition with $i = 0$ follows because the lower central subgroups are normal. If $\log P$ has ordinary total degree at most D , its coefficient at a monomial of degree j belongs to $\mathfrak{g}_j$, since $\mathfrak{g}_j = \mathfrak{g}$ for $j \leq D$.

Here is a direct verification of polynomiality, including the effect of noncommutativity. Suppose the coefficient at degree j in $A(x) = \log Q(x)$ lies in $\mathfrak{g}_{j+k}$. An ordinary difference $A(x+h) - A(x)$ has its degree- j coefficient in $\mathfrak{g}_{j+k+1}$, because it arises from degrees at least $j+1$. Using the Baker–Campbell–Hausdorff (BCH) formula, expand

$$\log(Q(x+h)Q(x)^{-1}) = \text{BCH}(A(x) + [A(x+h) - A(x)], -A(x)).$$

The terms containing no occurrence of $A(x+h) - A(x)$ cancel, since $\text{BCH}(A, -A) = 0$. Every remaining bracket contains at least one difference. Filtration indices add in brackets, so its coefficient at degree j belongs to $\mathfrak{g}_{j+k+1}$. The BCH series is finite. Induction starting at $k = 0$ proves that the k -fold group difference takes values in G_k . Its degree is bounded in terms of s, D throughout. Thus arbitrary coefficients in a bounded-degree logarithmic polynomial are allowed in Theorem 6.1, using the fixed filtration (6.4). The same argument works with any fixed number of ordinary variables.

6.2 An interpolation fact with smooth denominators

The arithmetic reason that a rough step can be removed is the following fact. Its analytic constants do not depend on the length of the progression. Rational denominators may depend on that length, which is always fixed before taking the asymptotic limit.

Lemma 6.2 (Exact division after interpolation). *Fix $e \geq 1$, $q \geq e$, and $A > 0$. For a fixed integer $H \geq 2q + 2$, let*

$$t_z = t_0 + mz \in [S, 2S), \quad 0 \leq z < H,$$

where m is a positive integer with all prime factors at most w , and $\gcd(t_z, W) = 1$ for every z , with $W = \prod_{p \leq w} p$. Suppose $w, S \rightarrow \infty$ and $Z/S^a \rightarrow \infty$ for every fixed $a > 0$. Let $\theta(z)$ be a real polynomial such that $\deg(t_z^e \theta(z)) \leq q$ and

$$\|t_z^e \theta(z)\|_{\mathbb{R}/\mathbb{Z}} \leq A(S/Z)^e \quad (0 \leq z < H). \quad (6.5)$$

For all sufficiently large w there is a rational polynomial $M(z)$, of degree at most $q - e$, integral at every index $0 \leq z < H$, such that

$$|\theta(z) - M(z)| \leq C_q A Z^{-e} \quad (0 \leq z < H). \quad (6.6)$$

Its coefficient denominators have only prime factors at most w . The constant in (6.6) is independent of H . The threshold for w is permitted to depend on H .

Proof. Write $T(z) = (t_0 + mz)^e \theta(z)$ and $\epsilon = A(S/Z)^e$. Choose $q + 1$ integer nodes $z_0, \dots, z_q$ in $[0, H - 1]$ with pairwise successive gaps at least $(H - 1)/(2q)$, for example the nearest integers to $j(H - 1)/q$. Let n_j be a nearest integer to $T(z_j)$, and let R be their degree-at-most- q Lagrange interpolant. On rescaling the index interval to $[0, 1]$, the interpolation nodes remain separated by $1/(2q)$. The Lagrange basis polynomials and all their coefficients in the rescaled variable are therefore bounded in terms of q alone. Since T already has degree at most q , interpolation of the errors gives

$$\sup_{0 \leq z \leq H-1} |R(z) - T(z)| \leq C_q \epsilon. \quad (6.7)$$

There is a positive integer D_H , depending only on the chosen nodes, such that $D_H R \in \mathbb{Z}[z]$. One may take the product of the nonzero node differences occurring in the Lagrange denominators. In particular, D_H is independent of w, t_0, m and the polynomial coefficients.

Let $a = -t_0/m$. The polynomial T has a zero of multiplicity at least e at a . The coefficient bounds in the rescaled Lagrange formula give, for $0 \leq j < e$,

$$|R^{(j)}(a)| = |(R - T)^{(j)}(a)| \leq C_q \epsilon (H - 1)^{-j} \left(1 + \frac{|a|}{H - 1}\right)^q \leq C_q \epsilon (1 + 2S)^q.$$

On the other hand $R^{(j)}(a)$ is rational with denominator dividing $D_H m^q$. We have $m \leq S$ because the progression lies in $[S, 2S)$ and has at least two terms. Consequently a nonzero such rational has absolute value at least $D_H^{-1} S^{-q}$. For fixed H , the displayed upper bound is smaller than this number for large w , by the assumed domination of every power of S by Z . Thus $R^{(j)}(a) = 0$ for all $j < e$, exactly. Polynomial division now gives

$$R(z) = (t_0 + mz)^e M(z), \quad M \in \mathbb{Q}[z], \quad \deg M \leq q - e. \quad (6.8)$$

In division by $(t_0 + mz)^e$, the only denominators introduced, besides those already dividing D_H , come from its leading coefficient m^e . All their prime factors are therefore at most w once w exceeds the prime factors of the fixed integer D_H .

For every integer $z \in [0, H - 1]$, (6.5) and (6.7) place $R(z)$ within $(C_q + 1)\epsilon$ of an integer. Since $R(z) \in D_H^{-1} \mathbb{Z}$, it is itself an integer for large w . Write $M(z) = a_z/b_z$ in lowest terms. Every prime factor of b_z is at most w , whereas $\gcd(t_z, W) = 1$. The integrality of $t_z^e M(z) = R(z)$ then forces $b_z = 1$. Finally divide (6.7) by $t_z^e \geq S^e$. This yields (6.6) with a constant depending on q alone. Only the rational exactness steps, not this constant, required a threshold depending on H . $\square$

6.3 Uniform progression comparison

Call an integer w -smooth if all its prime factors are at most w . For a nonempty finite set A we use $\mathbb{E}_A$ for normalized counting. The next Proposition compares two sampling laws for the same polynomial family. Its joint polynomial dependence on t and x is essential; the box, residue class, and observable may be chosen separately for each t .

Proposition 6.3 (Removal of a rough step). *Fix G/Γ , a number $d \geq 1$ of spatial variables, a degree bound D , and constants $c, C, B > 0$. Along an arbitrary sequence $w \rightarrow \infty$, let L, S, Z satisfy*

$$W \mid L, \quad L \text{ is } w\text{-smooth}, \quad S \rightarrow \infty, \quad S/L \rightarrow \infty, \quad Z/S^a \rightarrow \infty \quad \text{for every fixed } a > 0. \quad (6.9)$$

Fix a residue r modulo L with $\gcd(r, W) = 1$, and put $\mathcal{T} = [S, 2S) \cap (r + L\mathbb{Z})$. Let $P_t(x)$ be G -valued, with logarithmic coordinates polynomial jointly in $(t, x) \in \mathbb{R} \times \mathbb{R}^d$ of total degree at most D . These polynomials and their coefficients may change with w .

For every $\eta > 0$, the proportion of $t \in \mathcal{T}$ for which there exist a box $\mathcal{Q} \subseteq [-CZ, CZ]^d$ with all sides at least cZ , a vector $b \in \mathbb{Z}^d$, and an observable $F : G/\Gamma \rightarrow \mathbb{C}$ with $\|F\|_{\text{Lip}} \leq B$, such that

$$|\mathbb{E}_{x \in \mathcal{Q} \cap \mathbb{Z}^d} F(P_t(x)\Gamma) - \mathbb{E}_{x \in \mathcal{Q} \cap (b+t\mathbb{Z}^d)} F(P_t(x)\Gamma)| \geq \eta \quad (6.10)$$

tends to zero.

This conclusion holds for every sequence of the allowed data. In particular the exceptional proportion is uniform over the polynomial coefficients, the residue r , and all the boxes, residue vectors and observables in (6.10). The latter three may be chosen separately for every t without any polynomial dependence. The same conclusion holds for a fixed finite list of nilmanifolds and fixed complexity bounds.

Proof. The induction will turn a discrepancy on G/Γ into one on the kernel of a horizontal character. To preserve polynomial dependence during this reduction, we work on long progressions of exceptional values of t with smooth difference. Interpolation on such a progression will split the horizontal polynomial into an integer part and a slowly varying error. We first specify the stronger progression assertion to which this dimension induction applies.

A *bad progression family* consists of fixed bounds $d, D, c, C, B, \eta > 0$ and a fixed nilmanifold such that, for every prescribed integer H , there are sequences with $w, S \rightarrow \infty$ and $Z/S^a \rightarrow \infty$ for every fixed a , and progressions

$$t_z = t_0 + mz \in [S, 2S), \quad 0 \leq z < H, \quad m \text{ is } w\text{-smooth}, \quad \gcd(t_z, W) = 1, \quad (6.11)$$

on which every index has a discrepancy at least η as in (6.10). Here the logarithms of $P_z(x)$ need only be jointly polynomial in (z, x) with the fixed degree bound; the boxes, classes and tests may vary with z . The sequences may depend on H . It is equivalent to ask for arbitrarily large prescribed lengths: a longer progression can be restricted. We will prove that no such family exists, by induction on $\dim G$.

First, failure of the Proposition would produce such a family. Indeed, suppose the proportion of bad values is at least $\alpha > 0$ along a subsequence. In the index coordinate $t = r + Lu$, the set $\mathcal{T}$ is an integer interval of length tending to infinity. For prescribed H , finite Szemerédi's Theorem [33] supplies a fixed integer $K = K(H, \alpha)$ such that every subset of $[K]$ of density at least $\alpha/2$ contains an H -term progression. Partition the index interval into blocks of length K and discard its final incomplete block. Some full block has bad density at least $\alpha/2$ for large w . Its bad progression has step jL with $1 \leq j \leq K$. This step is w -smooth once $w \geq K$. Joint polynomiality is preserved by the affine substitution for t . Thus (6.11) holds for each fixed H .

We shall repeatedly thin a bad progression to one color in a finite coloring. Finite van der Waerden's Theorem [35] makes this legitimate: to obtain a length H of one color, start with the fixed length prescribed by that Theorem for H and the number of colors. Under $z = a + bu$, the new step is bm ; b is bounded in terms of the initial fixed length, so it remains w -smooth for large w . The new step is still at most S , and all scale and degree hypotheses remain valid. A finite choice of group data can also be fixed: first pass to a subsequence for each length, then choose one of the finitely many possibilities occurring for unbounded lengths. Bounds in all these colorings will be independent of H .

For dimension zero the quotient is a point and every discrepancy is zero. Assume the assertion has been proved in smaller dimension, and suppose that a bad progression family on G/Γ exists. For each z , at least one of its two samples has Haar discrepancy at least $\eta/2$. Write its step as $p_z \in \{1, t_z\}$ and its residue representative as $b_z \in \{0, \dots, p_z - 1\}^d$. Substitute $x = p_z v + b_z$. The resulting box lies in a fixed multiple of $[-Z/p_z, Z/p_z]^d$ and has

comparable sides. Its common scale tends to infinity. Theorem 6.1, with the fixed stretched filtration (6.4), therefore gives a character from a fixed finite list. Thin to a common nonzero character ξ , including the fixed denominator multiplication in that choice. Write

$$\xi(P_z(x)) = \sum_I \theta_I(z) x^I. \quad (6.12)$$

The coefficients θ_I are polynomials of bounded degree in z . The coefficient obstruction in the v variables is

$$\left\| p_z^{|I|} \sum_{J \geq I} \binom{J}{I} b_z^{J-I} \theta_J(z) \right\|_{\mathbb{R}/\mathbb{Z}} \leq A_0(p_z/Z)^{|I|} \quad (|I| > 0), \quad (6.13)$$

where A_0 is independent of H, z and of every polynomial coefficient.

We claim, descending through the positive spatial degrees, that there are rational polynomials $m_I(z)$ of bounded degree such that

$$m_I(z) \in \mathbb{Z}, \quad |\theta_I(z) - m_I(z)| \leq A_1 Z^{-|I|} \quad (0 \leq z < H, |I| > 0), \quad (6.14)$$

with smooth coefficient denominators and a constant A_1 independent of H . Take H larger than the finitely many degree requirements of Lemma 6.2; this does not restrict the arbitrarily long progression assertion. Suppose higher degrees have already been treated and put $e = |I|$. Their integral parts in (6.13) contribute integers. Their error contribution has size at most

$$C \sum_{J > I} p_z^e p_z^{|J|-e} Z^{-|J|} \leq C'(p_z/Z)^e,$$

since $p_z \leq 2S = o(Z)$; here $J > I$ means $J \geq I$ and $J \neq I$. It follows that $\|p_z^e \theta_I(z)\|_{\mathbb{R}/\mathbb{Z}} \leq C'(p_z/Z)^e$. If $p_z = t_z$ this is already the needed bound. If $p_z = 1$, multiplication by the integer t_z^e gives the same conclusion:

$$\|t_z^e \theta_I(z)\|_{\mathbb{R}/\mathbb{Z}} \leq C''(S/Z)^e.$$

Lemma 6.2 gives m_I and the claimed bound. There are only boundedly many spatial degrees and coefficients, so the descending induction terminates with A_1 depending only on the fixed data. The length-dependent thresholds for rational exactness can be met simultaneously.

The positive-degree coefficients now have the required integer parts and small errors. To obtain a bounded horizontal error on the whole box, we must also split the spatially constant coefficient. This requires a finite coloring argument. Let q_0 bound $\deg \theta_0$. Color z by the interval containing $\{\theta_0(z)\}$ in a partition of $[0, 1)$ into intervals of length less than 2^{-q_0-2} . Thin to a long monochromatic progression and reparametrize it. On each consecutive $(q_0 + 2)$ -term segment of the new progression,

$$\Delta^{q_0+1} \lfloor \theta_0(z) \rfloor = -\Delta^{q_0+1} \{\theta_0(z)\}$$

is an integer of absolute value less than one: the constant part of the common fractional-part interval cancels, and the sum of the absolute difference coefficients is 2^{q_0+1} . It is therefore zero. Newton interpolation, or induction using this difference identity, shows that a rational polynomial $m_0(z)$ of degree at most q_0 agrees with $\lfloor \theta_0(z) \rfloor$ on the whole new progression. Its Newton coefficients are integer differences; hence its ordinary coefficient denominators divide $q_0!$. All previously constructed m_I retain (6.14) after the same affine reparametrization.

Set

$$m_z(x) = \sum_I m_I(z) x^I, \quad a_z(x) = \xi(P_z(x)) - m_z(x). \quad (6.15)$$

For integer z in the progression, m_z has integer coefficients as a polynomial in x . By (6.14), its error satisfies, throughout $[-CZ, CZ]^d$,

$$|a_z(x)| \leq C_1, |\partial_{x_j} a_z(x)| \leq C_1/Z \quad (1 \leq j \leq d), \quad (6.16)$$

where C_1 is independent of H . To see this explicitly, the constant term lies in $[0, 1)$, and an error coefficient at degree e contributes at most $A_1 Z^{-e} (CZ)^e$ to the first bound and at most $e A_1 Z^{-e} (CZ)^{e-1}$ to a derivative. Only boundedly many monomials occur.

Choose a rational vector $V \in \mathfrak{g}$ with $\xi(\exp V) = 1$. Such a vector exists because a nonzero horizontal character has nonzero rational differential. There is a fixed positive integer K with $\exp(KV) \in \Gamma$. For completeness, the Mal'cev coordinates of $\exp(tV)$ are rational polynomials in t with zero constant term, by the finite BCH formula. Taking K divisible by their finitely many denominators makes these coordinates integral, which places $\exp(KV)$ in the lattice. All choices depend only on the fixed character and rational group data.

Put $H_\xi = \ker \xi$ and

$$Q_z(x) = \exp(-a_z(x)V) P_z(x) \exp(-m_z(x)V). \quad (6.17)$$

Then $\xi(Q_z(x)) = 0$ identically, for real (z, x) , and

$$P_z(x) = \exp(a_z(x)V) Q_z(x) \exp(m_z(x)V). \quad (6.18)$$

The group H_ξ is connected and simply connected: the exponential map identifies it with the kernel of the linear differential of ξ . That kernel is rational, so H_ξ is a rational subgroup of dimension $\dim G - 1$. Formula (6.17) and the finite BCH formula show that $\log Q_z(x)$ is jointly polynomial in (z, x) with a degree bound depending only on the old degree and nilpotence step. No bound on its coefficients is needed.

The polynomial now takes values in a smaller group. To apply the induction hypothesis, we must also express both sampling laws and their test functions on a fixed compact quotient of H_ξ . Partition each test box into a bounded number of rectangular subboxes, each of side between $\lambda cZ/2$ and $2\lambda cZ$, where the sufficiently small constant $\lambda > 0$ is chosen below. For example divide each side into the same sufficiently large fixed number of intervals. On a subbox choose a point x_* and replace the left factor $\exp(a_z(x)V)$ in (6.18) by $\ell_{z,*} = \exp(a_z(x_*)V)$. The error in the test is at most $C_2 \lambda$: the derivative estimate (6.16) bounds the change in a_z , and the action map on the product of a fixed compact subset of G and G/Γ has bounded derivative. The constant C_2 depends on the original Lipschitz bound, but not on H, z or the choice of subbox. Choose λ so that twice this error is less than $\eta/8$.

Split each subbox further according to $x \bmod K$. At a fixed z and residue vector u , the integer polynomial $m_z(x)$ has a fixed value $j \bmod K$, and hence

$$\exp(m_z(x)V)\Gamma = \sigma_j \Gamma, \quad \sigma_j = \exp(jV), \quad 0 \leq j < K.$$

The resulting test on the smaller group is

$$h(H_\xi \cap \sigma_j \Gamma \sigma_j^{-1}) \longmapsto F_z(\ell_{z,*} h \sigma_j \Gamma). \quad (6.19)$$

The intersection is a lattice in H_ξ : conjugation by rational σ_j preserves the rational structure, and a connected rational subgroup meets a rational lattice in a lattice. This can also be seen by taking a rational Mal'cev basis of the subgroup and clearing the finite coordinate denominators. The displayed map from its compact quotient to G/Γ is smooth. Since there are only finitely many j and $\ell_{z,*}$ lies in a fixed compact set, all the tests (6.19) have one uniform Lipschitz bound.

We justify the sampling weights in this splitting. For an interval of length comparable to Z and any residue modulo q , its count is its length divided by q , with error at most two. Multiplying this formula over the fixed number of coordinates shows that the normalized weights of the boundedly many subboxes under the unrestricted and step- t_z samples differ in total by $O(t_z/Z)$. For $w \geq K$, $\gcd(t_z, K) = 1$. Thus both samples assign weight $K^{-d} + O(K^d t_z/Z)$ to each K -residue vector inside a subbox: in the step- t_z coordinate, reduction modulo K is a bijection. The total difference of the weights of all the pieces is consequently $o(1)$, uniformly in z and in the box endpoints.

After freezing, the original discrepancy is still at least $7\eta/8$. Write the two averages as convex combinations over these pieces and replace one set of combination weights by the other. The error is $o(1)$ times the uniform supremum bound on the tests. It follows from the triangle inequality that at least one piece has a conditional discrepancy at least $\eta/2$, for large w . This lower bound does not depend on the number of pieces: it follows from a convex combination, not from an unnormalized sum.

For each z choose such a piece. Its subbox and frozen observable may vary arbitrarily with z , as allowed in the induction assertion. Thin once more to make its residue vector $u \bmod K$ and its coset index j common. In this subbox substitute $x = u + Ky$, taking $u \in \{0, \dots, K-1\}^d$. The unrestricted sample becomes all integer y in a box of side comparable to Z/K , within a fixed multiple of that scale. The step- t_z sample becomes one residue vector modulo t_z , because K is invertible modulo t_z . The new map $Q_z(u + Ky)$ is still jointly logarithmically polynomial, and Z/K still dominates every fixed power of S . We have therefore obtained a bad progression family on

$$H_\xi / (H_\xi \cap \sigma_j \Gamma \sigma_j^{-1})$$

of strictly smaller dimension.

All the new degree, box, observable and discrepancy bounds are independent of the progression length. Characters, rational directions, residue vectors and cosets range over fixed finite lists; as explained at the start, one such quotient works for unbounded lengths. The sole length-dependent quantities were interpolation denominators and asymptotic thresholds. Those denominators enter only the unrestricted coefficients of the kernel polynomial. At every retained integer index its spatial integer part has integer coefficients, so the fixed residue splitting and the finite list of quotient lattices are unchanged. This contradicts the induction hypothesis and proves that bad progression families do not exist. Failure of the Proposition produced precisely such a family, so the Proposition follows, with the uniformity asserted in its statement. $\square$

6.4 Residual steps and conditional sampling

We record explicitly the form needed when the step being removed is one factor of a larger product. This separates the polynomial requirement from the arbitrary choice of actual sampling box.

Corollary 6.4 (Removing one factor while retaining the others). *In Proposition 6.3, let q be another positive integer, fixed while t varies but permitted to change with w . Let $a \in \{0, \dots, q-1\}^d$ also be independent of t . Suppose the spatial boxes have common scale Z_0 , with Z_0/q dominating every fixed power of S . Suppose $\log P_t(a + qy)$ is polynomial jointly in (t, y) with fixed degree bound, and the normalized boxes lie in fixed multiples of $[-Z_0/q, Z_0/q]^d$ with comparable sides. Then, outside an exceptional proportion tending to zero, averaging P_t over the class $a + q\mathbb{Z}^d$ agrees to any prescribed accuracy with averaging over any compatible class modulo qt , with the same uniformities in boxes and tests. No coprimality between q and t is assumed.*

Proof. Write the finer class as $b + qt\mathbb{Z}^d$ with $b \equiv a \pmod{q}$. Under $x = a + qy$, its condition is exactly

$$y \equiv (b - a)/q \pmod{t}.$$

This identity does not use invertibility of either factor. Apply Proposition 6.3 to the normalized polynomial and the scale Z_0/q . Its uniformity permits arbitrary compatible b and arbitrary endpoints of the normalized boxes. $\square$

In applying Corollary 6.4, the retained residue class must be chosen independently of the factor being removed. Section 8 verifies this condition for each factor of the tail product.

Finally, uniformity over additional conditioned parameters has a precise probabilistic consequence. Suppose k factors range in fixed dyadic intervals and smooth residue classes, and let ν be the product of their normalized uniform laws. Suppose a conditional law μ satisfies $\mu \leq C_0\nu$, with fixed C_0 . If, for every choice of the other $k - 1$ factors, the exceptional proportion for removing the remaining factor is at most $\epsilon_w \rightarrow 0$, then Fubini's Theorem gives $\nu(E_\ell) \leq \epsilon_w$ and hence

$$\mu\left(\bigcup_{\ell=1}^k E_\ell\right) \leq C_0 k \epsilon_w = o(1).$$

The uniform bound ϵ_w follows from Proposition 6.3: otherwise choices of the additional parameters with exceptional proportion bounded below would themselves form a violating sequence of permitted data. This gives the conditional form of the progression comparison used in Section 8.

7 Local cube laws and lifts of face symmetries

The comparison from Section 6 will give symmetries of individual marginals of a joint nilmanifold-valued state. We need to act on the joint state even when those symmetries do not preserve its other marginals. The purpose of this Section is to construct such actions on a universal cover. Their nilpotence class, rather than the dimension of the cover, is what will control the finite recurrence argument. For the cube formalism in higher-order ergodic theory, see [24]. We prove the algebraic and lifting statements in the precise forms used here.

Throughout this Section, a rational subgroup of a connected simply connected nilpotent Lie group means a connected subgroup whose Lie algebra is rational for the rational structure determined by the specified lattice. We use the basic rational Lie theory of nilmanifolds: such a subgroup is closed, its lattice intersection is cocompact, and its orbit in the nilmanifold is an embedded compact homogeneous space. A rational element is an element with rational logarithm in this structure. Conjugation by a rational element preserves the rational structure. For these rational-coordinate facts and the periodicity of rational polynomial sequences, see [12, Section 2 and Appendix A]. All filtrations below have connected rational terms and satisfy

$$H_0 = H_1 = H \supseteq H_2 \supseteq \cdots \supseteq H_s \supseteq H_{s+1} = \{1\}, \quad [H_i, H_j] \subseteq H_{i+j}. \quad (7.1)$$

Terms beyond s are trivial. We write $\mathfrak{h}_i = \log H_i$. In particular H has nilpotence class at most s .

7.1 Upper-face groups and the missing corner

For $q \geq 0$, identify the vertices of a q -cube with subsets of $[q]$. For $D \subseteq [q]$, let g_D be the group-valued array equal to g at vertices containing D , and equal to the identity elsewhere. The case $D = \emptyset$ is the constant array. Define

$$C^q(H_\bullet) = \text{HK}^q(H_\bullet) = \left\langle g_D : D \subseteq [q], \quad g \in H_{|D|} \right\rangle \subseteq H^{\{0,1\}^q}.$$

For $q = 0$ this is H .

Lemma 7.1 (Upper-face coordinates). *The group $C^q(H_\bullet)$ is connected, simply connected, and rational. Order the subsets of $[q]$ by increasing cardinality, breaking ties in any fixed way. Every element of $C^q(H_\bullet)$ has a unique expression*

$$\prod_{D \subseteq [q]} (g_D)_D, \quad g_D \in H_{|D|}, \quad (7.2)$$

in this order; the map from the face coefficients to the array is a diffeomorphism. An array in this group which is the identity except possibly at the all-ones vertex has its remaining entry in H_q . The group is invariant under permutations and reflections of cube coordinates. If Γ is a lattice in H , its image in $(H/\Gamma)^{\{0,1\}^q}$ is compact.

Proof. For $X \in \mathfrak{h}_{|D|}$ write X_D for the corresponding face array. The Boolean monomials $\prod_{j \in D} e_j$ are linearly independent, so

$$\mathfrak{c}^q = \bigoplus_{D \subseteq [q]} \mathfrak{h}_{|D|} \prod_{j \in D} e_j$$

is a direct sum of vector spaces. The entrywise Lie bracket satisfies $[X_D, Y_E] = [X, Y]_{D \cup E}$. The filtration places this bracket in the summand indexed by $D \cup E$. This remains true when a set is empty, because $H_0 = H_1$ and the filtration is decreasing. Thus $\mathfrak{c}^q$ is a rational Lie subalgebra.

Every suffix in the chosen order of summands is an ideal: a bracket with a summand indexed by E stays at E or moves to a strictly larger set. Successively passing to the quotients by these suffix ideals gives (7.2). Equivalently, the coefficient at E is recovered from the entry at vertex E after the coefficients at proper subsets of E have been removed. This also proves uniqueness and smoothness of the inverse. The connected subgroup with Lie algebra $\mathfrak{c}^q$ is closed and simply connected, as follows from the exponential diffeomorphism for a simply connected nilpotent group. It is exactly the group generated by the faces.

The same recovery at successive vertices proves the assertion about an array supported at the top vertex. Reflecting a coordinate replaces e_j by $1 - e_j$. The resulting face indicator is a linear combination of upper-face indicators of no greater cardinality; the decreasing filtration therefore preserves $\mathfrak{c}^q$, and hence its exponential group. Permutations are immediate. Finally rationality gives a cocompact lattice $C^q(H_\bullet) \cap \Gamma^{\{0,1\}^q}$, so the indicated image is compact. $\square$

We will also use the following explicit rational filtration on the cube group:

$$\mathfrak{c}_i^q = \bigoplus_{D \subseteq [q]} \mathfrak{h}_{\max(i, |D|)} \prod_{j \in D} e_j, \quad i \geq 0. \quad (7.3)$$

It has $C_0^q = C_1^q = C^q(H_\bullet)$ and degree at most s . Indeed the bracket of its D and E summands lies at filtration level at least both $i + j$ and $|D \cup E|$. Thus it satisfies the required bracket inclusions.

The following missing-corner constraint and its coordinate-product consequence appear in [14, Proposition 11.5, proof of Proposition 11.2, and Appendix E]. We give the face-coordinate proof, including the uniform approximation over bounded observable families needed in Section 5.

Lemma 7.2 (Continuous missing-corner reconstruction). *Let G be a connected simply connected nilpotent Lie group of step at most s , and let Γ be a lattice in G . Equip G with its lower central filtration, with $G_0 = G_1 = G$. In dimension $s + 1$, any $2^{s+1} - 1$ vertices of a cube in the image of $C^{s+1}(G_\bullet)$ determine the last vertex uniquely and continuously on the set of admissible partial cubes. Linear orbit cubes*

$$(g^{b_0 + \sum_{j=1}^{s+1} e_j b_j} x)_e$$

belong to this compact cube set.

Consequently, for every $\varepsilon > 0$ and every bounded Lipschitz observable F on G/Γ , its value at the missing vertex is uniformly within ε of a finite sum of products of bounded continuous functions of the other individual vertices. Those individual functions may be taken Lipschitz and bounded by one, with scalar coefficients outside the products. For a fixed finite list of nilmanifolds and fixed bounds on $\|F\|_\infty$ and $\text{Lip}(F)$, finitely many such approximation recipes suffice at each accuracy.

Proof. Reflect the missing vertex to the all-ones vertex. If two group cubes $a, b \in C^{s+1}(G_\bullet)$ agree modulo Γ at all other vertices, then $z = a^{-1}b$ is a group cube whose entries at all proper subsets of $[s + 1]$ belong to Γ . Recover its face coefficients by (7.2). Inductively the coefficient at a proper subset E lies in Γ , because its value is the entry at E multiplied by inverses of previously recovered lattice elements. The top coefficient belongs to $G_{s+1} = \{1\}$. The top entry of z is therefore also a product of lattice elements. The original two top vertices agree modulo Γ .

The deletion map from the compact cube set to the space of partial cubes is now a continuous injection into a Hausdorff space. It is a homeomorphism onto its image, proving continuity of reconstruction. For a representative $x_0 \in G$ of x , the displayed linear cube is represented by the product of the constant arrays g^{b_0} and x_0 , with the codimension-one arrays g^{b_j} between them. It therefore lies in the cube group.

On the compact image of the deletion map, products of continuous functions of individual coordinates form an algebra containing constants and separating points. Stone–Weierstrass approximates the reconstructed observable by finite sums from this algebra. Lipschitz functions are dense in the continuous functions on each compact nilmanifold, so the factors can be made Lipschitz and then normalized to have sup norm at most one. Finally the family of observables with the prescribed sup and Lipschitz bounds is compact in the uniform norm. A finite uniform net, followed by the construction for each member of that net, proves the last assertion. $\square$

7.2 Adapted factorization on a sequence of scales

The lifting argument will use face information in every dimension through $s + 1$. We therefore need one rational filtration whose cube groups describe all these local laws, together with the point law. To obtain it, we first factor the orbit. The left factor will vary on the scale of the full interval, the right factor will be periodic modulo the lattice, and the remaining factor will lie in a fixed rational subgroup. Its Taylor coefficients will satisfy the irrationality conditions that give equidistribution in all the associated cube groups. The smooth–equidistributed–periodic decomposition of Green and Tao [12, Theorem 1.19] was

refined using Taylor-coefficient irrationality and filtration descent in [16, Definitions A.5–A.6 and Lemma 2.9]. We prove the sequential limiting form needed for the joint cube laws.

A polynomial $P : \mathbb{R} \rightarrow H$ is *adapted* to $H_\bullet$ if

$$\log P(b) = \sum_{j=0}^s b^j U_j, \quad U_j \in \mathfrak{h}_j.$$

This definition is unchanged on passing from monomials to binomial polynomials. Products and inverses of adapted polynomials are adapted: in the finite Baker–Campbell–Hausdorff expansion, a term of polynomial degree j belongs to $\mathfrak{h}_j$. Terms of degree above s vanish. Constant factors cause no difficulty, since H_0 normalizes every level. This definition implies polynomiality by group differences, as required in Theorem 6.1. More explicitly, suppose the coefficient of degree j in $\log R(b)$ lies in $\mathfrak{h}_{j+v}$. Give a formal shift t degree one as well. BCH shows that every coefficient of total degree $j+k$ in $\log(R(b+t)R(b)^{-1})$ lies in $\mathfrak{h}_{j+k+v}$; brackets of two terms can only increase this filtration index. The expression is zero at $t=0$, so every surviving monomial uses at least one power of t . For each fixed t , its degree- j coefficient in b therefore belongs to $\mathfrak{h}_{j+v+1}$. Iterating from $v=0$ puts every k -fold difference in H_k . The same argument works with total degree in several variables.

Lemma 7.3 (Taylor coordinates and finite descent). *Fix a connected simply connected nilpotent group K , a lattice Λ , and a rational filtration $K_\bullet$ satisfying (7.1). Let $L \rightarrow \infty$ along any sequence, and let P_L be arbitrary adapted polynomials. After passing to a subsequence there exist a fixed rational filtration $H_\bullet$ with $H_i \subseteq K_i$, a positive integer d_0 , rational elements $\sigma_0, \dots, \sigma_{d_0-1}$ of K , and factorizations*

$$P_L(b)\Lambda = h_L(b)Q_L(b)\gamma_L(b)\Lambda, \quad Q_L(b) = \prod_{j=1}^s a_{j,L}^{(b)} \binom{b}{j}, \quad a_{j,L} \in H_j, \quad (7.4)$$

such that the following hold.

1. The maps $\beta \mapsto h_L(L\beta)$ converge locally uniformly on $\mathbb{R}$ to a smooth map $h : \mathbb{R} \rightarrow K$. They and their first derivatives are bounded on every fixed compact interval, uniformly in L .
2. For integer b , $\gamma_L(b)\Lambda = \sigma_{b \bmod d_0}\Lambda$.
3. If $1 \leq j \leq s$ and $\xi : H_j \rightarrow \mathbb{R}$ is a nonzero rational homomorphism annihilating H_{j+1} and every $[H_i, H_{j-i}]$, $1 \leq i < j$, then

$$L^j \|\xi(a_{j,L})\|_{\mathbb{R}/\mathbb{Z}} \rightarrow \infty. \quad (7.5)$$

Here rational homomorphisms include all nonzero rational multiples of one another. No boundedness of the Taylor coefficients $a_{j,L}$ is asserted or required.

Proof. We first justify Taylor coordinates. For any adapted R write $a_0 = R(0)$ and remove this constant on the left. Suppose that, after removing the factors of orders below j , the remaining adapted polynomial R_j is the identity at $0, \dots, j-1$. The j th forward difference of $\log R_j$ at zero equals $\log R_j(j)$: all its other evaluated terms vanish. Terms of degree less than j in the logarithm make no contribution, and all terms of degree at least j have coefficients in $\mathfrak{h}_j$. Thus $a_j = R_j(j) \in H_j$. Multiplication on the left by $a_j^{-\binom{b}{j}}$ leaves an adapted polynomial vanishing at the first $j+1$ integers. After order s , its logarithm has degree at most s and $s+1$ zeros, so it is identically zero. This proves the ordered Taylor representation.

Choose $\lambda_L \in \Lambda$ so that $c_L = P_L(0)\lambda_L$ lies in a fixed compact set of representatives. Normalize by $R_L(b) = c_L^{-1}P_L(b)\lambda_L$. It is adapted, has constant term the identity, and $P_L(b)\Lambda = c_LR_L(b)\Lambda$. Adaptation is preserved here either by the preceding BCH observation, or by noting that conjugation by λ_L preserves all terms of $K_\bullet$.

We describe one descent step for a fixed current rational filtration $H_\bullet$ and a normalized adapted polynomial with Taylor coefficients $a_{j,L}$. If (7.5) fails, fix one of its characters ξ and pass to a subsequence on which

$$\xi(a_{j,L}) = m_L + \lambda'_L, \quad m_L \in \mathbb{Z}, \quad |\lambda'_L| \leq CL^{-j}.$$

Choose a fixed rational $V \in \mathfrak{h}_j$ with $\xi(\exp V) = 1$. Replace the polynomial R_L by

$$R'_L(b) = \exp(-\lambda'_L \binom{b}{j} V) R_L(b) \exp(-m_L \binom{b}{j} V). \quad (7.6)$$

It remains adapted to the old filtration. Its Taylor coefficients of orders less than j are unchanged, since both multipliers are the identity at $0, \dots, j-1$. Modulo H_{j+1} , the order- j coefficient is the old coefficient with the two displayed corrections. To verify this even in the presence of lower Taylor factors, commute those factors past the left correction at $b = j$; the commutators lie in $[H, H_j] \subseteq H_{j+1}$. Its ξ -value is consequently $-\lambda'_L + \xi(a_{j,L}) - m_L = 0$.

Replace H_j by $H_j \cap \ker \xi$ and leave all other positive levels unchanged. When $j = 1$ replace H_0 by the same kernel. The kernel is a connected rational subgroup, since in exponential coordinates it is the kernel of a rational linear form. Nesting is preserved because ξ kills H_{j+1} . A bracket inclusion whose target is the changed level is preserved because ξ kills each $[H_i, H_{j-i}]$. Inclusions whose target has larger index already land in H_{j+1} , and the other inclusions are unchanged or have a smaller source. Thus this is again a filtration. The Taylor calculation shows that R'_L is adapted to it. In the case $j = 1$, all higher coefficients already belong to H_2 , so its entire image lies in the new H .

The removed factors give

$$R_L(b) = \exp(\lambda'_L \binom{b}{j} V) R'_L(b) \exp(m_L \binom{b}{j} V).$$

The left factor is bounded and has derivative $O(L^{-1})$ on every interval $|b| \leq AL$, with constants allowed to depend on A, C, V . The right factor is an integer power of the fixed rational element $\exp V$ at integer inputs. Repeating the procedure strictly reduces $\sum_{i=1}^s \dim H_i$ at every step. It therefore terminates after finitely many steps. If any fixed rational level character still failed (7.5) on the final subsequence, the same operation would give another strict reduction. Hence all the asserted character limits hold.

There is no conjugation of accumulated smooth factors by rational factors in this procedure: from $P = \epsilon_1 R_1 \gamma_1$ and $R_1 = \epsilon_2 R_2 \gamma_2$ one gets $P = (\epsilon_1 \epsilon_2) R_2 (\gamma_2 \gamma_1)$. Thus h_L is the product of c_L and finitely many of the bounded smooth factors just described. Pass to a subsequence on which c_L converges and every bounded scalar $L^j \lambda'_L$ converges. Since $L^{-j} \binom{L^\beta}{j}$ converges with all derivatives on compact sets to $\beta^j / j!$, this gives the first conclusion.

For completeness, the right factors have a common bounded period modulo Λ , even though the integers m_L need not be bounded. Finitely many fixed rational elements, together with a finite generating set of Λ , generate a discrete group Λ' containing Λ . Here is the denominator argument. Collect a word into integer powers of the finitely many generator and iterated-commutator types of length at most the nilpotence class. Collection terminates because a commutation error has strictly greater commutator length. The logarithms of these finitely many types have rational coordinates. Applying the finite BCH formula to their ordered powers bounds every coordinate denominator by one fixed integer, independent of the word and its integer exponents. This proves discreteness. A compact fundamental set for Λ then shows that $[\Lambda' : \Lambda] < \infty$.

The normal core $\Lambda'' = \bigcap_{g \in \Lambda'} g\Lambda g^{-1}$ has finite index in Λ' , so Λ'/Λ'' is a finite group. Let a be a common multiple of its element orders. Each $\binom{b}{j}$ modulo a has a fixed period, for example $as!$ for $0 \leq j \leq s$: Vandermonde's identity shows that $\binom{b+as!}{j} - \binom{b}{j}$ is divisible by a . Consequently every accumulated right factor is periodic in this finite quotient with one common period d_0 . There are only finitely many possible maps from its residue classes to Λ'/Λ'' . Pass to a subsequence on which this map is fixed and choose rational representatives σ_r . This gives the second conclusion and completes the proof. $\square$

7.3 The joint local point and cube laws

The factorization has fixed a subgroup, a filtration, and a finite period. We now identify the distribution of the remaining factor and its cubes. The key step is to detect any horizontal obstruction in a cube group in one of the Taylor coefficients controlled above.

For a rational element $\sigma \in K$, put

$$\Gamma_\sigma = H \cap \sigma\Lambda\sigma^{-1}, \quad \Gamma_\sigma^{[q]} = C^q(H_\bullet) \cap (\sigma\Lambda\sigma^{-1})^{\{0,1\}^q}.$$

These are lattices in their respective groups. By Haar measure on $hH\sigma\Lambda/\Lambda$ we mean the image of invariant probability on H/Γ_σ under $y \mapsto hy\sigma\Lambda$. Use the analogous convention for cube cosets.

Proposition 7.4 (Simultaneous local cube law). *Under the hypotheses of Lemma 7.3, retain the subsequence and its data. Define probability measures*

$$m_{\beta,r}^{[q]} = \text{Haar image on } h(\beta)^{\text{diag}} C^q(H_\bullet) \sigma_r^{\text{diag}} \Lambda^{\{0,1\}^q} / \Lambda^{\{0,1\}^q}. \quad (7.7)$$

The empirical point laws satisfy

$$\frac{1}{[L]} \sum_{0 \leq b < [L]} \delta_{P_L(b)\Lambda} \implies \frac{1}{d_0} \sum_{r=0}^{d_0-1} \int_0^1 m_{\beta,r}^{[0]} d\beta. \quad (7.8)$$

For every fixed cube dimension q , every $\beta \in (0,1)$, every residue $r \bmod d_0$, and every bounded Lipschitz test F on $(K/\Lambda)^{\{0,1\}^q}$,

$$\lim_{\alpha \downarrow 0} \limsup_{L \rightarrow \infty} \left| \mathbb{E}_{\substack{b_0/L \in [\beta, \beta+\alpha), \ b_0 \equiv r \pmod{d_0} \\ b_j/L \in [0, \alpha), \ b_j \equiv 0 \pmod{d_0}, \ 1 \leq j \leq q}} F((P_L(b_0 + \sum_{j=1}^q e_j b_j)\Lambda)_e) - \int F dm_{\beta,r}^{[q]} \right| = 0. \quad (7.9)$$

The inputs in this average are independent and uniform on the specified integer progressions. The same assertion holds for boxes with side lengths between $c\alpha L$ and $C\alpha L$, whose base coordinates are within $C\alpha L$ of βL and whose increments have absolute value at most $C\alpha L$, where $c, C > 0$ are fixed before either limit. For fixed bounds the assertions are uniform over the tests and these boxes. In particular, arbitrary fixed-modulus progression restrictions are permitted if the base has residue r and all increments have residue zero modulo d_0 . Every progression modulus is fixed before the length limit, which always precedes the locality limit. The same factorization supplies these conclusions for every fixed q .

Proof. We first prove equidistribution before inserting the smooth factor. Fix q, r and abbreviate $C = C^q(H_\bullet)$. The cube polynomial

$$\mathbf{Q}_L(\mathbf{b}) = (Q_L(b_0 + \sum_{j=1}^q e_j b_j))_e$$

takes values in C . To see this and record the needed coefficient information, expand

$$\binom{b_0 + \sum_{j=1}^q e_j b_j}{\ell} = \sum_{D \subseteq [q]} \left(\prod_{j \in D} e_j \right) F_{\ell,D}(\mathbf{b}). \quad (7.10)$$

Each $F_{\ell,D}$ has degree at most ℓ , and every one of its monomials uses exactly the increment variables indexed by D , each with positive exponent. In particular it vanishes for $|D| > \ell$. Thus the ℓ th Taylor factor on the cube is the product of the face arrays $(a_{\ell,L})_D^{F_{\ell,D}(\mathbf{b})}$ with $|D| \leq \ell$. These are in C , because $H_\ell \subseteq H_{|D|}$. This expansion also shows adaptation to (7.3): a coefficient of total polynomial degree k in a factor of order $\ell \geq k$ lies in $\mathfrak{h}_\ell \subseteq \mathfrak{h}_{\max(k, |D|)}$. BCH preserves this assertion on multiplying the Taylor factors.

We claim that $\mathbf{Q}_L$ is equidistributed on $C/\Gamma_{\sigma_r}^{[q]}$ on every box whose sides are fixed positive fractions of L , whose origin is $O(L)$, and with any fixed progression restrictions. The error tends to zero uniformly for fixed bounds on these data and on the Lipschitz tests. Otherwise Theorem 6.1, applied to the fixed rational nilmanifold and the filtration (7.3), supplies along a subsequence a nonzero horizontal character from a fixed finite list. Pass to a further subsequence on which this character, denoted $\chi : C \rightarrow \mathbb{R}$, is fixed. It annihilates commutators and takes integer values on $\Gamma_{\sigma_r}^{[q]}$.

For a face D , write $\chi_D(y) = \chi(y_D)$ wherever this is defined. Choose the largest $j \geq 1$ such that for some $|D| \leq j$ the restriction $\xi = \chi_D|_{H_j}$ is nonzero. Such a pair exists because the face subgroups generate C , including the empty face and $H_0 = H_1$. The restriction is a rational homomorphism: it is integer-valued on the lattice $H_j \cap \sigma_r \Lambda \sigma_r^{-1}$, and conjugation by σ_r preserves rationality. Maximality makes it vanish on H_{j+1} . For $1 \leq i < j$, choose $A, B \subseteq D$ with $A \cup B = D$, $|A| \leq i$, and $|B| \leq j - i$; one or both sets may be empty. If $u \in H_i$ and $v \in H_{j-i}$, their face arrays on A, B lie in C and $[u_A, v_B] = [u, v]_D$. Since χ kills this commutator, ξ kills $[H_i, H_{j-i}]$. Thus ξ is exactly a character covered by (7.5).

Applying χ to (7.10) gives the scalar polynomial

$$\chi(\mathbf{Q}_L(\mathbf{b})) = \sum_{\ell, D} \chi_D(a_{\ell,L}) F_{\ell,D}(\mathbf{b}).$$

All terms with $\ell > j$ vanish by maximality. In its degree- j part, the coefficient of

$$b_0^{j-|D|} \prod_{k \in D} b_k \quad \text{is} \quad \frac{\xi(a_{j,L})}{(j - |D|)!}. \quad (7.11)$$

Indeed the top part of the binomial polynomial is the j th power divided by $j!$. No other face can contribute to this monomial: its increment support must be exactly D . No lower Taylor order has degree j . This proves both the value and the absence of cancellation in (7.11).

On a fixed residue progression, substitute $b_\nu = r_\nu + dt_\nu$ with fixed d . Its top-degree coefficient is multiplied by d^j ; translations of the box origin do not alter it. The bounded scalar coefficient obstruction in Theorem 6.1, clearing the fixed binomial-to-monomial denominators if necessary, would therefore give

$$L^j \|a \xi(a_{j,L})\|_{\mathbb{R}/\mathbb{Z}} = O(1)$$

for a fixed nonzero rational number a . This contradicts (7.5) for the rational character $a\xi$. The claim follows. The argument also covers $q = 0$, when only the empty face occurs. If C is trivial, the claim is immediate.

In the averages in (7.9), all vertex inputs have the same residue $r \bmod d_0$. Consequently their right factors can be replaced exactly by $\sigma_r \Lambda$. For fixed α , the preceding claim gives the Haar law for the Q_L cube. At every vertex, $b/L = \beta + O_q(\alpha)$; Lemma 7.3 therefore gives

$$h_L(b) = h(\beta) + O(\alpha) + o_{L \rightarrow \infty}(1)$$

in any fixed local metric, uniformly on the sampled box. Multiplication by these bounded elements and passage to the compact nilmanifold changes a bounded Lipschitz test by $O_F(\alpha) + o(1)$, uniformly over the remaining entries. The latter uniformity follows from smoothness of the action on the compact nilmanifold. First letting $L \rightarrow \infty$ and then $\alpha \downarrow 0$ proves (7.9), including its stated uniform versions.

Finally partition $[0, 1]$ into finitely many intervals of length at most α . On each interval and each residue class, apply the same argument with $q = 0$ and freeze h at an endpoint. The proportions of the residue classes tend to $1/d_0$, and the interval proportions tend to their lengths. Letting $\alpha \downarrow 0$ gives the Riemann integral in (7.8). This integral is well defined because the Haar images depend continuously on β . No further subsequence depending on q was used: (7.5) proves the equidistribution assertion for each fixed dimension. $\square$

7.4 Lifting a symmetry of one marginal

The local cube law describes the joint state as a mixture of Haar measures on compact orbits, with one filtration describing the cubes on each orbit. In Section 8, the progression comparison will show that an automorphism of one component preserves the projections of these cube cosets when applied on a face. The next proposition converts precisely that information into a transformation of the joint cover. We will then show that any family of these lifts generates a group of nilpotence class at most s , as required for recurrence.

Choose linear coordinates on $\mathfrak{h}$ by choosing, for every j , a complement of $\mathfrak{h}_{j+1}$ in $\mathfrak{h}_j$. Assign weight j to coordinates on that complement. The weighted degree of a monomial is the sum of its variable weights, with multiplicity; constants have weight zero. A *polynomial shear* will mean a map

$$T(u)_{j,k} = u_{j,k} + p_{j,k}(u), \quad \deg_{\text{wt}} p_{j,k} < j. \quad (7.12)$$

In particular the displacement at weight j uses only coordinates of smaller weight.

Proposition 7.5 (A marginal face symmetry has a joint lift). *Let $K = K' \times K''$ be a connected simply connected nilpotent Lie group with lattice $\Lambda = \Lambda' \times \Lambda''$, and let π denote the first group projection and its induced map of nilmanifolds. Let $H \subseteq K$ be a connected rational subgroup with a connected rational filtration $H_\bullet$ satisfying (7.1). Let $h \in K$, and let $\sigma \in K$ be rational. Let S be a fixed automorphism of K' preserving Λ' . Suppose that, for every $1 \leq q \leq s+1$, applying S to the entries on any upper codimension-one face preserves setwise the projected cube coset*

$$\pi \left(h^{\text{diag}} C^q(H_\bullet) \sigma^{\text{diag}} \Lambda^{\{0,1\}^q} / \Lambda^{\{0,1\}^q} \right). \quad (7.13)$$

Then there is a polynomial shear $T : \mathfrak{h} \rightarrow \mathfrak{h}$ such that, for every $u \in \mathfrak{h}$,

$$\pi(h \exp(Tu) \sigma \Lambda) = S(\pi(h \exp(u) \sigma \Lambda)). \quad (7.14)$$

The lift is a bijection of the cover. Its action on the other component is unrestricted. This conclusion applies separately to every point coset from Proposition 7.4; no measurable choice of lifts as a function of β, r is required.

Proof. Put $J_i = \pi(H_i)$ and $J = J_1$. These are connected rational groups; their Lie algebras are rational images of those of H_i . Their filtration satisfies (7.1), and projection of the face generators shows $\pi(C^q(H_\bullet)) = C^q(J_\bullet)$. In the remainder of the proof use h, σ, Λ for the projected elements and lattice. The projected point coset $hJ\sigma\Lambda/\Lambda$ is preserved by S : each of its points can occur at a vertex on the specified face of a constant cube, and face

preservation, also for the inverse map, gives equality of the point sets. Choose $c \in J$ such that

$$S(h\sigma)\Lambda = hc\sigma\Lambda. \quad (7.15)$$

The ambient automorphism

$$\Psi(y) = h^{-1}S(h)S(y)S(h)^{-1}h$$

satisfies $S(hy\sigma)\Lambda = h\Psi(y)c\sigma\Lambda$ for $y \in J$. We next verify on the group cover that $\Psi(J) = J$.

The map $y \mapsto y\sigma\Lambda$ on J has stabilizer $\Gamma_J = J \cap \sigma\Lambda\sigma^{-1}$. Rationality makes J/Γ_J a compact embedded orbit. Given a path $y(t)$ in J from the identity to y , the path $\Psi(y(t))c\sigma\Lambda$ stays in this orbit. Lift it there starting at $c \in J$. It is also a lift under the ambient covering $K' \rightarrow K'/\Lambda$ after multiplying on the right by σ . Uniqueness of lifts under this covering forces the ambient lift $\Psi(y(t))c$ to lie in J . Thus $\Psi(J) \subseteq J$; equality follows because its differential is injective and both connected groups have the same dimension. Consequently

$$\Phi(y) = c^{-1}\Psi(y)c$$

is an automorphism of J , and the action on the point coset is represented by the affine map $y \mapsto c\Phi(y)$.

We claim that its linear part is the identity on each associated graded space:

$$(\Phi_* - \text{id})j_i \subseteq j_{i+1}, \quad 1 \leq i \leq s. \quad (7.16)$$

Fix $U \in j_i$ and work in dimension $i+1$. Put $\exp(tU)$ on the face with its first i coordinates equal to one, and the identity elsewhere. This is a path $z(t)$ in $C^{i+1}(J_\bullet)$. Apply the affine map $c\Phi$ on the upper face of the last coordinate. The resulting array $z'(t)$ has its image in the same cube coset by hypothesis. At $t=0$ it is the constant- c upper-face array, which itself belongs to $C^{i+1}(J_\bullet)$.

This quotient assertion lifts to $z'(t) \in C^{i+1}(J_\bullet)$ for every t . Indeed this cube group is rational with lattice $C^{i+1}(J_\bullet) \cap \Gamma_J^{\{0,1\}^{i+1}}$, so its quotient is an embedded compact subspace of the product point orbit. Lift the path through that quotient beginning at $z'(0)$ and use uniqueness in the ambient product covering, exactly as in the preceding paragraph.

Multiply $z'(t)$ on the left by the inverse constant- c last-face array, and then on the left by $z(t)^{-1}$. The resulting group cube is the identity except at the top vertex, whose value is

$$\exp(-tU)\Phi(\exp(tU)).$$

Lemma 7.1 puts this element in J_{i+1} . Differentiating at $t=0$ proves (7.16). In particular Φ preserves every filtration level.

Choose filtration-adapted coordinates on j . The polynomial

$$D(Y) = \log(c\Phi(\exp Y)) - Y \quad (7.17)$$

has weighted degree strictly less than the output weight in each coordinate. For its linear part this follows directly from (7.16): an input from level i moves only to levels greater than i . For the remaining terms use BCH with $\log c$ and Φ_*Y . Every nonconstant bracket correction contains at least one copy of the fixed vector $\log c$, which spends at least one unit of filtration weight but contributes no polynomial degree in Y . A bracket contributing to output weight j therefore has input weighted degree at most $j-1$. The constant term also has the required degree.

The linear projection $p = \pi_* : \mathfrak{h} \rightarrow \mathfrak{j}$ has a filtration-preserving linear right inverse R . To construct one, choose an adapted basis downstairs and lift each basis vector of weight i to $\mathfrak{h}_i$, which is possible because $p(\mathfrak{h}_i) = \mathfrak{j}_i$. Set

$$T(u) = u + RD(pu). \quad (7.18)$$

This is a shear of the form (7.12). Explicitly, a filtration-preserving linear map can send an input coordinate of weight i only to an output of weight at least i . Substitution in D and then application of R thus preserve the strict inequality between input degree and output weight. Moreover $p(Tu) = pu + D(pu) = \log(c\Phi(\exp(pu)))$, proving (7.14). Finally a map of (7.12) is inverted successively in increasing weight, so it is a polynomial bijection of the cover. $\square$

Remark 7.6. Invariance of the point marginal alone would not suffice. On $\mathbb{R}^2/\mathbb{Z}^2$, the automorphism $S(x, y) = (x + y, y)$ preserves Haar measure but is not a translation. For $a = (0, 1/3)$, applying S to the upper second face of the additive square $(0, a, 0, a)$ gives $(0, a, 0, Sa)$, whose alternating corner sum is $Sa - a = (1/3, 0) \neq 0$ in the torus. It therefore fails the two-dimensional face condition. The analogous test in dimension $i + 1$ forces the graded identity (7.16) in the proof above.

7.5 Bounded-step shears and Haar approximation

The lifts of different marginal symmetries may move the same coordinates. The next Lemma controls the group they generate and gives every fixed word in these lifts the same limiting distribution on the joint orbit.

Lemma 7.7 (Nilpotence and expanding weighted boxes). *On any finite-dimensional real vector space with coordinate weights in $\{1, \dots, s\}$, all the shears (7.12) form a group of nilpotence class at most s . This bound does not depend on the dimension or the coefficients of the shears. A direct product with any group of ordinary translations has the same bound.*

Let $\mathfrak{h}$ carry the adapted coordinates above, and let λ_E be normalized Lebesgue measure on

$$B_E = \{u : |u_{j,k}| \leq E^j \text{ for all } j, k\}, \quad E \longrightarrow \infty. \quad (7.19)$$

For every fixed finite set $\mathcal{T}$ of shears,

$$\max_{T \in \mathcal{T}} \|T_* \lambda_E - \lambda_E\|_{\text{TV}} \longrightarrow 0. \quad (7.20)$$

For every fixed $h \in K$ and rational $\sigma \in K$, the images of λ_E under $u \mapsto h \exp(u) \sigma \Lambda$ converge weakly to the Haar image on $hH\sigma\Lambda/\Lambda$. The same limit holds after any fixed shear, and hence after each word in any fixed finite list of words in the lifts from Proposition 7.5.

Proof. Composition preserves (7.12), and inversion does so by solving one weight at a time. Let V_d be the space of polynomials of weighted degree at most d , including the constants, and put $V_{-1} = 0$. Substitution by a shear preserves V_s , and substitution minus the identity maps V_d into V_{d-1} . Indeed every term in the difference of a substituted monomial uses at least one strict lower-degree displacement in place of its coordinate. Use inverse substitution if necessary to obtain a group representation rather than an opposite representation. It is faithful because all coordinate functions belong to V_s .

Let $\mathcal{I}$ denote the algebra of linear operators on V_s which send each V_d into V_{d-1} . Then $\mathcal{I}^{s+1} = 0$, and the representation lies in $1 + \mathcal{I}$. The elementary identities for inverses in a nilpotent algebra give $[1 + \mathcal{I}^a, 1 + \mathcal{I}^b] \subseteq 1 + \mathcal{I}^{a+b}$. Thus every $(s + 1)$ -fold group commutator

is the identity. Extra translation coordinates can all be given weight one, which proves the direct-product assertion as well.

For a fixed shear T , its displacement in a weight- j coordinate is $O_T(E^{j-1})$ on B_E . Its derivative matrix is block triangular by weight with identity diagonal, so its Jacobian determinant is one. The image $T(B_E)$ is contained in the box with side bounds $E^j + C_T E^{j-1}$. The latter box exceeds B_E in relative volume $O_T(E^{-1})$. Since T preserves Lebesgue measure, the symmetric difference of $T(B_E)$ and B_E has the same bound. This proves (7.20), uniformly over a fixed finite list by taking the largest of its constants. The assertion is deliberately for fixed shears; no uniform coefficient bound over all shears is needed.

For $a \in H$ fixed, left multiplication in logarithmic coordinates is

$$u \mapsto \log(a \exp u).$$

BCH makes this another shear: every displacement bracket uses the fixed vector $\log a$ and hence has input weighted degree strictly below its output weight. Equation (7.20) therefore implies that every weak subsequential limit of the projected boxes on $H/(H \cap \sigma \Lambda \sigma^{-1})$ is invariant under all left translations by H . This compact homogeneous space has a unique such probability measure, its Haar probability. Compactness supplies subsequential limits, and uniqueness shows that the entire sequence converges. Translating on the left by h and on the right by $\sigma \Lambda$ gives the stated point-coset Haar law. Applying (7.20) once more proves the identical limit after each fixed shear or word. $\square$

The locality and finiteness qualifications in this Section are useful in Section 8. There the marginal face symmetries will be proved for every fixed residue modulus before d_0 is selected. The finite recurrence argument will then use only finitely many fixed words on each limiting coset. Lemma 7.7 supplies their common Haar limit without imposing either a dimension bound on those covers or measurable dependence of the lifts on the limiting coset.

8 Finite alignment of the models

We prove Principle 2.4. The argument separates a finite coloring construction from its realization by integer variables. The finite construction fixes all its choices before the threshold and the model complexity are known. The integer realization uses the marginal cube symmetries from the preceding two Sections; its passage from existence to positive probability loses only the number of those finite choices.

Throughout this Section, a target is a block $B = T \cup \{i\}$ with $\mathcal{E}_B \neq \emptyset$. A pair at pivot i means $e = (B, A)$, where

$$B = T \cup \{i\}, \quad A = P \cup \{j\}, \quad \emptyset \neq P < T < \{j\} < \{i\}.$$

All lists below are finite. Their sizes may depend on n, r, s , but never on w, τ , or the complexity of the models.

The integer meaning of one transformation is elementary. At a center tp , where t is a tail product and p is the pivot variable, an integer displacement q need not be divisible by t . Since $(t, M) = 1$, choose an integer ρ with $t\rho \equiv q \pmod{M}$ and put $\Delta = (q - t\rho)/M$. Then

$$t(p + m\rho) + mM\Delta = tp + mq \quad (m \in \mathbb{Z}).$$

The change $p \mapsto p + m\rho$ supplies the correct residue modulo M ; the remaining change $m\Delta$ is in the progression index of the model on that residue class. We will realize these two

changes by a transformation of the model state. A transformation chosen for one target can move the states for other targets as well. This is why the finite construction uses words in possibly noncommuting transformations and protects the comparisons already obtained for earlier targets. A general word need not translate the integer variable p . The precise residue choices and the states on which these transformations act are constructed after the finite plan.

8.1 A finite plan for words and rational scales

Put $p_\ell = 2^{n-\ell}$ and $p_I = \sum_{\ell \in I} p_\ell$. At an update targeting $B_0 = T_0 \cup \{i\}$, define the multiplier tuple $\kappa_{i,T_0}(k)$, for $k \in \mathbb{N}$, by

$$\kappa_\ell(k) = \begin{cases} k^{p_\ell}, & \ell < i, \\ k^{-p_{T_0}}, & \ell = i, \\ 1, & \ell > i. \end{cases} \quad b' = b\kappa(k). \quad (8.1)$$

Here multiplication of tuples is coordinatewise. Thus $b'_{B_0} = b_{B_0}$. For every pair $e = (U, A)$ at this pivot, with $U = C \cup \{i\}$,

$$\frac{b'_A}{b'_U} = \frac{b_A}{b_U} k^{\lambda_e}, \quad \lambda_e = p_A - p_C + p_{T_0} > 0. \quad (8.2)$$

Indeed, if $a = \min P$, then $p_a > \sum_{\ell > a} p_\ell \geq p_C$, so already $p_P > p_C$. This proves positivity for every pair at the pivot, including those with $C \neq T_0$. These compensating rescalings follow the mechanism in [1, proof of Proposition 4.1]. Here the transformations for different targets may not commute, so the comparisons that must survive an update will be indexed by words.

For each pivot i , let a group G_i of nilpotence class at most s act on a set $\mathcal{X}_i$. Associate to every pair $e = (B, A)$ at that pivot an element $L_e \in G_i$. For every target B at that pivot and every positive rational a , let $C_{B,a} : \mathcal{X}_i \rightarrow \{0, 1\}^r$ be an arbitrary map, called its palette. The finite choices below must work for every such collection of actions and maps.

A word template is a word in letters (e, γ) and their inverses, where $\gamma \in \mathbb{Q}$. At scale b it is interpreted as a group word by

$$(e, \gamma)[b] = L_e^{q_0 \gamma b_A / b_U}, \quad e = (U, A). \quad (8.3)$$

The integer q_0 is chosen after the finite plan; all exponents actually used will then be integers. Words act on the left, so wJx means first apply J , then w . The empty word is allowed. At a center $x \in \mathcal{X}_i$, a requirement for a block B , leading-multiplier list $\mathcal{L}$, and word list $\mathcal{W}$ is

$$C_{B,ab_B}(x) = C_{B,ab_B}(w[b]x) \quad (a \in \mathcal{L}, w \in \mathcal{W}). \quad (8.4)$$

Nilpotent polynomial recurrence was developed by Leibman [25] and by Bergelson and Leibman through polynomial Hales–Jewett theorems [4, 5]. We use the finite-coloring consequence of nilpotent polynomial recurrence in [37, Corollary 3.7]. In the form needed here, for finitely many ordered words

$$w_j(k) = a_{j,1}^{P_{j,1}(k)} \cdots a_{j,l_j}^{P_{j,l_j}(k)}, \quad P_{j,l} \in \mathbb{Z}[k], \quad P_{j,l}(0) = 0,$$

in a nilpotent group of bounded step, every finite coloring admits $k > 0$ and J for which $J, w_1(k)J, \dots, w_q(k)J$ have the same color. Only the subgroup generated by the finitely many letters is involved; it is countable, as required by that Corollary. The recurrence

statement permits these noncommuting ordered products. For the handedness, apply its left-translate formulation to the inverse words and to the coloring $g \mapsto C(g^{-1})$, and then invert the resulting pattern. To check the polynomial hypothesis of that specialization, let $D \geq 1$ bound the degrees of the exponents and stretch the lower central series to the filtration $H_0 = G$, $H_j = \gamma_{\lceil j/D \rceil}(G)$ for $j \geq 1$. The map $\alpha \mapsto a^{P(|\alpha|)}$ is an IP polynomial for this filtration, since repeated derivatives are ordinary finite differences of its exponent. Product and inverse closure, as in [37, Theorem 2.5], includes the required ordered words. Include the identity word in the recurrence family. Here $k = |\alpha|$ for a nonempty finite index set α , so the parameter is positive. Stretching the filtration verifies the recurrence hypothesis; the acting group still has nilpotence class at most s .

There is a useful uniform finite version of this statement. Give every letter type its own generator in the free nilpotent group of step s on these finitely many generators. On the compact space of its colorings with a fixed finite palette, each successful pair (k, J) defines an open cylinder set. The preceding recurrence statement shows that these cylinders cover the space. A finite subcover gives finitely many pairs (k, J) . Pulling a coloring back by the homomorphism that interprets the generators proves the same assertion in every nilpotent group of step at most s . The finite alternatives depend on the word templates, polynomial exponents, palette size, and s , and on no numerical values assigned to the generators.

Lemma 8.1 (Finite word plan). *For given n, r, s , there are a positive integer q_0 and finite lists of scale updates and word jumps with the following property, uniformly over the actions and palette maps just specified and the initial centers at all pivots. Start the scale tuple at $(1, \dots, 1)$, process the pivots increasingly, and fix an order of their targets. At each pivot one can choose updates (8.1) and jumps from these lists so that, at the resulting scale b and center, the terminal palette comparisons hold simultaneously for all its targets. These comparisons are (8.4) with leading multiplier d_B and word*

$$\prod_{A \in D} ((B, A), d_A/d_B)[b], \quad (8.5)$$

in a fixed order, for every target B , every $D \subset \mathcal{E}_B$, and every multiplier tuple d obtainable from later prescribed updates. When later pivots are processed, the center at this pivot is left fixed: these terminal requirements already include all the later scale changes.

All these lists depend only on n, r, s . There are finite lists $\mathcal{B} \subset \mathbb{Q}_{\geq 0}^n$ and $\mathcal{A} \subset \mathbb{Q}_{> 0}$ containing, respectively, the final scale tuples and every palette index consulted during the construction. All intermediate scales belong to finite lists as well. The list $\mathcal{A}$ also contains every required product of entries of a tuple in $\mathcal{B}$. All powers used in (8.3) are integral. If each L_e also translates an auxiliary integer coordinate v_e by 1, all auxiliary indices visited from $v = 0$ lie in a prescribed finite slot list.

Proof. We give the backward recursion, including the requirements protecting an earlier target. At a fixed pivot let its ordered targets be $B_1, \dots, B_q$. Suppose the required lists after update j have already been specified for $B_1, \dots, B_j$; denote them by $\mathcal{L}_{j,B}$ and $\mathcal{W}_{j,B}$. At $j = q$ these are the terminal lists in the statement. Taking a full product of the leading and word lists only strengthens the requirements.

Target B_j at update j . Its leading product does not change. For a word w , let $I_k(w)$ denote the old-scale template obtained by replacing each letter (e, γ) by $(e, \gamma k^{\lambda_e})$. Equation (8.2) gives the exact identity

$$w[b\kappa(k)] = I_k(w)[b]. \quad (8.6)$$

For B_j and a fixed current center x , color the acting group by $g \mapsto (C_{B_j, ab_{B_j}}(gx))_{a \in \mathcal{L}_{j,B_j}}$. Its size is at most $2^{r|\mathcal{L}_{j,B_j}|}$. Regard each rational old-scale letter needed in $\mathcal{W}_{j,B_j}$ as a

separate abstract generator. The words $I_k(w)$ are ordered products of these generators to powers $\pm k^{\lambda_e}$, with all exponents vanishing at zero. The finite version of recurrence therefore supplies a finite set Ω_j of alternatives (k, J) satisfying the required equalities for B_j at $J[b]x$ and scale $b\kappa(k)$. This set is independent of the incoming numerical scale.

For each previously processed $B = B_l$, $l < j$, define predecessor lists by including, for every $(k, J) \in \Omega_j$,

$$\mathcal{L}_{j-1,B} \supset \{a\kappa_B(k) : a \in \mathcal{L}_{j,B}\}, \quad (8.7)$$

$$\mathcal{W}_{j-1,B} \supset \{J\} \cup \{I_k(w)J : w \in \mathcal{W}_{j,B}\}. \quad (8.8)$$

The unions over all alternatives are finite. If these predecessor requirements hold at (x, b) , then for each new leading index $ab_B\kappa_B(k)$ they compare the same old center x both with $J[b]x$ and with $I_k(w)[b]J[b]x$. Equality through the old center and (8.6) give

$$C_{B,ab'_B}(J[b]x) = C_{B,ab'_B}(w[b']J[b]x).$$

These are exactly the requirements after the update. Thus (8.7)–(8.8) prove preservation, with no assumption that the jump fixes earlier targets. At $j = 1$ there is no earlier target to protect. Backward induction defines the whole plan, and forward induction executes it.

Next plan the pivots in decreasing order. Once the plans at pivots larger than i are fixed, take all coordinatewise products of their possible multiplier tuples; call this finite list $\mathcal{D}_i$. Use $\mathcal{D}_i$ as the list of d 's in the terminal requirements at pivot i . The preceding construction returns multiplier options depending only on that list and the fixed data, so there is no dependence on an unknown incoming scale. The last pivot starts with the singleton future list $\{(1, \dots, 1)\}$.

Starting all scales at 1, enumerate all intermediate and terminal paths through these finite alternatives. This gives finite scale lists and finitely many rational palette indices ab_B at every checkpoint. Define $\mathcal{B}$ to contain the final scales and $\mathcal{A}$ to contain all these indices and all needed subset products of final scales. Now choose q_0 divisible by the denominators of every rational number $\gamma b_A/b_U$ used in any evaluated word on any enumerated path. This does not affect the abstract recurrence choices: q_0 merely changes the homomorphism interpreting each old-scale generator.

Finally enumerate all prefixes of all words, their cumulative jumps, and the terminal tests, with the now integral exponents. Their sums in the auxiliary translation coordinates give a finite slot list, which we enlarge to contain 0. Intermediate word evaluations use only this list. One may assign arbitrary palettes at other slots when defining the coloring on the whole acting group. Every step of the construction used only n, r, s . $\square$

8.2 Integer residue shifts and polynomial arrays

Fix a pivot i having targets, and condition initially on the earlier raw variables. Write $p = t_i$, set $L = MW^w$, and, for every pair $e = (U, A)$ at this pivot, write $U = C \cup \{i\}$ and define

$$q_e = \frac{h_A}{q_0 h_U} t_A, \quad t_C r_e \equiv q_e \pmod{L}, \quad 0 \leq r_e < L, \quad \Delta_e = \frac{q_e - t_C r_e}{M}. \quad (8.9)$$

These are integers for all sufficiently large w . In fact h_A/h_U is a multiple of W^w , which is eventually divisible by $q_0 W$, and t_C is a unit modulo the smooth modulus L . Consequently

$$W \mid q_e, \quad W \mid r_e, \quad W^w \mid \Delta_e. \quad (8.10)$$

The last assertion follows from the full modulus MW^w in the congruence. In particular every fixed positive integer eventually divides Δ_e . For an integer slot vector v put $s(v) = \sum_e v_e r_e$. The identity realizing one own-target letter is

$$t_C(p + mr_e) + mM\Delta_e = t_C p + mq_e \quad (m \in \mathbb{Z}). \quad (8.11)$$

It uses no divisibility of q_e by t_C .

Let $D_i^- = \prod_{\ell < i} X_\ell^2$ and use microcells in the pivot variable of length

$$R = M \left\lfloor H_i^{1/2}/M \right\rfloor, \quad N_{\text{loc}} = R/M. \quad (8.12)$$

Admissibility implies that R dominates every fixed power of $M + D_i^-$, that $D_i^- R/H_i \rightarrow 0$, and that $R/X_i \rightarrow 0$. Since $W^w \leq M$, all slot shifts and all bounded array evaluations below have size at most a fixed power of $M + D_i^-$ and are $o(R)$ in the pivot variable.

After discarding partial endpoint cells, condition on a microcell and on $r_0 = p \bmod M$. Its first point of that residue is p_* , and

$$p = p_* + M\ell, \quad 0 \leq \ell < N_{\text{loc}}.$$

Because r_0 is a W -unit, every point of this progression is a W -unit. Its harmonic weights have ratio $1 + O(R/X_i)$, so the conditional law of ℓ differs in total variation by $o(1)$ from the uniform law, uniformly in the earlier variables.

For $B = T \cup \{i\}$, put $t = t_T$. For every prescribed slot and every prescribed bounded integer array index $u = (u_e)_{e \in (B,A)}$, the numerical arguments we will use are

$$t(p + s(v)) + M \sum_{e \in (B,A)} u_e \Delta_e. \quad (8.13)$$

Except on a set of conditionings of probability $o(1)$, all these arguments, for the whole microcell, lie in one H_i -interval for each B . Here and below the constants may depend on the fixed finite word lists. To verify the boundary assertion uniformly, in a dyadic pivot range $[Y, 2Y)$ the preimages of H_i boundaries have $O(Yt/H_i + 1)$ neighborhoods of length $O(R)$. An interval of length a has $a\phi(W)/W + O(\phi(W))$ W -units. Relative to the W -units in $[Y, 2Y)$, these neighborhoods therefore cost

$$O\left(\frac{t(R+W)}{H_i} + \frac{R+W}{Y}\right) = o(1).$$

Harmonic weights change this bound by at most a constant on each dyadic range. Averaging the ranges and using $t \leq D_i^-$ proves the assertion. Endpoint cells have still smaller cost.

On a good conditioning choose the nilsequence formula for $S_{B,a,c}$ at the common interval and the residue $r_v = t(p + s(v)) \bmod M$. For each v, a, c , write that formula as $F(g^k x \Gamma)$, using a group representative x and absorbing the fixed origin of k into x . Retain the entire group-valued polynomial array

$$u \mapsto g^{(t(p+s(v))-r_v)/M + \sum_{e \in (B,A)} u_e \Delta_e} x. \quad (8.14)$$

Only finitely many evaluations must agree with the model; the array itself is defined for every u .

We record why these arrays lie in fixed compact nilmanifolds. If G has lower central Lie algebra series $\mathfrak{g}_j$, let $\mathfrak{a}_q(G)$ consist of polynomials in q real variables whose constant coefficient is in $\mathfrak{g}$ and whose coefficient of total degree $j > 0$ is in $\mathfrak{g}_j$. Pointwise brackets preserve this space: a bracket of degrees j, k lies in $\mathfrak{g}_{j+k}$ when both are positive, and a

bracket with a constant coefficient lies in a still deeper allowed level. It is a finite-dimensional rational nilpotent Lie algebra of step at most s . Its simply connected group $\mathcal{P}_q(G)$ consists of the corresponding arrays, with pointwise multiplication.

$$\mathcal{P}_q(\Gamma) = \{a \in \mathcal{P}_q(G) : a(z) \in \Gamma \text{ for every } z \in \mathbb{Z}^q\}$$

is a lattice. For discreteness, sufficiently many integer evaluations determine every coefficient of the logarithm; a convergent sequence of lattice-valued arrays is eventually constant at those evaluations and hence constant. For cocompactness, choose a rational basis respecting the lower central series. The change between logarithmic coordinates and ordered Mal'cev coordinates is a rational polynomial with zero constant term. Clearing its finitely many denominators shows that the exponential of a sufficiently divisible integral coefficient vector is lattice-valued at every integer input. In a rational Mal'cev basis for $\mathfrak{a}_q(G)$, fixed integer multiples of the basis vectors therefore exponentiate into $\mathcal{P}_q(\Gamma)$. Successively reducing the ordered coordinates modulo these elements, starting with the quotient by the next ideal in the Mal'cev flag, places every coset in a bounded coordinate box. This proves cocompactness.

Integer input shifts $u \mapsto u + \mathbf{e}$ preserve the Lie algebra and this lattice, and evaluation at a fixed integer input induces a smooth map of the quotients. Formula (8.14) belongs to this group: in its BCH logarithm a term of degree j in u contains at least j occurrences of $\log g$, and hence lies in $\mathfrak{g}_j$.

For each target take the product over all v, a, c of these array quotients, denoting it by K_B/Λ_B , and put

$$K/\Lambda = \prod_B K_B/\Lambda_B.$$

The choices range over a fixed finite list because the original model complexity is bounded. Increasing ℓ by 1 multiplies each array on the left by the constant array g^t . Thus the joint state is a linear orbit on K/Λ , adapted to its lower central filtration and of step at most s . Denote its state at ℓ by $Y(\ell)$. For an own pair $e = (B, A)$ write $\mathcal{S}_e$ for the lattice-preserving automorphism of the whole component K_B shifting the corresponding input coordinate by 1.

8.3 Typical marginal cube invariance

The state $Y(\ell)$ now stores every model reading needed at this pivot. We must show that, for typical conditionings, its marginal cube laws permit the array shifts $\mathcal{S}_e$. Removing the rough tail factors from their sampling steps will supply these symmetries.

Lemma 8.2 (Conditional face invariance). *Fix the bounded-complexity model family and the finite lists above. For a good conditioning on the earlier variables, the pivot microcell, and its residue, form the joint orbit $Y(\ell)$ just constructed. Fix a cube dimension $a \leq s + 1$, a positive integer d , constants $0 < c < C$, and a bound for the supremum and Lipschitz norms of tests. Uniformly over boxes for $(\ell_0, \dots, \ell_a)$ in $[-CN_{\text{loc}}, CN_{\text{loc}}]^{a+1}$ with sides at least cN_{loc} , all fixed residue vectors modulo d , all such tests of one marginal cube, and every own-pair automorphism, the difference between the marginal cube average and its image under that automorphism on an upper codimension-one face tends to zero in probability over the conditioning.*

Consequently, from any sets of conditionings of probability bounded away from zero one can select a sequence on which, simultaneously for all these fixed data, the marginal face invariances hold. On a further subsequence to which Proposition 7.4 applies, every projected local cube coset has the corresponding face symmetry in dimensions through $s + 1$. The modulus in this last assertion may be the period selected by that Proposition.

Proof. Fix a target $B = T \cup \{i\}$ and write $t = t_T$. Formulae outside the conditioned cell mean the extensions of the already selected nilsequence formulae, so all bounded-multiple boxes in the statement are defined. We will compare them with averages whose step has no factor t .

An exposure retaining polynomial dependence. We prove the assertion in probability by changing the order of exposure. We must freeze the target's model formula while leaving its tail variables available for progression comparison. A fixed pivot microcell alone does not do this as the tail varies. Instead we condition on a short bin for tp ; the eventual microcell origin will enter the base coordinate x_0 , not a coefficient of the polynomial family. Expose the earlier variables outside T , the residues modulo $L = MW^w$ of those inside T , dyadic bins $[S_\ell, 2S_\ell)$ for $\ell \in T$, the pivot residue $r_0 \bmod M$, and a bin for tp of width $t_{10}R$, where $t_{10} = \prod_{\ell \in T} S_\ell$. Do not expose the exact pivot microcell at this stage. Then $t_{10} \leq t \leq 2^{|T|}t_{10}$. Every r_e is fixed: earlier residues determine $q_e \bmod L$ and $t_C \bmod L$. For own pairs $e = (B, A)$ the full integer q_e is fixed, because $A \cap T = \emptyset$. Hence all slot shifts $s(v)$ are fixed. So are

$$r_* = tr_0 \bmod M, \quad r_v = t(r_0 + s(v)) \bmod M.$$

We may discard exposed bins whose entire preimage in p , for every t in the indicated product of dyadics, is not inside $[X_i + 2R, X_i^2 - 2R]$. Such a discard forces the actual p into a fixed multiplicative neighborhood of a cutoff endpoint, up to an $O(R)$ enlargement. Its harmonic probability is $O_n(1/\log X_i) + o(1)$. Also discard tp bins within a sufficiently large fixed multiple of $t_{10}R$ of an H_i boundary. The boundary count preceding the Lemma gives probability $o(1)$ for this discard. The constant is chosen to cover the given bounded-multiple boxes, all slots, and all required bounded array indices. On the remaining bins the interval and residue selecting every formula for this target are fixed throughout the exposure. Thus all its parameters F, g, x are fixed as the variables in T vary.

For later use, the conditional law of those variables is dominated by a constant, depending only on n , times the product of uniform laws on their dyadic bins and prescribed L residues. Before conditioning on the tp bin, their harmonic densities in each dyadic differ from uniform by at most a factor 2 per variable. If the exposed bin is $[Q, Q + t_{10}R)$, its preimage at a given t has length $t_{10}R/t$, between $2^{-|T|}R$ and R . It lies fully inside the pivot cutoff. On the residue $r_0 \bmod M$ the unnormalized pivot mass there is

$$(1 + o(1)) \frac{1}{M} \int_{Q/t}^{(Q+t_{10}R)/t} \frac{du}{u} = (1 + o(1)) \frac{1}{M} \log \left(1 + \frac{t_{10}R}{Q} \right),$$

uniformly in t . The discretization error is relatively $O(M/R)$, and the common normalization cancels. In particular conditioning on this bin changes the preceding product comparison by a bounded factor. This is the required domination; no exact microcell has been fixed in deriving it.

Putting the tail product in the sampling step. Let p_* be the first point of $r_0 \bmod M$ in the actual microcell eventually selected. Put $Y_* = \lfloor Q/M \rfloor$ and change the base and increment coordinates by

$$x_0 = \frac{t(p_* + M\ell_0) - r_*}{M} - Y_*, \quad x_j = t\ell_j \quad (1 \leq j \leq a). \quad (8.15)$$

The unknown cell origin has entered x_0 rather than a polynomial coefficient. Since p_* is within R of the actual pivot, these coordinates lie in boxes in a fixed multiple of

$$Z = \frac{t_{10}R}{M},$$

with all sides comparable to Z . The original residue restrictions on $\ell_0, \dots, \ell_a$ become one residue vector modulo dt in x . Changing an endpoint by less than dt has vanishing relative counting cost, so these are precisely the permitted normalized box averages.

At vertex $\omega \in \{0, 1\}^a$, set $X = x_0 + \sum_{j=1}^a \omega_j x_j$. Its array exponent is

$$X + Y_* + \frac{r_* + ts(v) - r_v}{M} + \sum_{e=(B,A)} u_e \frac{q_e - tr_e}{M}. \quad (8.16)$$

Every parameter here except X and t was fixed by the exposure. The displayed exponent is a polynomial of degree at most two, with its only possible quadratic terms of the form tu_e . BCH then shows that the logarithms of the arrays, viewed in the fixed array group, are ordinary polynomials jointly in x and the variables t_ℓ , $\ell \in T$, of bounded degree. There is no bound required on their coefficients. This joint polynomiality is the hypothesis that permits Proposition 6.3.

Removing each factor of the step. We compare the sampling in a class modulo dt with the sampling in the same class modulo d , removing the factors of t in a fixed order. Consider removal of t_ℓ , and condition on all other factor values. Let w' be the product of the factors still in the step after this removal. The current step is $dw't_\ell$. There is a representative a_0 for its base coordinate modulo dw' which can be chosen independently of t_ℓ , after allowing all the finitely many residues modulo d . Indeed the base congruence modulo w' is

$$M(x_0 + Y_*) + r_* \equiv 0 \pmod{w'}, \quad (8.17)$$

and the increment congruences are $x_j \equiv 0 \pmod{w'}$. Since w' is a product of W -units, $(M, w') = 1$, and also $(d, w') = 1$ for large w . The Chinese remainder theorem therefore gives representatives $a_j \in [0, dw')$ determined by these congruences and the chosen d residues, independently of t_ℓ .

Substitute $x_j = a_j + dw'y_j$. The current restriction becomes a single class vector modulo t_ℓ in y , while removing it gives unrestricted integer y . No coprimality of w' and t_ℓ is used: both original classes are nested congruences, and division of $x_j - a_j$ by the integer dw' is exact. The new common box scale is $Z/(dw')$. It dominates every fixed power of S_ℓ , uniformly over the other factor values, by the choice of R . The substitution has coefficients independent of t_ℓ , so joint log-polynomiality in (y, t_ℓ) is retained. The variable t_ℓ lies in a fixed residue modulo $L = MW^w$ in $[S_\ell, 2S_\ell)$, with

$$S_\ell/L \longrightarrow \infty.$$

All hypotheses of Proposition 6.3 hold.

That Proposition bounds the exceptional proportion uniformly in the polynomial coefficients, all permitted boxes, class vectors, and tests. Its uniformity in endpoints is essential: the exact microcell and hence the endpoints may depend on the pivot sampled after the exposure. For each choice of other factors the exceptional proportion in t_ℓ is $o(1)$ with a uniform bound. Integrate first against the independent uniform dyadic/residue laws and then use the conditional domination already proved. Each removal therefore fails with probability $o(1)$. A union bound over the finitely many factors and the triangle inequality give agreement, to error tending to zero in probability, between the original marginal cube law and the law with only its d -residue restrictions. The same reasoning applies to a test composed with the fixed face automorphism; its Lipschitz bound is still fixed.

In this latter law, applying $\mathcal{S}_e$ on the upper face $\omega_j = 1$ replaces x_j by $x_j + \Delta_e$ in (8.16), at all slots and for all observables of this component simultaneously. By (8.10), $d \mid \Delta_e$

eventually. Moreover $|\Delta_e|/Z \rightarrow 0$ uniformly, since its size is bounded by a fixed power of $M + D_i^-$ whereas R dominates all those powers. Translation of the corresponding box changes normalized counting averages by $O(|\Delta_e|/Z + d/Z) = o(1)$. Comparing on both sides establishes the asserted face invariance. Taking the finite union over targets and own pairs is harmless. This argument proves marginal invariance; it makes no assertion about the other components.

The countable diagonal and the period. Enumerate the conditions consisting of a positive integer modulus, a cube dimension at most $s + 1$, bounds $C \in \mathbb{N}$, lower side bounds $c = 1/j$, Lipschitz bounds in $\mathbb{N}$, and tolerances $1/j$. The assertion just proved holds uniformly over the uncountably many endpoints and tests within each such condition. Each of the first finitely many conditions fails with probability tending to zero. Given events of probability at least $\epsilon > 0$ along a subsequence, choose an increasing number of initial conditions whose union of failures has probability less than $\epsilon/2$, and select one conditioning from the event outside that union. By making the initial list and the accuracy increase successively, we obtain a sequence satisfying every fixed condition with error tending to zero.

Now fix the finite choices of ambient groups along a subsequence and apply Proposition 7.4. Its rational filtration, period d_0 , and rational representatives may depend on this sequence. The face invariances have already been secured for every fixed positive integer modulus, so they hold for d_0 . For any fixed small relative box size $\alpha > 0$, they hold for base boxes near a given macroscopic point β and increment boxes near zero, with the residue restrictions specified in that Proposition. First take the sequence limit and then let $\alpha \downarrow 0$. The projected local cube Haar measure is therefore invariant under the face automorphism. Since it has full support on its compact cube coset and the automorphism is a homeomorphism, that coset is preserved. This proves the last assertion with the required order of limits. $\square$

8.4 From the word plan to positive conditional mass

Fix one of the finitely many incoming scale tuples at pivot i . Lemma 8.1 gives finitely many complete options, each recording a cumulative jump word, a resulting scale tuple, and its slot. Enlarge their number to one common bound $Q_i \geq 1$, and let $V_i \geq 1$ bound the number of possible slots, uniformly over incoming scales. These constants depend only on n, r, s .

For given earlier variables and an integer pivot value $x \in \mathbb{N}$, let $\mathcal{P}_i(b; x)$ be the following event: some allowed outcome b' from the incoming scale b satisfies, for every target $B = T \cup \{i\}$, every $d \in \mathcal{D}_i$, color c , and $D \subset \mathcal{E}_B$,

$$S_{B, b'_B d_B, c}(t_T x) > 2\tau \implies S_{B, b'_B d_B, c} \left(t_T x + \sum_{A \in D} \frac{h_A b'_A d_A}{h_B b'_B d_B} t_A \right) > \tau. \quad (8.18)$$

For the large w under consideration, every displayed offset is an integer. This follows either from admissibility and the finite rational lists, or directly from the integer powers and (8.9).

Lemma 8.3 (Conditional alignment mass). *For every fixed $\tau > 0$ and every permitted bounded-complexity family of models, outside a set of earlier-variable/microcell/residue conditionings of probability tending to zero, the conditional uniform pivot probability that*

$$\mathcal{P}_i(b; p + s(v)) \quad \text{holds for some prescribed slot } v$$

is at least $1/(2Q_i)$. The same statement holds for the conditional harmonic probability, after replacing the constant by $1/(3Q_i)$ if necessary. The constants are independent of the threshold and model complexity.

Proof. Suppose the uniform assertion fails on sets of conditionings of probability bounded below along a subsequence. By Lemma 8.2, choose a sequence in those sets with all its empirical face invariances. Pass to a subsequence fixing the ambient array nilmanifolds and using Proposition 7.4. There are only finitely many observable formulae in a conditioning. Their common supremum and Lipschitz bounds give, by compactness, a further subsequence on which all of them converge uniformly. Denote the limiting observables by $F_{B,a,c,v}$.

For each stored factor and integer input u , define the reading of a joint state $y \in K/\Lambda$ by

$$\mathcal{R}_{B,a,c,v,u}(y) = F_{B,a,c,v}(\text{ev}_u \pi_{B,a,c,v} y). \quad (8.19)$$

Here $\pi_{B,a,c,v}$ selects the indicated array factor and ev_u evaluates it at the integer input u . The coordinates of u are indexed by the own pairs (B, A) ; when adding u to a slot vector, insert zeros at all other pairs. These readings are continuous functions of y and are defined before choosing any lifts on a point-law coset.

Consider a point-law coset of that Proposition,

$$h(\beta)H\sigma_r\Lambda/\Lambda,$$

with its Haar image measure $\mu_{\beta,r}$. The preceding Lemma and Proposition 7.5 lift each own-pair array shift to a polynomial shear of the joint cover coordinates $z \in \mathfrak{h}$. Such a lift may move other target components. Adjoin its translation $v \mapsto v + \mathbf{e}_e$ and call the result L_e . Lemma 7.7 puts these transformations in a nilpotent group of step at most s , including the ordinary slot translations. Write $\Theta(z) = h(\beta)\exp(z)\sigma_r\Lambda$. At slot v , color a state by the palette

$$C_{B,a}(z, v) = (\mathbf{1}_{\{\mathcal{R}_{B,a,c,v,0}(\Theta(z)) > 3\tau/2\}})_{c \in [r]}. \quad (8.20)$$

These palettes have at most 2^r values, as required by the finite plan.

Apply that plan pointwise starting at any $(z, 0)$. It returns one of the Q_i options, a jumped center, and terminal palette equalities there. Only own letters occur in a terminal word for B . Write $m = (m_e)_e$ for their integer powers. Their projection on this entire component is exactly the commuting array shift $u \mapsto u + (m_e)_e$, while their slot effect is $v \mapsto v + m$, with the zero extension just specified. Consequently at a center state y and slot v the two readings are $\mathcal{R}_{B,a,c,v,0}(y)$ and $\mathcal{R}_{B,a,c,v+m,m}(y)$. This statement holds even if earlier jumps involved letters belonging to other targets.

To check the numerical meaning, use the corresponding prelimit observables and evaluate the same readings on an actual array state $Y(\ell)$, with $p = p_* + M\ell$. The center value is the model at $t(p + s(v))$. The translated value is the model at

$$t \left(p + s(v) + \sum_e m_e r_e \right) + M \sum_e m_e \Delta_e = t(p + s(v)) + \sum_e m_e q_e. \quad (8.21)$$

Each occurrence uses the correct residue-specific observable at its new slot. For the terminal word at outcome b' and future multiplier d , the powers are

$$m_e = q_0 \frac{b'_A}{b'_B} \frac{d_A}{d_B} \quad (e = (B, A), A \in D), \quad m_e = 0 \quad (A \notin D).$$

They are integers by the plan, and $m_e q_e = (h_A b'_A d_A / (h_B b'_B d_B)) t_A$. The leading index is $b'_B d_B$. Therefore (8.21) gives exactly (8.18) with $x = p + s(v)$. This numerical interpretation is used for terminal own-letter words. An arbitrary cumulative jump need not be a translation of the actual pivot orbit.

For an option o with scale outcome b' and slot v , let $F_o \subset K/\Lambda$ consist of the states y satisfying every required comparison. Explicitly, for each B, d, c, D , put $a = b'_B d_B$ and take m as above, and impose

$$\mathcal{R}_{B,a,c;v,0}(y) \leq 3\tau/2 \quad \text{or} \quad \mathcal{R}_{B,a,c;v+m,m}(y) \geq 3\tau/2. \quad (8.22)$$

These finitely many conditions are closed by continuity of the readings. They depend only on the state, the option's finite data, and the limiting formulae. In particular they do not depend on the choice of lifts. For every starting cover coordinate, the pointwise palette construction puts the K/Λ -valued coordinate of the jumped state into F_o for at least one option o . Equality of palettes implies (8.22), including at equality with the threshold.

Sample z from the expanding weighted boxes of Lemma 7.7. For each fixed option o , let Y_o be the K/Λ -valued coordinate of the state after its cumulative jump. The jump acts on the cover coordinate z by a fixed shear, so the law of Y_o converges to $\mu_{\beta,r}$. Pointwise coverage gives

$$1 \leq \sum_o \Pr\{Y_o \in F_o\}.$$

Using the closed-set direction of Portmanteau separately for the finitely many summands yields

$$1 \leq \sum_o \mu_{\beta,r}(F_o), \quad \mu_{\beta,r}\left(\bigcup_o F_o\right) \geq \frac{1}{Q_i}. \quad (8.23)$$

The last implication also follows by summing the indicator inequality $\sum_o \mathbf{1}_{F_o} \leq Q_i \mathbf{1}_{\bigcup_o F_o}$. This is the only quantitative loss in passing from pointwise recurrence to the point-law measure. Although the shears may depend on the coset, the sets F_o do not. Therefore (8.23) integrates over the macroscopic-position/residue mixture without a measurable choice of lifts.

Replace each comparison (8.22) by the open comparison

$$\text{center} < 7\tau/4 \quad \text{or} \quad \text{translated value} > 5\tau/4, \quad (8.24)$$

and take the same finite intersections and union. This defines an open set containing $\bigcup_o F_o$, so it has mixture measure at least $1/Q_i$. Proposition 7.4 gives weak convergence of the actual uniform conditional state laws to that mixture. Open-set Portmanteau therefore gives lower limit at least $1/Q_i$ for this open event. For sufficiently late terms, uniform convergence of all observables has error less than $\tau/4$. If the actual center is greater than 2τ , its limiting reading is greater than $7\tau/4$, forcing the second alternative in (8.24); the actual translated reading is then greater than τ . The good boundary condition ensures that these actual readings use exactly the selected model formulae. By (8.21) the open event implies success at some $p + s(v)$.

This contradicts the selected conditional probabilities being less than $1/(2Q_i)$. Hence the exceptional sets have probability tending to zero. The uniform-to-harmonic total variation estimate following (8.12) proves the last statement. The compactness subsequence and how late the estimates hold may depend on τ and the fixed model complexity. The constants Q_i and the displayed lower bounds do not. $\square$

8.5 Returning to the raw variables and successive pivots

The conditional conclusion permits a bounded list of shifts of the pivot. We now remove those shifts before proceeding to the next pivot. This step conditions only on earlier raw variables; translation invariance is not claimed inside a fixed microcell.

Let T_a denote translation by a . For the raw pivot law μ_i , put $X = X_i$. Lemma 3.1(2) gives, for $a \in W\mathbb{Z}$ with $|a| = o(X)$,

$$\|T_a\mu_i - \mu_i\|_{\text{TV}} \leq \frac{|a|}{X(\log X - W/X)} \leq (1 + o(1)) \frac{|a|}{X \log X}. \quad (8.25)$$

Here probability total variation is one half of the total-mass norm used in Lemma 3.1.

In our application each prescribed slot satisfies

$$W \mid s(v), \quad |s(v)| \leq CMW^w \leq CM^2 = o(X_i),$$

uniformly in all earlier raw variables. Thus (8.25) gives $o(1)$ uniformly for every such slot. In particular it remains valid after integrating against an arbitrary event depending on earlier variables.

Proof of Principle 2.4. For the given n, r, s take the lists and all budgets from Lemma 8.1. Let I be the set of pivots having targets and define

$$\delta = \prod_{i \in I} \frac{1}{3Q_i V_i} > 0, \quad (8.26)$$

with the empty product equal to 1. This definition precedes the choice of admissible parameters, models, and τ . Now fix any such parameters, any family of models of step at most s and bounded complexity in the sense of Definition 2.2, and any fixed $\tau > 0$.

Suppose E is an event of the earlier raw variables on a given finite path, with incoming scale b and all the earlier pivot requirements secured. Lemma 8.3, integrated over the pivot microcell and residue, implies

$$\frac{1}{3Q_i} \mathbb{P}(E) - o(1) \leq \sum_v \mathbb{P}(E, \mathcal{P}_i(b; p + s(v))). \quad (8.27)$$

The exceptional set of conditionings has probability $o(1)$ unconditionally, so its intersection with E also costs $o(1)$. Conditional on the earlier variables each $s(v)$ is fixed. Equation (8.25) therefore changes each summand by $o(1)$ when $p + s(v)$ is replaced by p . There are at most V_i slots, and consequently

$$\mathbb{P}(E, \mathcal{P}_i(b; p)) \geq \frac{1}{3Q_i V_i} \mathbb{P}(E) - o(1). \quad (8.28)$$

This estimate is uniform over the finite incoming-scale list and the finite path events. No conditional density assumption on E has been used.

On the successful extension choose the first successful scale outcome in a fixed enumeration. This is a measurable choice from a finite list and partitions that event into finitely many new path events. They depend only on raw variables through the current pivot. Apply (8.28) to each path at the next pivot. Pivots without targets impose no condition and can be skipped. Summing the finitely many path estimates and iterating yields total success probability at least $\delta - o(1)$.

For completeness, consider explicitly a requirement secured at pivot i when its resulting scale was $b^{(i)}$. Every later update multiplies the scale by one of the finite tuples used to construct $\mathcal{D}_i$. On any ensuing path, their complete product is some $d \in \mathcal{D}_i$. Requirement (8.18) at pivot i was required for this very d , at the original earlier raw variables and the original raw pivot. Later steps change none of those raw variables. Therefore at the final scale $b^{\text{fin}} = b^{(i)}d$ its leading index is exactly b_B^{fin} , and every offset ratio is exactly

$h_A b_A^{\text{fin}} / (h_B b_B^{\text{fin}})$. This verifies preservation between pivots in addition to the within-pivot preservation proved in (8.7)–(8.8).

Every successful final path thus gives $b^{\text{fin}} \in \mathcal{B}$ such that, simultaneously for every block B , color c , and $D \subset \mathcal{E}_B$,

$$S_{B, b_B^{\text{fin}}, c}(t_B) > 2\tau \implies S_{B, b_B^{\text{fin}}, c}\left(t_B + \sum_{A \in D} \frac{h_A b_A^{\text{fin}}}{h_B b_B^{\text{fin}}} t_A\right) > \tau.$$

For blocks with no adding pairs the sole comparison is the identity, which holds automatically. Taking the ordinary lower limit as $w \rightarrow \infty$ gives probability at least δ . The lists and δ depend only on n, r, s , whereas the rate of convergence may depend on the fixed threshold and model family. These are exactly the quantifiers of Principle 2.4. $\square$

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